

CBWeb3 Platform — Scenario A User Manual

Enhanced Correspondent Banking (dual-layer HTLC PvP settlement)

2026-08-05

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About this document

This manual compiles the per-portal user manuals for **Scenario A — Enhanced Correspondent Banking** into a single document, in the order used by the portal index: Bank, Governance, Treasury, Supervisor, NOC, and finally the cross-portal end-to-end walkthrough.

The chapters are the Markdown manuals kept in the repository under `docs/user-manuals/`. Each manual’s own table of contents has been replaced by the single table of contents at the front of this document; the text is otherwise unchanged. Cross-references between manuals resolve to the corresponding chapter of this PDF.

Generated on 2026-08-05.

Implementation status — read this first

Every portal below is wired to a live backend through the API Gateway. The manuals only describe features that exist in the shipped product, and each manual states at the top exactly where its data comes from. A portal’s screens are populated only when the backend services it calls are reachable; in an offline or demo environment with the backends down, screens show empty tables or zero counters rather than fabricated data.

Portal	Scenario A	Scenario B	Data source today
Bank	Live backend	Live backend	Real API (API Gateway)
Governance	Live backend (Dashboard + Registry exposed)	Live backend	Real API; Scenario A retains a development mock toggle
Treasury	Live backend (approvals exposed)	Live backend	Real API (API Gateway)
Supervisor	Live backend (connectivity-dependent)	Live backend (connectivity-dependent)	Real API; every screen calls live endpoints. See the note below for auth/data caveats
NOC	Live backend	Live backend	Real API — connects to the live NOC backend service

Important: The **Supervisor** portal’s screens call real backend endpoints through the API Gateway — it is **not** a hardcoded mock-data or fake-login build, and its frontend holds no sample data. Two caveats remain, documented per screen in each manual:

- **Data freshness.** When the backend services are offline (as in most local/demo setups) screens show empty tables or zero counters, and in a development environ-

ment the endpoints may return demonstration data rather than a live production data set.

- **Authentication.** Scenario B authenticates against the platform's OIDC provider (Keycloak). In Scenario A, production Keycloak enforcement is not yet active, so the login is not a production security boundary.

Do not treat figures shown in an offline or development environment as authoritative for any real supervisory decision.

Conventions used in every manual

- **Route** — the in-app URL path (the part after the portal's base address).
- **Sidebar label** — the name as it appears in the portal's left navigation.
- **Status / state tables** — the exact status values you will see in the UI and what each one means.
- **Planned / not yet available** — a feature that exists in the design but is not exposed in the current build. These are called out explicitly and never documented as if they work.

Bank Portal — User Manual

(Scenario A)

Audience: Commercial bank operators participating in a CBWeb3 spoke. **Data source:** Live backend (real API Gateway). **Scenario:** Enhanced Correspondent Banking — dual-layer HTLC PvP settlement.

Overview

The Bank Portal is the day-to-day operational interface for a commercial bank in **Scenario A — Enhanced Correspondent Banking**. Through it you manage the full digital-currency lifecycle:

- Obtain tCeBM (tokenised Central Bank Money) by requesting issuance from the Central Bank.
- Convert approved fiat reserves into on-chain tCeBM tokens (reserve tokenisation).
- Redeem tCeBM back to fiat reserves when no longer needed for settlement.
- Propose and accept FX trade agreements with counterparty banks on other spokes.
- Execute atomic PvP (Payment vs. Payment) cross-border settlements using Hash Time Lock Contracts (HTLCs).

Actor	Portal	Role
Commercial Bank Operator	Bank Portal	Manages balances, requests issuance/redemption, runs FX trades and PvP settlements

Access & Login

URL: The Bank Portal URL provided by your system administrator (e.g., `http://localhost:5173` in a local deployment).

Credentials: A **Username** and **Password** for your institutional operator account. The username must be at least 3 characters; the password at least 6 characters.

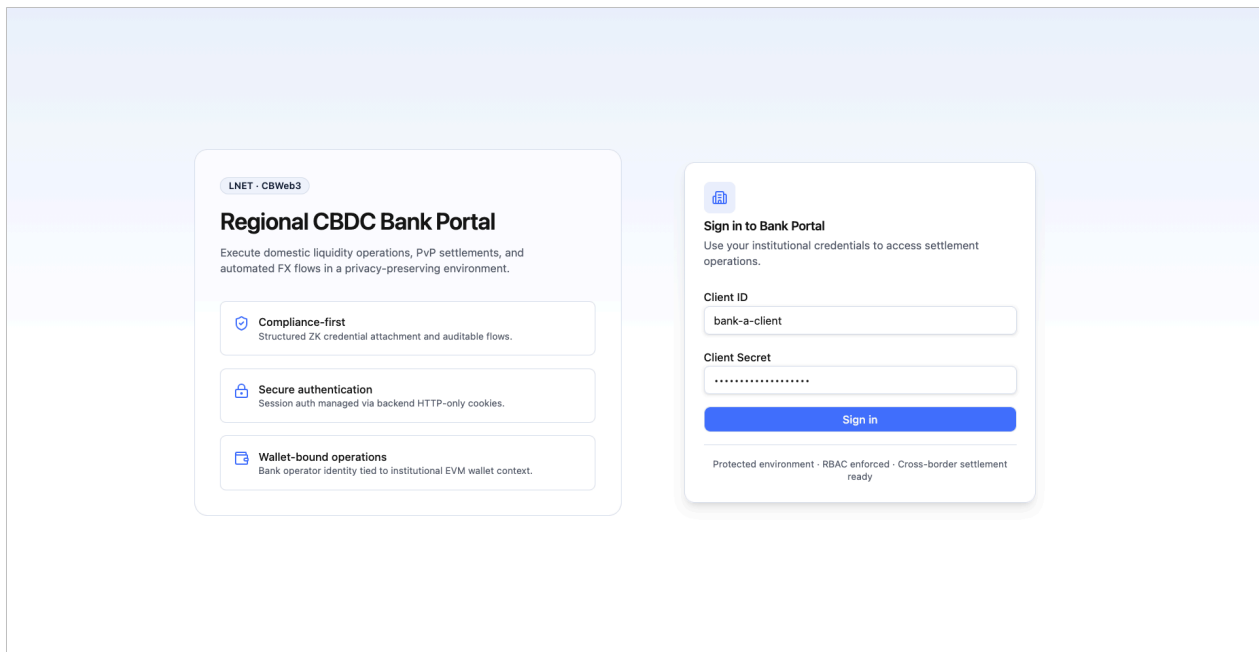


Figure 1: Bank Portal Login Screen

Signing in

1. Open the Bank Portal URL in your browser.
2. Enter your **Username** and **Password**.
3. Click **Sign in**.

On success you are redirected to the Dashboard. A **Signed in** toast appears in the top-right corner (with the description “Welcome back to the Bank Portal.”).

First login / new institution: If you see an “unauthorized” error immediately after entering correct credentials, your institution may not yet be fully onboarded. Complete the Onboarding flow first (see [Onboarding](#)).

Session management: Authentication is handled via backend HTTP-only cookies. The session is tied to your institutional EVM wallet context.

Navigation

The left-hand sidebar exposes the following pages. All items below are active in the current build.

Sidebar label	Route	Purpose
Dashboard	/	Balances, pending counts, recent HTLC activity
Statement	/statement	Chronological ledger of tokenized-fiat and tCeBM movements
Issuance Requests	/deposits	Request tCeBM issuance from the Central Bank
Reserve Tokenisation	/escrows	Convert approved fiat reserves into tCeBM

Sidebar label	Route	Purpose
Redeems	<code>/redeems</code>	Redeem tCeBM back to fiat reserves
Trade Agreements	<code>/agreements</code>	Manage FX agreements with counterparty banks
PvP Settlement	<code>/htlc</code>	Initiate and track atomic cross-border HTLC transfers
Onboarding	<code>/onboarding</code>	Register a new institution with the Central Bank

Not available in this build (Scenario A): Liquidity Transfers, AMM Trading, Compliance Center, and Settings are present in the codebase but are not exposed in the sidebar for Scenario A. Do not attempt to access them directly.

Screens

Dashboard (/)

The Dashboard provides a real-time snapshot of the institution's financial position and active settlement contracts. Data is fetched from the live backend each time the page loads.

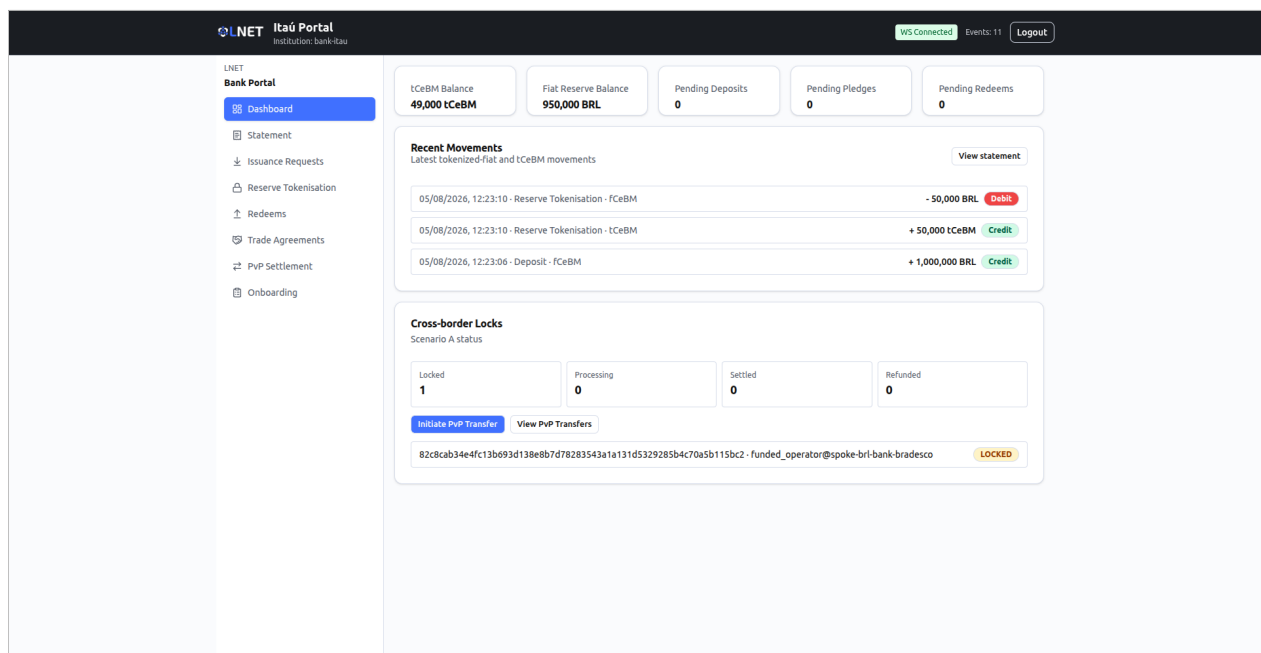


Figure 2: Bank Portal Dashboard

Balance widgets

Widget	Description
tCeBM Balance	Current tCeBM token balance held by the institution's EVM wallet.
Fiat Reserve Balance	Current fiat reserve balance registered in the spoke.

Pending transaction counters

Counter	What it counts	Where to act
Pending Deposits	Issuance requests awaiting Central Bank approval.	Issuance Requests (/deposits)
Pending Pledges	Reserve tokenisation requests awaiting approval.	Reserve Tokenisation (/escrows)
Pending Redeems	Redemption requests awaiting approval.	Redeems (/redeems)

Cross-Border Locks

A summary card shows HTLC contract counts by lifecycle state:

Counter	Corresponding HTLC state
Locked	HTLC_STATE_LOCKED — contract active, awaiting counterparty claim
Processing	HTLC_STATE_SETTLING or HTLC_STATE_REFUNDING — on-chain operation in progress
Settled	HTLC_STATE_SETTLED — atomic swap completed
Refunded	HTLC_STATE_REFUNDED — funds returned after expiry

Below the counters, the 5 most recent HTLC contracts are listed with their Contract ID, receiver identity, and state badge. Two quick-action buttons are available: **Initiate PvP Transfer** (goes to [/htlc/new](#)) and **View PvP Transfers** (goes to [/htlc](#)).

Recent Movements

A **Recent Movements** card lists the 5 latest tokenized-fiat and tCeBM movements (timestamp, type, token, amount, and a Credit/Debit badge). A **View statement** button opens the full [Statement](#) page.

Statement ([/statement](#))

The Statement page provides a consolidated, chronological record of tokenized-fiat (fCeBM) and tCeBM movements received and sent by your institution.

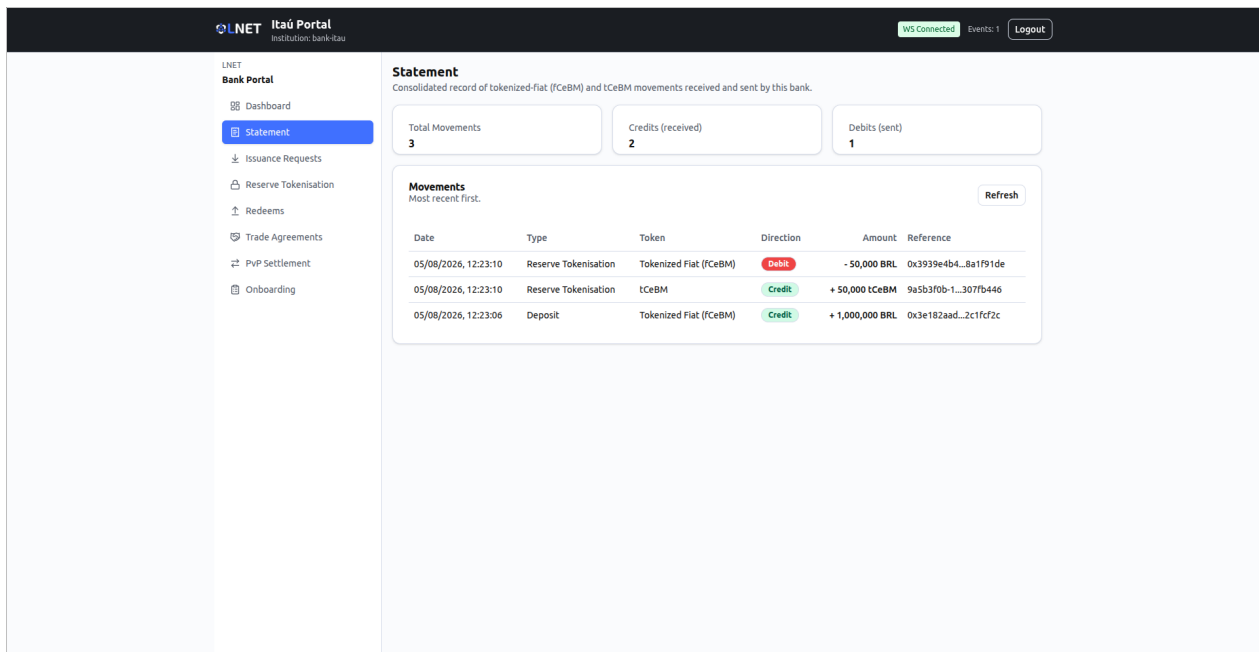


Figure 3: Statement Page

Three summary counters are shown at the top: **Total Movements**, **Credits (received)**, and **Debits (sent)**.

Use the **Refresh** button to reload the movements list (the page does not auto-refresh).

Movements table

Column	Description
Date	Movement timestamp
Type	Movement kind — Deposit, Reserve Tokenisation, Redeem, or PvP Settlement
Token	Tokenized Fiat (fCeBM) or tCeBM
Direction	Credit (received) or Debit (sent) badge
Amount	Signed amount (+ for credits, - for debits)
Reference	On-chain reference hash (truncated; hover for full value)

Onboarding (/onboarding)

Before using any operational features, a new commercial bank must complete the onboarding process. This registers the institution with the Central Bank and provisions an EVM wallet.

The wizard progresses through four steps (the stepper labels them Operator, Institution, KYC, Complete):

1. Operator Confirmation → 2. Institution Data → 3. KYC Approval → 4. Complete

Step 1: Operator Confirmation

The screen displays your current Operator ID. Verify it matches your institutional account and click **Start Onboarding**.

Step 2: Institution Data

Fill in your institution's registration details.

Field	Constraints
Institution Name	Minimum 2 characters
Bank Code	1–8 characters; lowercase letters (a–z), digits (0–9), or hyphens (–) only
Country	ISO 3166-1 alpha-2 code (e.g., BR); exactly 2 characters
Role	Select Commercial Bank or Treasury Bank
Username	Minimum 3 characters; lowercase letters, digits, underscores, or hyphens only
Email	Valid email address

Click **Submit Onboarding Request**. Inline validation errors appear below each field if constraints are violated.

Step 3: KYC Approval

The request enters **PENDING** state. The portal polls the backend every **5 seconds** automatically; you can also click **Refresh status** manually.

Element shown	Description
Status badge	PENDING , CREDENTIAL_REQUESTED , or APPROVED
Request ID	Keep this for reference in communications with the Central Bank
Wallet	EVM wallet address (may show – until the Central Bank processes the request)
Elapsed	Time since submission

Keep this browser tab open while waiting. The wizard will automatically advance when the Central Bank acts.

A **Central Bank Governance operator** must log into the Governance Portal, navigate to the Registry, and click **Approve Onboarding** for your pending request. They must provide a reason of at least 10 characters.

Step 4: Complete

Once approved, the portal finalises onboarding: it exchanges the approval for a PKI credential and activates the institution account on-chain.

The completion screen shows:

- **Wallet Address** — the EVM wallet permanently associated with this institution.
- **Transaction Hash** — the on-chain activation transaction.
- **PKI login badge** — **PKI login validated** or **PKI login pending**.

After completion, all other portal features become available.

Onboarding status reference

Status	Meaning
NONE	No request submitted; wizard starts at Step 1
PENDING	Submitted and awaiting Central Bank review
CREDENTIAL_REQUESTED	System is internally requesting the PKI credential (shown as Pending to the operator)
APPROVED	Central Bank approved; final activation in progress
KYC_APPROVED	Central Bank approved (KYC cleared); treated the same as APPROVED — final activation in progress
ACTIVE	Fully onboarded; all features accessible
REJECTED	Central Bank rejected the request; contact governance support
REVOKED	Access revoked after activation; contact governance support
FROZEN	Account frozen by Central Bank governance; contact governance support

Issuance Requests (/deposits)

Use this page to request the Central Bank to **issue tCeBM** against your institution's fiat collateral. Tokens are only minted after the Central Bank approves the request.

Bank Portal
Institution: 32c42881-b6ec-427f-8645-1f7ac8f9d3ca

WS Connected Events: 22 Logout

Bank Portal

- Dashboard
- Issuance Requests**
- Reserve Tokenisation
- Redeems
- Trade Agreements
- PvP Settlement
- Onboarding

tCeBM Balance
3,000 tCeBM

Fiat Reserve Balance
7,000 BRL
Mirrors commercial bank fiat reserves

Pending Issuance Requests
0

Request Token Issuance
Submit fiat collateral proof to request tCeBM issuance by the central bank.

Amount (BRL)
5000

[Review Issuance Request](#) [Refresh](#)

Issuance Requests
Total records: 2

ID	Fiat Amount	Status	Issuance Reference	Rejection Reason	Created At
197397a8-5ec1-43b1-9c69-dbed7c245caf	10,000 BRL	APPROVED	0xe2cc874e...1e4eaaee	-	28/05/2026, 12:27:05
5b163e00-d9c2-47c4-9a0e-4865e006e399	15,000 BRL	REJECTED	-	10000 issuance was approved	28/05/2026, 12:27:26

Figure 4: Deposits Page

Balance widgets

The page displays your current **tCeBM Balance**, **Fiat Reserve Balance**, and the count of **Pending Issuance Requests**.

Submitting an issuance request

1. Enter the **fiat amount** in the amount field. Must be a **positive whole number** (decimals are not accepted).
2. Click **Review Issuance Request** to trigger client-side validation.
3. A confirmation card appears showing the formatted amount.
4. Click **Confirm Request** to submit to the Central Bank.

A success toast displays the new request ID. The request appears in the table with status **PENDING**.

Use the **Refresh** button to reload the table at any time (the page does not auto-refresh).

Issuance Requests table

Column	Description
ID	Unique identifier for the request
Fiat Amount	The fiat amount submitted
Status	Current status (see Status Reference)
Issuance Reference	On-chain mint transaction hash (truncated; hover for full value) — populated after approval
Rejection Reason	Central Bank's reason if the request was rejected
Created At	Submission timestamp

Reserve Tokenisation (/escrows)

After a deposit request reaches **APPROVED** status, use this page to **convert the approved fiat reserve into tCeBM tokens** (a tokenisation request, also called a pledge). The Central Bank executes a burn-and-mint operation on approval: old fiat-backed tokens are burned and new tCeBM is minted to your institution's wallet.

Only submit a tokenisation request after a Deposit has been approved. Submitting without a corresponding approved deposit will be rejected.

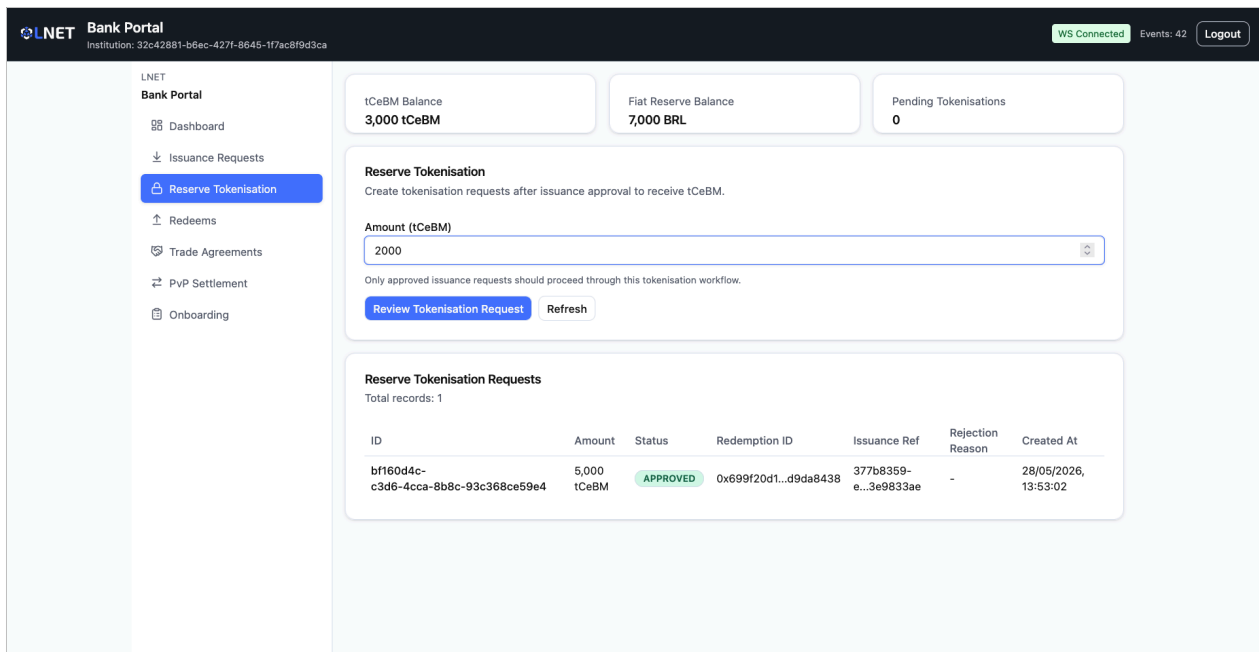


Figure 5: Escrows Page

Submitting a tokenisation request

1. Enter the **tCeBM amount** in the amount field. Must be a positive whole number.
2. Click **Review Tokenisation Request**.
3. Review the confirmation card and click **Confirm Request**.

Reserve Tokenisation Requests table

Column	Description
ID	Unique identifier
Amount	tCeBM amount requested
Status	Current status
Redemption ID	Burn transaction hash (populated after approval)
Issuance Ref	Mint transaction hash (populated after approval)
Rejection Reason	Central Bank's reason if rejected
Created At	Submission timestamp

Redeems (/redeems)

Use this page to **convert tCeBM back to fiat reserves**. The system automatically executes a Zeto (privacy-preserving) token transfer before forwarding the request to the Central Bank for fiat release approval.

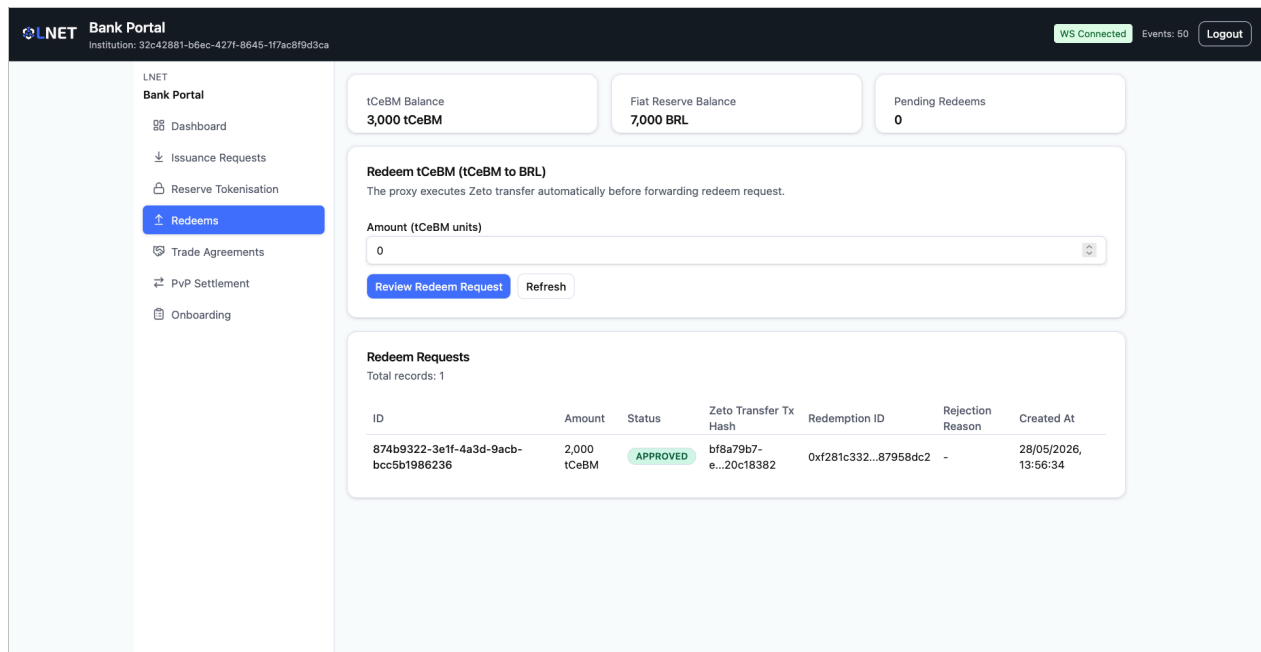


Figure 6: Redeems Page

Submitting a redemption request

1. Enter the **tCeBM amount** to redeem. Must be a positive whole number.
2. Click **Review Redeem Request**.
3. Review the confirmation card (the Central Bank fiat release approval message is shown).
4. Click **Confirm Request** to submit.

The system executes the Zeto transfer automatically. Monitor the status column for progress.

Redeem Requests table

Column	Description
ID	Unique identifier
Amount	tCeBM amount redeemed
Status	Current status
Zeto Transfer Tx Hash	Privacy-preserving token transfer hash (populated by the system automatically)
Redemption ID	Fiat release transaction hash (populated after Central Bank approval)
Rejection Reason	Central Bank's reason if rejected
Created At	Submission timestamp

Trade Agreements

Agreement Inbox (/agreements)

Lists all FX trade agreements associated with your institution, both proposed by you and received from counterparties.

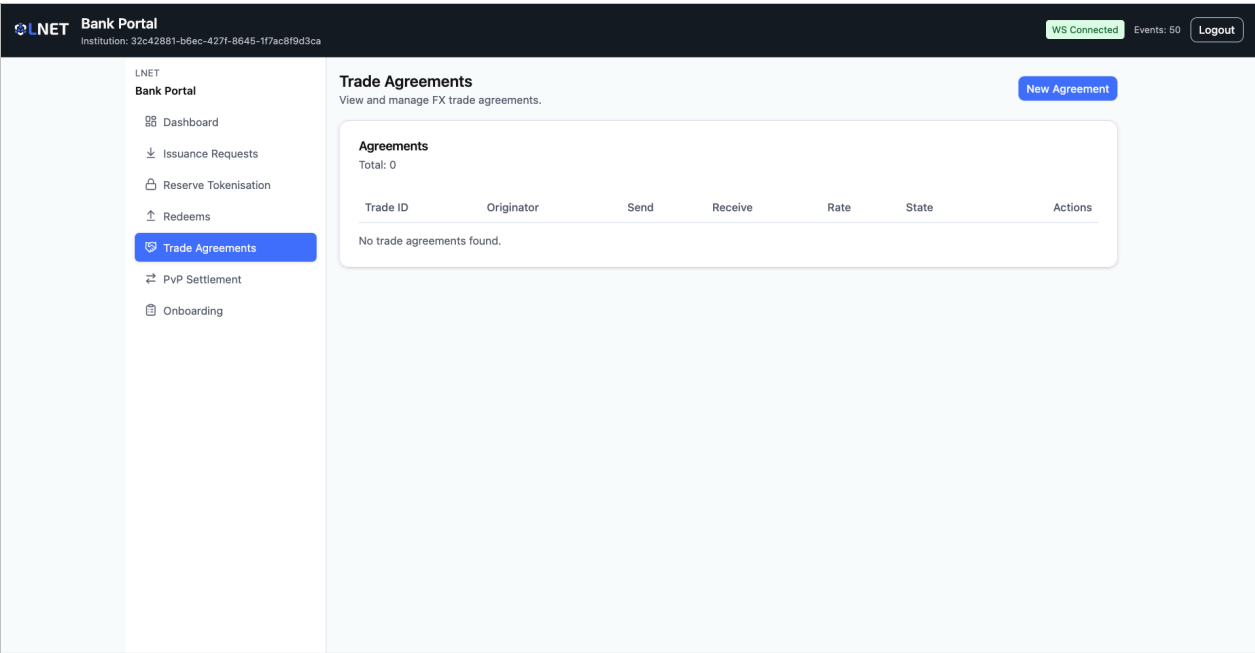


Figure 7: Trade Agreements Inbox

Column	Description
Trade ID	Unique identifier (truncated; click to open detail)
Originator	Institution that proposed the agreement
Send	Amount and currency the originator sends
Receive	Amount and currency the originator receives
Rate	Auto-calculated exchange rate
State	Current agreement state (see Status Reference)
Actions	View button — opens the Agreement Detail page

Click any row or the **View** button to open the detail. Click **New Agreement** (top right) to create a proposal.

Propose Agreement (/agreements/new)

Use this page to propose a new FX trade agreement with a counterparty on another spoke.

Bank Portal
Institution: 32c42881-b56c-427f-8645-1f7ac8f9d3ca

WS Connected Events: 50 Logout

Bank Portal

- Dashboard
- Issuance Requests
- Reserve Tokenisation
- Redeems
- Trade Agreements**
- PvP Settlement
- Onboarding

New Trade Agreement
Propose a new FX trade agreement with a counterparty.

[Back to agreements](#)

Parties
Specify the counterparty and intermediary addresses.

Counterparty B Identity
funded_operator@spoke-b-bank-d

Settlement Agent Identity
funded_operator@spoke-a-cb

Custodian Identity
funded_operator@spoke-b-bank-d

Beneficiary Identity
funded_operator@spoke-b-bank-b

Spoke-A HTLC Receiver
funded_operator@spoke-a-bank-c

Spoke-B HTLC Receiver
funded_operator@spoke-b-bank-b

Trade Terms
Define the amounts, currencies, and exchange rate.

Send Amount Send Currency

Figure 8: New Agreement Form

Paladin Identity format: Several fields require a Paladin Identity — the network-level identifier for a participant node. The format is:

`funded_operator@<spoke>-<entity>`

Examples for the standard test environment:

Institution	Paladin Identity
Central Bank — Spoke A	funded_operator@spoke-a-cb
Commercial Bank A — Spoke A	funded_operator@spoke-a-bank-a
Financial Correspondent — Spoke A	funded_operator@spoke-a-bank-c
Central Bank — Spoke B	funded_operator@spoke-b-cb
Commercial Bank B — Spoke B	funded_operator@spoke-b-bank-b
Financial Correspondent — Spoke B	funded_operator@spoke-b-bank-d

Contact your system administrator if you are unsure of the correct identity for a counterparty.

Parties section — Counterparty B, Settlement Agent, Custodian, and Beneficiary are required; the four spoke/receiver fields are optional. All party fields are dropdowns populated from the network’s Paladin identity roster:

Field	Description
Counterparty B Identity	(Required) Paladin identity of the receiving institution
Settlement Agent Identity	(Required) Paladin identity of the settlement intermediary
Custodian Identity	(Required) Paladin identity of the custodian
Beneficiary Identity	(Required) Paladin identity of the end beneficiary
Source Spoke ID	(Optional) Spoke identifier of the originating spoke (selected from the roster)

Field	Description
Destination Spoke ID	(Optional) Spoke identifier of the counterparty spoke (selected from the roster)
Source Receiver	(Optional) HTLC receiver identity on the originating spoke — used to auto-fill the PvP form when you initiate as Originator
Destination Receiver	(Optional) HTLC receiver identity on the counterparty spoke — used to auto-fill the PvP form when you continue as Counterparty

Trade Terms section:

Field	Description
Send Amount	The amount your institution will transfer
Send Currency	Read-only; fixed to this spoke's own currency (set at build time via <code>VITE_FIAT_SYMBOL</code> , e.g. BRL)
Receive Amount	The amount your institution expects to receive
Receive Currency	Currency of the receive leg, selected from the Latin American currency list: ARS, BOB, BRL, CLP, COP, CRC, CUP, DOP, GTQ, HNL, MXN, NIO, PAB, PEN, PYG, USD, UYU, VES
Exchange Rate	Auto-calculated from the two amounts (read-only; displayed as “1 {send} = {rate} {receive}”)
Expiry Date	Date and time after which the agreement can no longer be settled — must be at least 5 minutes in the future

After filling all required fields:

1. Click **Review Agreement** — client-side validation runs; inline errors appear for any invalid fields.
2. A confirmation card shows all parties and trade terms.
3. Click **Confirm Proposal** to submit.

On success, you are navigated to the Agreement Detail page for the new agreement, which is now in `PROPOSED` state.

Agreement Detail (/agreements/:tradeId)

Opened by clicking any agreement row in the inbox or after creating a proposal.

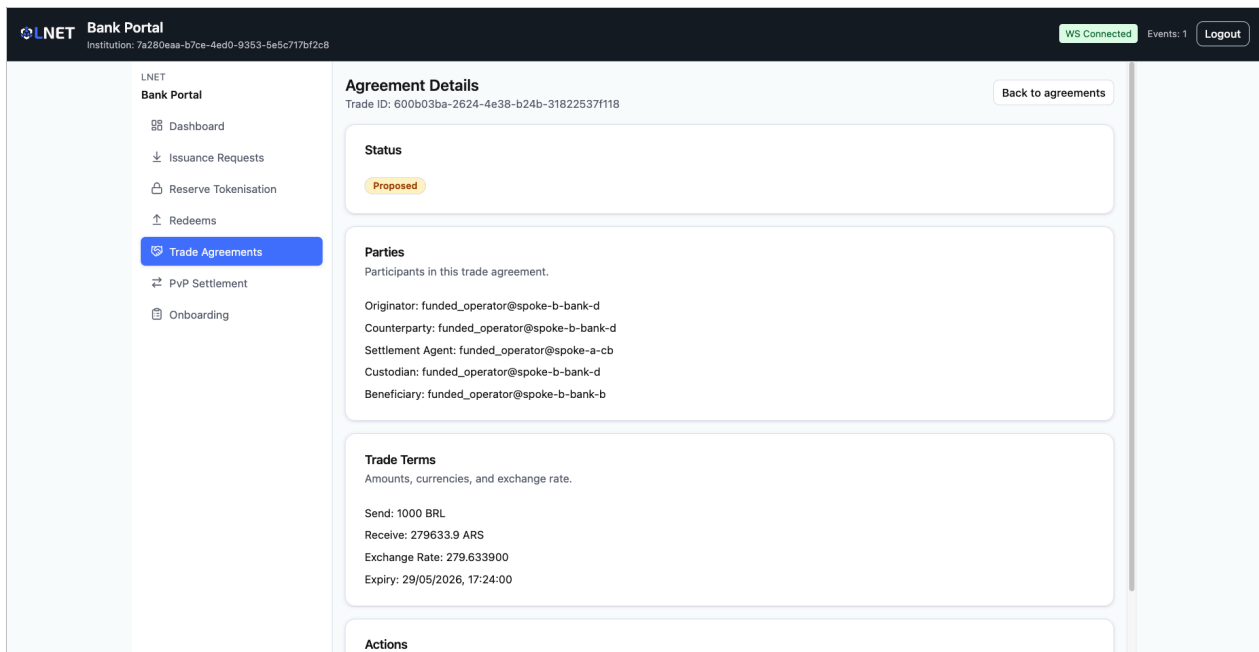


Figure 9: Agreement Detail Page

The page shows three sections:

- **Status** — current state badge.
- **Parties** — Originator, Counterparty, Settlement Agent, Custodian, Beneficiary.
- **Trade Terms** — Send/Receive amounts and currencies, exchange rate, expiry timestamp.

Available actions by state:

Agreement state	Available buttons
PROPOSED (you are the originator)	Cancel only
PROPOSED (you are the counterparty)	Accept, Reject, Cancel
ACCEPTED	Initiate PvP Transfer, Continue PvP Transfer
All other states	No actions available

Note: **Accept** and **Reject** are counterparty-only actions — the institution that proposed the agreement (the originator) sees only **Cancel**. A confirmation dialog always appears before executing.

- **Accept** — Moves the agreement to **ACCEPTED**; it is now ready for PvP settlement.
- **Reject** — Declines the proposal. Moves to **REJECTED**. Irreversible.
- **Cancel** — Cancels the proposal. Moves to **CANCELLED**. Irreversible.
- **Initiate PvP Transfer** — Navigates to `/htlc/new` pre-filled with your role as Originator (Spoke-A). The agreement ID and receiver are auto-populated from the agreement's **Source Receiver** field.
- **Continue PvP Transfer** — Navigates to `/htlc/new` pre-filled with your role as Counterparty (Spoke-B). The agreement ID and receiver are auto-populated from the agreement's **Destination Receiver** field.

PvP Settlement (HTLC)

PvP (Payment vs. Payment) settlements use **Hash Time Lock Contracts (HTLCs)** to perform atomic cross-border transfers. Either both legs complete or neither does — partial settlement is not possible.

How the two-leg process works

1. **Originator (Spoke-A):** Locks tCeBM in an HTLC with a hash lock and optional timelock. A Settlement Code is generated.
2. **Counterparty (Spoke-B):** Uses the Settlement Code to lock their tCeBM on their spoke.
3. **Settlement:** Once both locks are in place, the originator completes the settlement, which atomically delivers funds to both receivers.
4. **If the settlement window expires:** Either party can revoke their lock to recover their funds.

PvP Settlement History (/htlc)

Lists all HTLC contracts linked to your institution. The in-page title is **PvP Settlement History**.

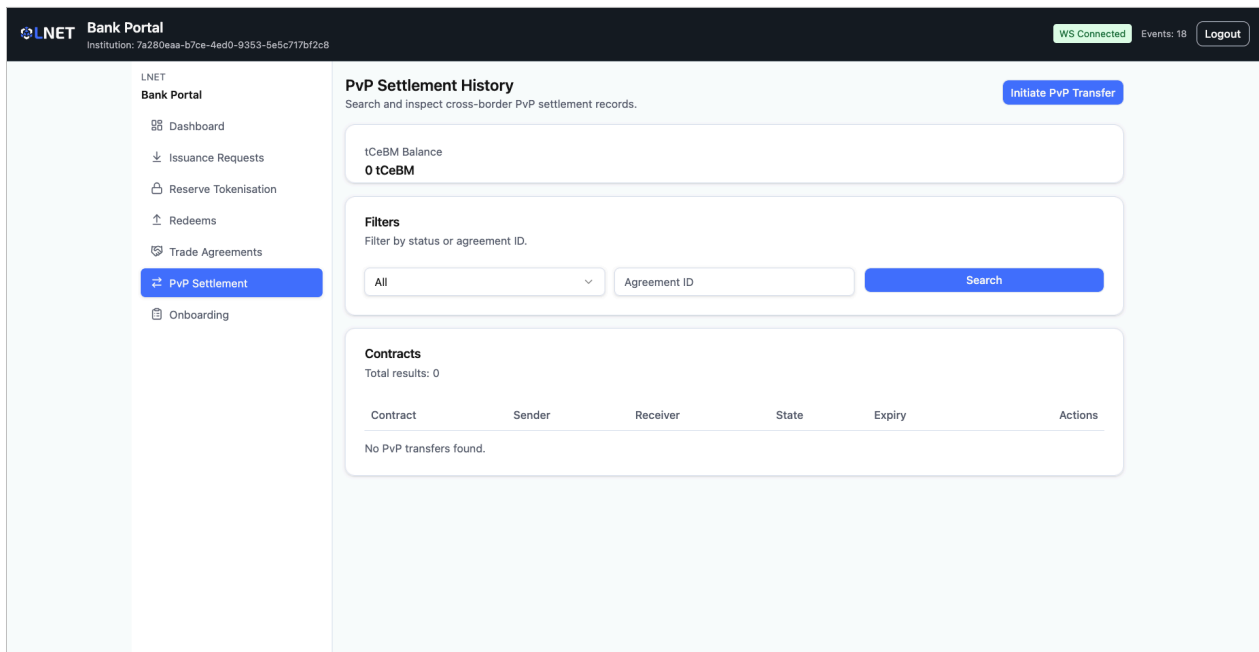


Figure 10: HTLC Settlement History

Filters:

Filter	Options
Status	All, Pending Settlement (LOCKED), Processing (SETTLING), Settled, Revoking (REFUNDING), Revoked (REFUNDED)
Agreement ID	Free-text search for contracts linked to a specific trade agreement

Click **Search** to apply filters. The table shows total matching results.

Contracts table:

Column	Description
Contract	HTLC contract ID
Sender	Paladin identity of the initiating institution
Receiver	Paladin identity of the receiving institution
State	Current contract state badge
Expiry	Settlement window expiry timestamp
Actions	View button — opens the Settlement Detail page

The **Initiate PvP Transfer** button (top right) navigates to `/htlc/new`.

New Transfer (`/htlc/new`)

This single page handles both roles in a PvP settlement. Use the **Choose PvP Action** selector to switch between the two modes.

The screenshot displays the LNET Bank Portal interface. The top header includes the LNET logo, 'Bank Portal', and a long alphanumeric institution ID. On the right of the header, it shows 'WS Connected', 'Events: 23', and a 'Logout' button. The left sidebar contains a 'Bank Portal' section with links to Dashboard, Issuance Requests, Reserve Tokenisation, Redeems, Trade Agreements, **PvP Settlement** (highlighted), and Onboarding. The main content area is titled 'PvP Settlement' and includes a 'Back to history' button. Below the title, there's a section for 'tCeBM Balance' showing '3,000 tCeBM'. The 'Link to FX Agreement (optional)' section has a dropdown menu showing '600b03ba... — 1000 BRL ↔ 279633.9 ARS' and a 'Clear' button. The 'SELECT YOUR ROLE IN THIS TRADE' section has two buttons: 'I am the Originator (Spoke-A)' (selected) and 'I am the Counterparty (Spoke-B)'. Below this, it says 'Pre-filling: receiver = —, amount = 1000 BRL'. The 'Choose PvP Action' section has two buttons: 'Initiate PvP Transfer' (selected) and 'Continue PvP Transfer'. The 'Initiate PvP Transfer' section has a sub-header and a description: 'Start the PvP settlement process. Receiver and amount are required.' It also shows a pre-filled agreement ID and a message: 'Pre-filled from agreement 600b03ba.... You may edit any field.'

Figure 11: Initiate PvP Transfer Form

Bank Portal
Institution: 7a280eaa-b7ce-4ed0-9353-5e5c717bf2c8

WS Connected Events: 0 Logout

PvP Settlement
Use one form to initiate a new transfer or continue with an existing settlement reference.

Back to history

tCeBM Balance
0 tCeBM

Link to FX Agreement (optional)
Select an accepted FX agreement to pre-fill the form below.

600b03ba... — 1000 BRL ↔ 279633.9 ARS Clear

SELECT YOUR ROLE IN THIS TRADE
☐ I am the Originator (Spoke-A) ☒ I am the Counterparty (Spoke-B)
 Pre-filling: receiver = —, amount = 279633.9 ARS

Choose PvP Action
Each action has its own dedicated form to avoid confusion.

Initiate PvP Transfer Continue PvP Transfer

Continue PvP Transfer
Continue the flow from the receiver side using an existing settlement reference.

Pre-filled from agreement 600b03ba.... You may edit any field.

Figure 12: Continue PvP Transfer Form

Optional: Link to an FX Agreement At the top of the page, select an accepted FX agreement from the dropdown. Then select your role (**I am the Originator** or **I am the Counterparty**). The form below is auto-filled with the receiver identity and amount from the agreement. You can still edit any field.

Mode 1: Initiate PvP Transfer (you are Originator / Spoke-A) Use this mode when your institution is locking funds first.

Field	Required	Description
Agreement ID	No	FX trade agreement ID this settlement is linked to (alphanumeric, hyphens, underscores; max 64 chars)
Receiver (Paladin identity)	Yes	The Paladin identity of the receiving institution on your spoke
Amount (tCeBM)	Yes	Positive integer; amount to lock in the HTLC
Timelock Duration	No	No explicit timelock, 1 hour, 6 hours, 24 hours, or a custom date/time. Must be at least 5 minutes in the future. If not set, the backend applies its default.

Steps:

1. Fill in all required fields.
2. Click **Review PvP Transfer** — validation runs.
3. A confirmation card shows Agreement ID, Receiver, Amount, and Timelock.
4. Click **Confirm PvP Transfer**.

On success, you are navigated to the Settlement Detail page. A **Settlement Code** (hash lock) is generated automatically. You must share this 64-character hex code with the counterparty institution so they can lock funds on their spoke.

Mode 2: Continue PvP Transfer (you are Counterparty / Spoke-B) Use this mode when the originator has already locked funds and has sent you the Settlement Code.

Field	Required	Description
Settlement Code	Yes	The 64-character hexadecimal hash received from the originating institution (with or without 0x prefix)
Receiver (Paladin identity)	Yes	The Paladin identity that will receive tCeBM on your spoke
Amount (tCeBM)	Yes	Positive integer; the amount specified for the counterparty leg

The Continue form has no Agreement ID input. When you reach this form via **Continue PvP Transfer** from an accepted agreement (or the FX agreement picker), the agreement ID is carried through automatically; it is not entered manually here.

Steps:

1. Enter the Settlement Code exactly as received from the originator.
2. Fill in the Receiver identity and Amount.
3. Click **Review PvP Continuation**.
4. Confirm the details and click **Confirm PvP Continuation**.

On success, you are navigated to the Settlement Detail page for the new contract.

Settlement Details (/htlc/:contractId)

Opened by clicking **View** on any row in the Settlement History, or navigated to automatically after creating a new transfer. The in-page title is **Settlement Details**.

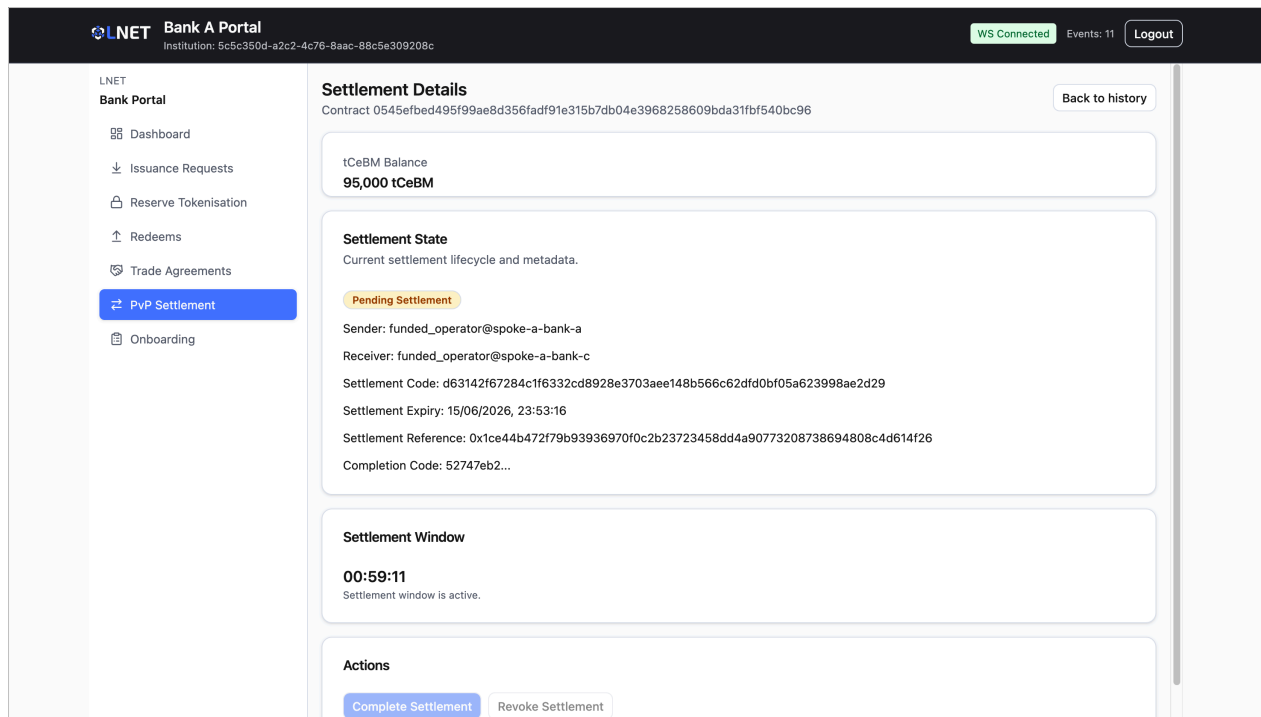


Figure 13: HTLC Details

The page auto-polls the backend every **5 seconds** while the contract is in an active state (LOCKED, SETTLING, REFUNDING). Polling stops automatically once the contract reaches SETTLED or REFUNDED.

Settlement State card:

Field	Description
Status badge	Current contract state
Sender	Initiating institution's Paladin identity
Receiver	Receiving institution's Paladin identity
Settlement Code	The hash lock securing the atomic swap
Settlement Expiry	Timestamp after which the settlement window closes
Settlement Reference	On-chain Zeto lock reference (populated after settlement completes)
Completion Code	The pre-image secret (first 8 characters shown when available; "not available" while waiting for the counterparty)

Settlement Window card (only shown while contract is LOCKED):

A live countdown shows time remaining. When expired: "Settlement window has expired. Revocation is available."

Actions:

Button	When enabled	What it does
Complete Settlement	Contract is LOCKED, the Completion Code is available, and the counterparty has also locked funds (<code>counterparty_locked = true</code>)	Finalises the atomic swap. Delivers tCeBM to both receivers. Irreversible.
Revoke Settlement	Contract is LOCKED and the settlement window has expired	Returns the locked funds to the sender. Irreversible.

Both actions display a confirmation dialog before execution. If a network timeout occurs during settlement, the page polls for up to 30 seconds to confirm whether the settlement completed on-chain.

Typical Workflows

Workflow 1: Get tCeBM into your wallet (issuance)

1. Go to **Issuance Requests** (/deposits).
2. Enter the fiat amount and click **Review Issuance Request**, then **Confirm Request**.
3. Wait for the Central Bank to approve the request in the Governance Portal. The status changes from PENDING to APPROVED.
4. Go to **Reserve Tokenisation** (/escrows).
5. Enter the same amount and click **Review Tokenisation Request**, then **Confirm Request**.
6. Wait for the Central Bank to process the tokenisation. On approval, the tCeBM balance on the Dashboard increases.

Workflow 2: Initiate a cross-border PvP payment (as Originator)

Prerequisite: Your institution has tCeBM in its wallet. You have an accepted FX trade agreement with the counterparty.

1. Go to **Trade Agreements** (/agreements), open the accepted agreement.
2. Click **Initiate PvP Transfer**. The form at /htlc/new opens pre-filled.
3. Verify the receiver identity and amount. Set a timelock if required.
4. Click **Review PvP Transfer**, then **Confirm PvP Transfer**.
5. Copy the **Settlement Code** shown on the Settlement Detail page.
6. Share the Settlement Code with the counterparty bank's operator out-of-band (e.g., by secure message).
7. Wait for the counterparty to lock their funds on Spoke-B.
8. Once `counterparty_locked` is true, the **Complete Settlement** button becomes active.
9. Click **Complete Settlement**, confirm the dialog. Both legs of the transfer complete atomically.

Workflow 3: Complete a cross-border PvP payment (as Counterparty)

Prerequisite: You have received a 64-character Settlement Code from the originator and an accepted FX agreement.

1. Go to **Trade Agreements** (/agreements), open the accepted agreement.
2. Click **Continue PvP Transfer**. The form at /htlc/new opens pre-filled.
3. Paste the Settlement Code into the **Settlement Code** field.
4. Verify the receiver identity and amount.

5. Click **Review PvP Continuation**, then **Confirm PvP Continuation**.
6. Monitor the Settlement Detail page. Once the originator calls Complete Settlement, your contract moves to **SETTLED** and tCeBM is delivered to the receiver.

Workflow 4: Accept an incoming FX agreement

1. Go to **Trade Agreements** (/agreements). Look for agreements in **PROPOSED** state.
2. Click the row to open the Agreement Detail.
3. Review all parties and trade terms carefully.
4. Click **Accept**, then confirm. The agreement moves to **ACCEPTED**.
5. You can now initiate or continue a PvP settlement against this agreement.

Workflow 5: Redeem tCeBM back to fiat

1. Go to **Redeems** (/redeems).
2. Enter the tCeBM amount and click **Review Redeem Request**, then **Confirm Request**.
3. The system automatically executes the Zeto privacy-preserving token transfer.
4. Wait for the Central Bank to approve the fiat release. On approval, the redemption ID is populated.

Status Reference

Deposit / Escrow / Redeem Request Statuses

Status	Meaning
PENDING	Submitted and awaiting Central Bank approval
APPROVED	Approved; on-chain operation complete
REJECTED	Denied by the Central Bank — check the Rejection Reason column
MINT_FAILED	Approval was granted but the on-chain mint transaction failed — contact your Central Bank operator

FX Trade Agreement States

State	Display label	Meaning
FX_STATE_PROPOSED	Proposed	Submitted and awaiting counterparty response
FX_STATE_ACCEPTED	Accepted	Counterparty accepted; ready for PvP settlement
FX_STATE_REJECTED	Rejected	Counterparty declined the proposal
FX_STATE_CANCELLED	Cancelled	Cancelled by the originator before acceptance
FX_STATE_SETTLED	Settled	The associated PvP settlement completed successfully

HTLC Contract States

State (internal)	Display label	Meaning
HTLC_STATE_LOCKED	Pending Settlement	Contract is locked; waiting for the receiver to lock their side and complete the settlement
HTLC_STATE_SETTLING	Processing	Settlement is being executed on-chain
HTLC_STATE_SETTLED	Settled	Atomic swap completed successfully; funds delivered
HTLC_STATE_REFUNDING	Revoking	Timelock expired; revocation is being processed on-chain
HTLC_STATE_REFUNDED	Revoked	Funds returned to the sender after settlement window expiry

Onboarding States

See the [Onboarding status table](#) in the Onboarding section above.

Troubleshooting

Login fails with “unauthorized”

- Verify the **Username** and **Password** are correct and have not been rotated.
- Confirm the institution’s onboarding status is **ACTIVE**. New accounts must complete onboarding first.
- Confirm the spoke backend services are running (`make spoke-up` or check `docker compose ps`).

Amount field shows an error on submission

- Amounts on the Deposits, Escrows, and Redeems pages must be **positive integers** (whole numbers). Decimals are not accepted.
- Verify the amount is greater than zero.

Deposit, Escrow, or Redeem is stuck in PENDING

- The Central Bank operator must log into the Governance Portal and approve the request.
- Confirm that governance operators are aware of the pending request.
- Click **Refresh** on the page to reload the latest status.

HTLC settlement is not progressing — “Complete Settlement” button is greyed out

- The **Complete Settlement** button requires two conditions: the Completion Code must be available (your lock generated it) **and** the counterparty must have also locked funds on their spoke (`counterparty_locked = true`). Check the Settlement Code was shared correctly with the counterparty.
- The Settlement Detail page auto-polls every 5 seconds — wait for the counterparty to submit their continuation before the button enables.

HTLC settlement window has expired

- Once expired, the only available action is **Revoke Settlement** to return your locked funds.

- If you still wish to settle, both parties must cancel their existing contracts and create a new agreement with a fresh timelock.

Counterparty cannot find the FX agreement proposal

- Confirm the **Counterparty B Identity** in the agreement form exactly matches the counterparty's Paladin identity (format: `funded_operator@spoke-<x>-<entity>`).
- Ask the counterparty to click **Refresh** or reload the Trade Agreements page — the inbox does not auto-refresh on that page.

Onboarding form submission fails

- Check that all fields satisfy the constraints listed in the [Institution Data](#) table.
- Verify the Bank Portal can reach the API Gateway (check backend connectivity).

The institution shows **ACTIVE** in the Governance Portal but the wizard is still at Step 3

- Click **Refresh status** on the Step 3 (KYC Approval) screen to force an immediate status check.
- If the session has expired, log out, log back in, and return to the Onboarding page.

Settlement Code validation error

- The Settlement Code must be exactly **64 hexadecimal characters** (0–9, a–f, A–F). An `0x` prefix is accepted.
- Ensure no spaces or line breaks were introduced when copying the code from the originator.

Governance Portal — User Manual

(Scenario A)

Audience: Central bank governance operators. **Data source:** Live backend by default (`VITE_USE MOCKS=false`). A local mock mode is available for development — if the portal is running in mock mode, data shown is synthetic and no changes reach the backend. Confirm with your system administrator which mode is active in your environment.

1. Overview

The Governance Portal is the Central Bank's control surface for **Scenario A — Enhanced Correspondent Banking**. It gives governance officers sovereign-level controls over:

- **KYC / participant admission** — approving commercial banks that have submitted onboarding requests
- **Account intervention** — freezing or unfreezing participant accounts with an auditable reason
- **Audit trail** — immutable log of all governance actions with filtering by category, severity, and date

All actions performed in the portal are confirmation-gated and permanently recorded in the governance audit log. The portal communicates exclusively with the Central Bank's API Gateway over HTTPS; there is no direct connection to the blockchain or Paladin sidecar.

Access is restricted to accounts holding the **ROLE_GOVERNANCE** role — the highest RBAC tier in the system.

2. Access and login

Portal URLs (local development)

Instance	URL	API Gateway
Central Bank A	<code>http://localhost:5177</code>	<code>http://localhost:38080/api/v1</code>
Central Bank B	<code>http://localhost:5178</code>	<code>http://localhost:60080/api/v1</code>

In a production or staging deployment, the URL is provided by your network administrator. Always confirm you are connecting to the correct spoke's portal.

Signing in

1. Open the portal URL.
2. On the **Sign in to <Institution> Portal** page (e.g. “Sign in to Central Bank A Portal” — the institution name comes from the VITE_INSTITUTION_NAME environment variable), enter your **Username** and **Password**.
 - The Username is your governance operator account identifier (minimum 3 characters; example: `admin@brasil.governance.gov`).
 - The Password is the corresponding credential provisioned by the network administrator (minimum 6 characters).
3. Click **Sign in**.
4. On success, you are redirected to the Dashboard.

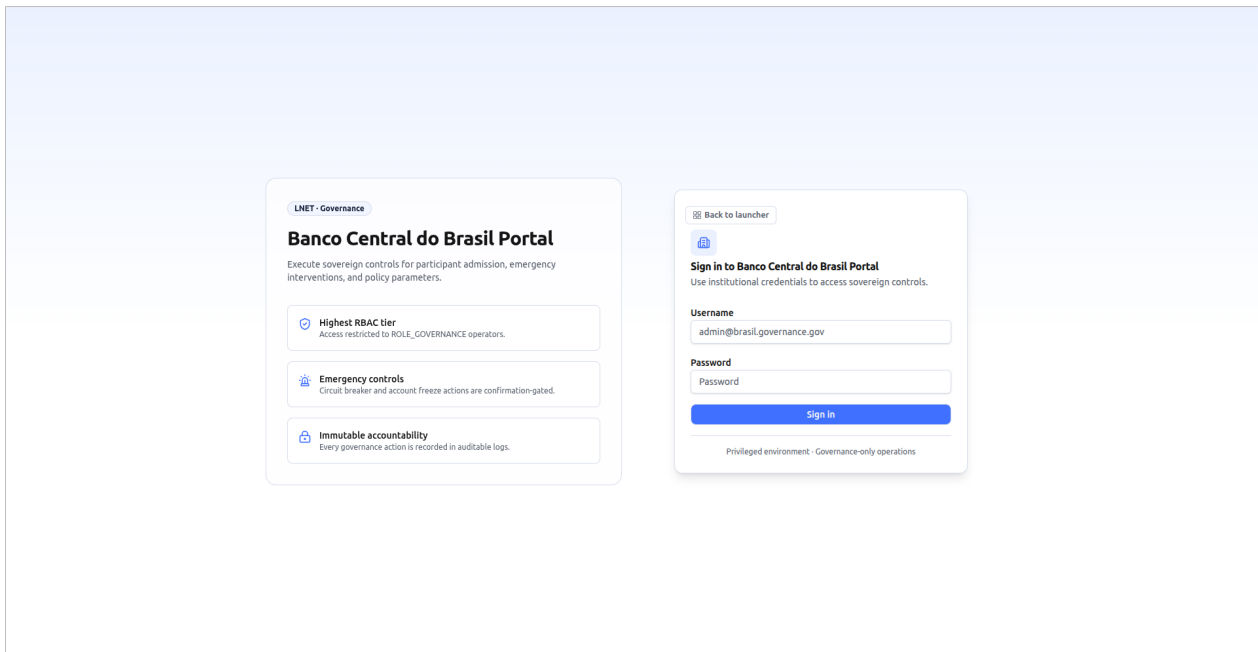


Figure 14: Governance Portal Login Screen

Access denied immediately after sign-in? Your account must hold the `ROLE_GOVERNANCE` role in the Keycloak realm for your spoke. Contact your network administrator to verify your role assignment.

3. Navigation

The portal uses a left-hand sidebar with the following items. All routes require authentication.

Sidebar label	Route	Purpose
Dashboard	/	At-a-glance health summary and recent events
Registry	/registry	KYC approval queue for pending commercial banks
Accounts	/accounts	Freeze or unfreeze participant accounts
Audit	/audit	Full audit trail with filters
Settings	/settings	Operator display preferences

Unauthenticated requests to any protected route redirect to `/login`. Unknown routes redirect to the Dashboard.

4. Screens

4.1 Dashboard (/)

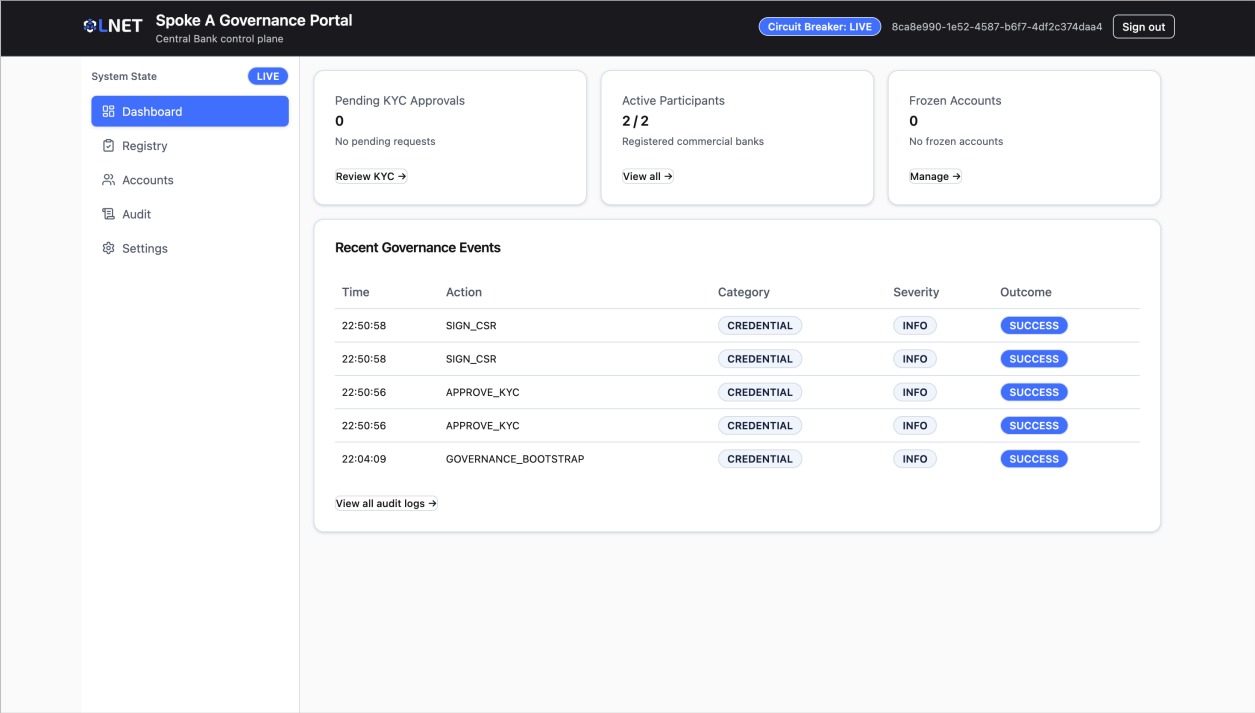


Figure 15: Governance Dashboard

The Dashboard is the landing page after login. It loads data from three live data sources in parallel — the participant registry, the accounts list, and the audit log — and presents a summary without requiring manual refresh.

Summary cards

Card	What it shows	Action
Pending KYC Approvals	Number of commercial banks awaiting Central Bank validation. Highlighted in yellow when greater than zero with a “NEEDS ATTENTION” badge.	Links to Registry
Active Participants	Count of active commercial banks vs. total registered commercial banks.	Links to Accounts
Frozen Accounts	Number of accounts currently frozen by governance. Highlighted in red when greater than zero.	Links to Accounts

Recent Governance Events table shows the five most recent entries from the audit log, with columns: Time, Action, Category, Severity, and Outcome. Use the “View all audit logs” link to open the full Audit page.

When the portal starts up, if Pending KYC Approvals is greater than zero, review and act on those requests before proceeding with other tasks.

4.2 Registry (/registry)

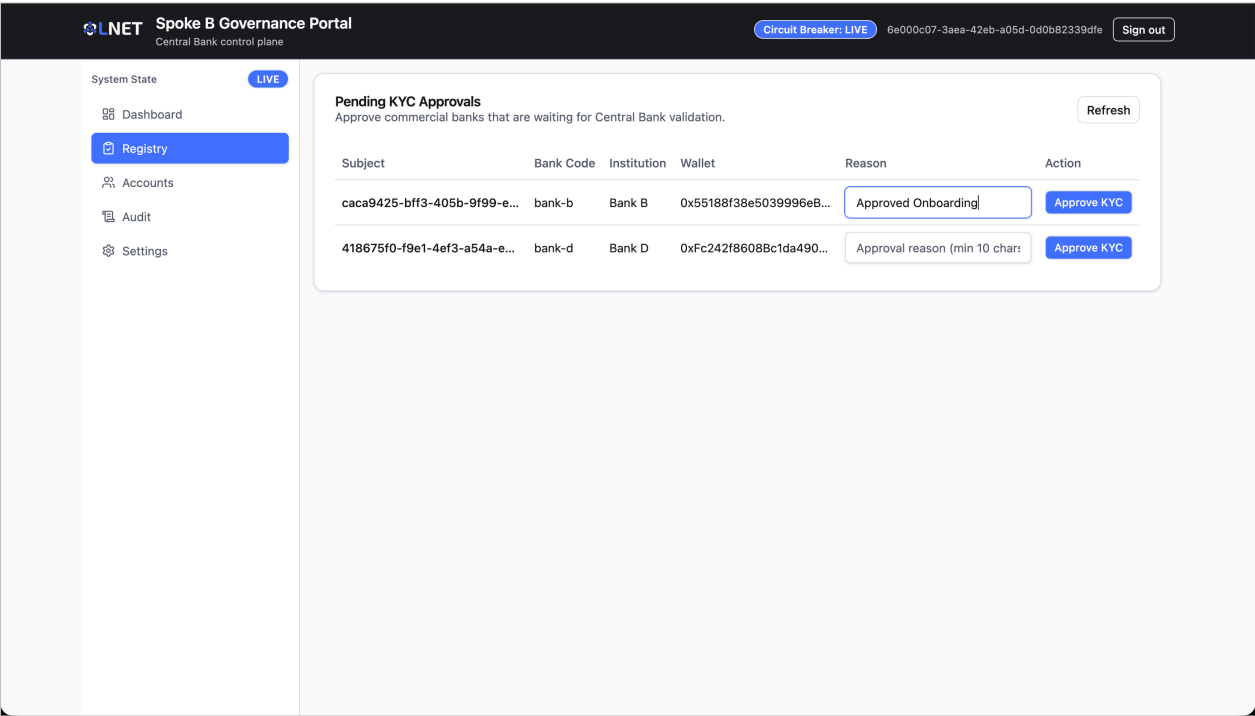


Figure 16: Registry Overview

The Registry screen is the primary admission gate for new commercial banks. Its current active section is the **Pending KYC Approvals** table.

Note on commented-out sections. The full participant list (compliance registry) and the manual credential issuance form exist in the codebase but are currently disabled pending a UI redesign. The only active functionality on this page is the KYC approval queue.

Pending KYC Approvals table

The table lists every commercial bank that has submitted an onboarding request with status `CREDENTIAL_REQUESTED`. The page **auto-refreshes this list every 15 seconds**; you can also trigger an immediate refresh with the **Refresh** button.

Column	Description
Subject	The bank’s unique identity identifier (UUID). Truncated in the display; hover for the full value.
Bank Code	Short alphanumeric code for the institution (e.g., horizonte).

Column	Description
Institution	Human-readable institution name (e.g., “Banco Horizonte S.A.”).
Wallet	The institution’s registered on-chain wallet address. Truncated; hover for the full address.
Reason	Free-text field where you enter your approval justification (minimum 10 characters).
Action	Approve KYC button — confirms the institution and moves it to KYC_APPROVED status.

When there are no pending requests, the table displays “No pending KYC requests.”

For the approval workflow, see [Approve a pending KYC request](#).

4.3 Accounts (/accounts)

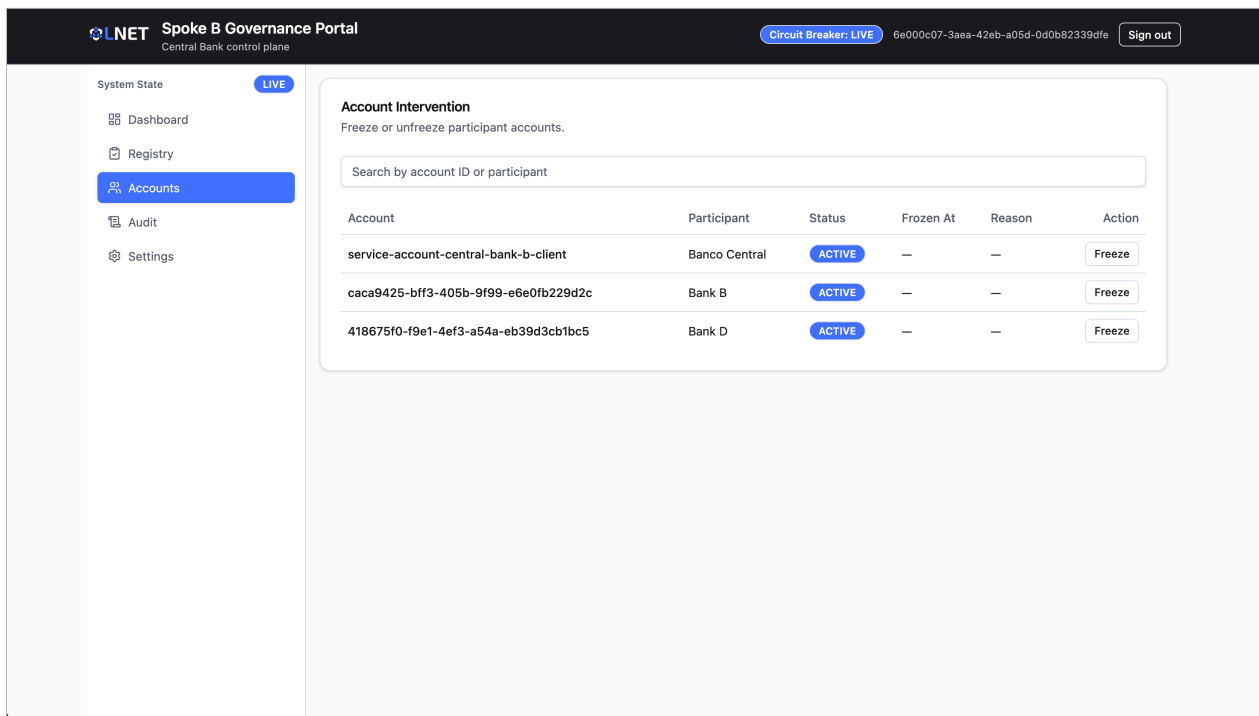


Figure 17: Registry Overview

The Accounts page (titled **Account Intervention** in the UI) allows governance operators to freeze or unfreeze participant accounts. This is a high-privilege action — it directly blocks or restores a participant’s ability to transact.

Account list

Use the **Search** field to filter by account ID or participant name.

Column	Description
Account	The account identifier.

Column	Description
Participant	The name of the institution that holds the account.
Status	ACTIVE or FROZEN .
Frozen At	Timestamp when the account was frozen. Shown as – if the account is not frozen.
Reason	The reason provided when the account was frozen. Shown as – if not frozen.
Action	Freeze (for active accounts) or Unfreeze (for frozen accounts).

Confirmation panel

Clicking **Freeze** or **Unfreeze** on a row opens a confirmation panel below the table:

1. A **Reason** text area appears (required, minimum 10 characters).
2. Click **Confirm Freeze** or **Confirm Unfreeze** to apply. The button shows “Submitting...” while the request is in flight.
3. Click **Cancel** to dismiss without taking action.

Both freeze and unfreeze actions are recorded in the audit log under the **FREEZE** category with **CRITICAL** severity.

4.4 Audit (/audit)

The screenshot displays the 'Governance Audit Trail' section of the 'Banco Central do Brasil Portal'. The header includes the LNET logo and 'Central Bank control plane'. A sidebar on the left contains navigation links: System State (LIVE), Dashboard, Registry, Accounts, Audit (selected), and Settings. The main content area shows a table of audit logs with the following data:

Timestamp	Actor	Action	Category	Severity	Outcome
05/08/2026, 13:31:12	6070ed07-30d9-41bd-b052-ac317d25a34d	APPROVE_KYC	CREDENTIAL	INFO	SUCCESS
05/08/2026, 13:31:08	6070ed07-30d9-41bd-b052-ac317d25a34d	APPROVE_KYC	CREDENTIAL	INFO	SUCCESS
05/08/2026, 13:16:44	service-account-central-bank-brazil-client	GOVERNANCE_BOOTSTRAP	CREDENTIAL	INFO	SUCCESS

Filters at the top of the table include 'All categories', 'All severities', 'From date', and 'To date'. An 'Apply Filters' button is also present.

Figure 18: Registry Overview

The Audit page (titled **Governance Audit Trail** in the UI) provides an immutable history of all governance actions and their outcomes. Use it to investigate past decisions and fulfill compliance or regulatory reporting requirements.

Filters

Filter	Options
Category	All categories · Credential · Circuit Breaker · Freeze · Parameter (Circuit Breaker is a Scenario-B carryover retained in the shared audit type; it typically returns no results in Scenario A)
Severity	All severities · Info · Warning · Critical
From date	Date picker — start of the date range
To date	Date picker — end of the date range

Set the desired filters and click **Apply Filters**. The button shows “Applying...” while results load.

Audit log table

Column	Description
Timestamp	Date and time the event was recorded (locale-formatted).
Actor	The operator or system that performed the action.
Action	A short description of what was done (e.g., “Account frozen”, “KYC approved”, “Parameters updated”).
Category	The governance domain. See the Audit category reference table below.
Severity	INFO, WARNING, or CRITICAL.
Outcome	SUCCESS or FAILED.

Audit category reference

Category	Events included
CREDENTIAL	KYC approvals, credential issuance
CIRCUIT_BREAKER	Circuit-breaker events (Scenario-B carryover retained in the shared audit type; not emitted in Scenario A)
FREEZE	Account freeze and unfreeze events
PARAMETER	Parameter update events

4.5 Settings (/settings)

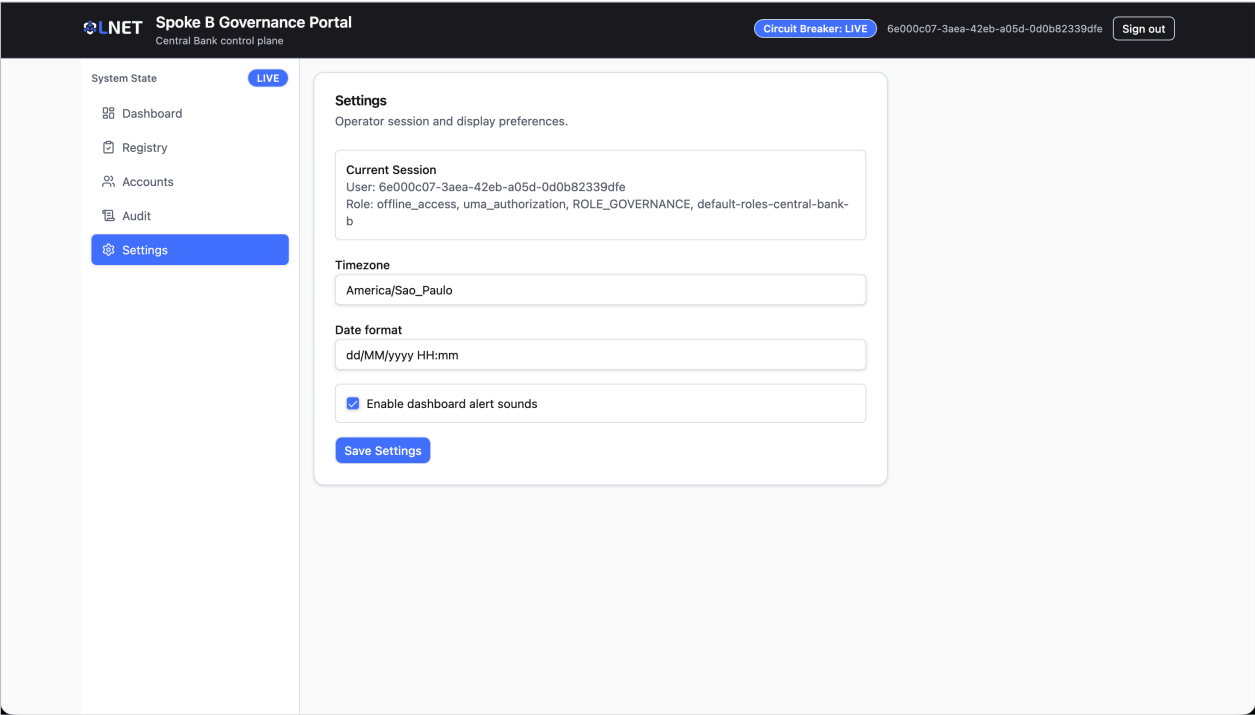


Figure 19: Registry Overview

The Settings page controls operator display preferences for the current session. It does not affect backend state.

Setting	Description
Current Session	Read-only. Shows your authenticated user subject and assigned roles.
Timezone	The timezone used to format dates in the portal. Default: <code>America/Sao_Paulo</code> .
Date format	The display format for dates and times. Default: <code>dd/MM/yyyy HH:mm</code> .
Enable dashboard alert sounds	Checkbox — enables audio alerts for dashboard notifications.

Click **Save Settings** to apply. A confirmation toast appears on success.

Note: Timezone and date format settings are currently stored in component state only. They reset when you navigate away or reload the page. Persistent preference storage is planned for a future release.

5. Typical workflows

5.1 Approve a pending KYC request

Use this workflow when a commercial bank has completed onboarding and their request appears in the Registry.

1. Open the **Dashboard**. If the **Pending KYC Approvals** card shows a count greater than zero, navigate to **Registry**.
 2. On the **Registry** page, locate the pending institution in the **Pending KYC Approvals** table.
 3. Verify the **Subject**, **Bank Code**, **Institution**, and **Wallet** fields match the institution's onboarding documentation.
 4. In the **Reason** field on the same row, enter a justification for approval (minimum 10 characters, e.g., "AML screening passed — approved per board resolution 2026-06-15").
 5. Click **Approve KYC**.
 6. A success toast confirms "KYC approved for [subject]".
 7. The row is removed from the pending list on the next refresh (within 15 seconds, or immediately if you click **Refresh**).
 8. Navigate to **Audit** and confirm a **CREDENTIAL** / **INFO** / **SUCCESS** entry was recorded.
-

5.2 Freeze a participant account

Use this workflow when a compliance event requires immediate blocking of a participant.

1. Navigate to **Accounts**.
 2. Use the search field to find the institution by account ID or participant name.
 3. Confirm the account **Status** is **ACTIVE**.
 4. Click **Freeze** on the relevant row.
 5. In the confirmation panel that appears below the table, enter a detailed reason in the **Reason** field (minimum 10 characters).
 6. Click **Confirm Freeze**.
 7. The account **Status** changes to **FROZEN** and the **Frozen At** and **Reason** columns populate.
 8. Navigate to **Audit** and confirm a **FREEZE** / **CRITICAL** / **SUCCESS** entry was recorded.
-

5.3 Unfreeze a participant account

Use this workflow when a previously frozen account is cleared to resume operations.

1. Navigate to **Accounts**.
 2. Locate the frozen account (Status badge shows **FROZEN** in red).
 3. Click **Unfreeze** on the relevant row.
 4. In the confirmation panel, enter a reason explaining why the freeze is being lifted (minimum 10 characters).
 5. Click **Confirm Unfreeze**.
 6. The account **Status** returns to **ACTIVE** and the **Frozen At** and **Reason** columns clear.
 7. Navigate to **Audit** and confirm an unfreeze entry was recorded.
-

5.4 Review the audit trail for a governance action

Use this workflow when investigating a past governance action for compliance or incident review.

1. Navigate to **Audit**.
2. Set the **Category** filter to the relevant domain (e.g., **FREEZE** to review account freeze actions).
3. Optionally set **Severity** (e.g., **CRITICAL** for high-impact events).
4. Optionally set a **From date** and **To date** to narrow the time range.
5. Click **Apply Filters**.

6. Review the results. Each row shows the timestamp, actor, action description, category, severity, and outcome.
7. Cross-reference the **Actor** and **Action** fields with the expected operator and action description.

6. Status reference

Account status

Status	Meaning	Displayed in
ACTIVE	Account is operational; participant can transact.	Accounts page, Dashboard
FROZEN	Account has been suspended by governance; participant cannot transact.	Accounts page, Dashboard

KYC / participant status

Status	Meaning	Displayed in
CREDENTIAL_REQUESTED	The institution has submitted an onboarding request awaiting governance approval.	Registry — Pending KYC table
KYC_APPROVED	Governance has approved the KYC request; credential is being finalized.	Audit log
ACTIVE	Institution is fully onboarded and authorized to transact.	Dashboard participant count
FROZEN	Institution has been suspended by governance.	Dashboard, Accounts

Audit severity

Severity	Meaning
INFO	Routine governance action (e.g., KYC approval).
WARNING	Action that may require follow-up attention.
CRITICAL	High-impact action: freeze or parameter update.

Audit outcome

Outcome	Meaning
SUCCESS	The action completed successfully on the backend.
FAILED	The action failed; review the action description and contact backend support.

Audit category

Category	Governance domain
CREDENTIAL	KYC approvals and credential issuance
CIRCUIT_BREAKER	Circuit-breaker control (Scenario-B carryover retained in the shared audit type; not emitted in Scenario A)
FREEZE	Account freeze and unfreeze
PARAMETER	System parameter updates

7. Troubleshooting

Sign-in fails with correct credentials

- Confirm your Keycloak account is enabled in the realm for your spoke.
- Confirm your account holds the `ROLE_GOVERNANCE` role. Other roles (e.g., `ROLE_COMMERCIAL_BANK`, `ROLE_TREASURY`) do not grant access to this portal.
- Confirm the Central Bank backend services (API Gateway, auth service) are running for your spoke.
- Confirm you are pointing at the correct portal URL for your entity (Central Bank A vs. Central Bank B).

Dashboard shows zeros for all cards

- The Dashboard loads data from the live backend on each page load. If all cards show zero immediately after login, check that the backend registry and audit services are reachable.
- If running in mock mode (`VITE_USE MOCKS=true`), synthetic data should populate immediately. If it does not, check the browser console for errors.

Pending KYC table is empty but you expect requests

- The table auto-refreshes every 15 seconds. Wait for a refresh cycle or click **Refresh**.
- Confirm the institution has completed their onboarding flow and submitted a credential request. The institution must have status `CREDENTIAL_REQUESTED` in the compliance registry for the entry to appear here.
- Confirm the portal is connected to the correct spoke's API Gateway.

“Reason must contain at least 10 characters” error

- All confirmation-gated actions (KYC approval, account freeze/unfreeze) require a reason of at least 10 characters. Enter a descriptive justification before confirming.

Accounts list is empty or stale

- The Accounts page loads data on mount. Reload the page to refresh.
- Confirm the compliance service is accessible via the API Gateway.

Audit log is empty

- Audit log entries are created by backend governance actions. If no governance actions have been performed yet, the log will be empty.
- Confirm the audit service is running and accessible.

- Use the **Apply Filters** button with all filters set to **ALL** to ensure no active filter is hiding results.

Treasury Portal — User Manual

(Scenario A)

Audience: Central bank treasury and issuance operators. **Data source:** Live backend (real API Gateway) for all approval and monitoring screens.

1. Overview

The Treasury Portal is the Central Bank's **operational control plane** for a CBWeb3 spoke. Treasury operators review and approve all actions that require Central Bank authority: minting tokenised fiat against bank deposit requests, converting tokenised fiat into tCeBM (the wholesale CBDC reserve token), authorising tCeBM redemptions back to fiat, and monitoring cross-border HTLC atomic settlements. The portal also provides a per-participant transfer-limit management screen.

Role in the system

Actor	Portal	Responsibilities
Central Bank Treasury Operator	Treasury Portal	Approve or reject issuance (deposit) requests, tokenisation (escrow) requests, and redemption (redeem) requests; monitor HTLC settlement lifecycle; manage transfer limits

Every approval action in this portal is the counterpart to a request submitted by a commercial bank in the Bank Portal. The bank requests — the Central Bank approves — and only then does the relevant on-chain operation execute.

Currency flow summary

Bank submits deposit request

- Treasury approves (Issuance Approval) - tokenised fiat minted
- Bank submits tokenisation (pledge) request
 - Treasury approves (Tokenisation Approval) - tokenised fiat burned, tCeBM minted
 - tCeBM in circulation

Bank submits redemption request

- Treasury approves (Redeems Approval) - tCeBM burned, fiat released

Simulation note: CBWeb3 operates in a simulation environment where real fiat currency does not exist on-chain. Tokenised fiat is a stand-in for the fiat collateral a commercial bank would deposit with the Central Bank in production. tCeBM is the actual CBDC token, always minted by burning tokenised fiat in a single atomic burn-to-mint operation. This maintains a 1:1 backing ratio at all times within the simulation.

2. Access and Login

Open the **Treasury Portal URL** provided by your system administrator in your browser and enter your institutional credentials.

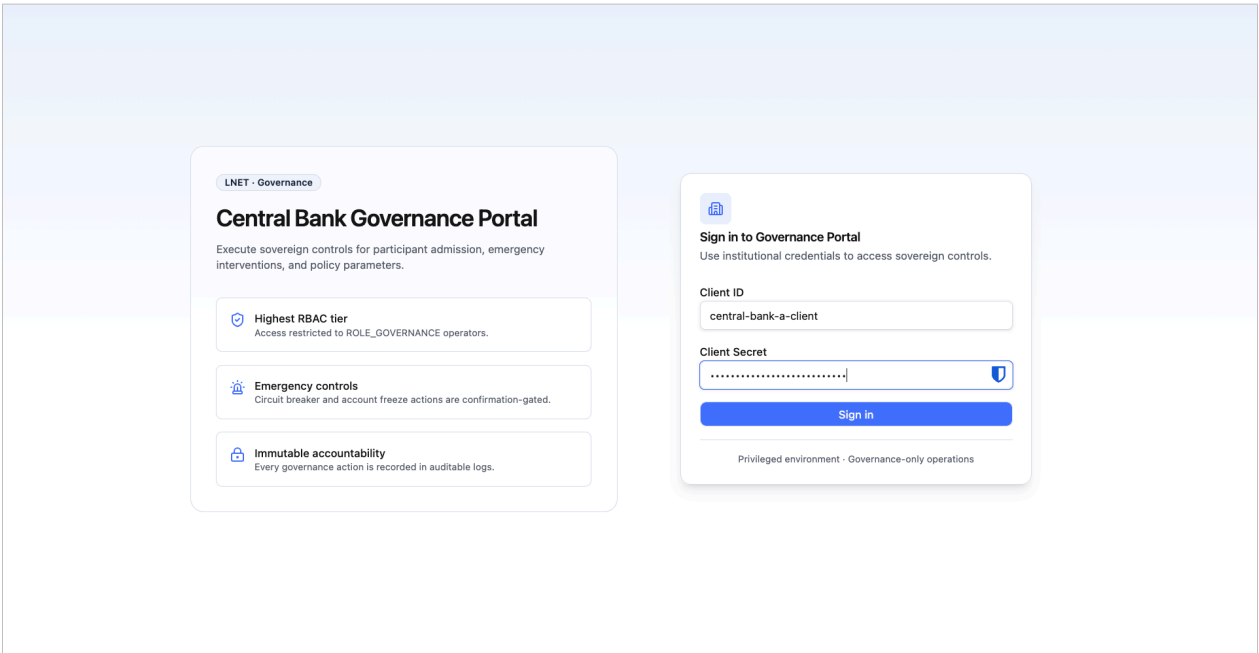


Figure 20: Treasury Portal Login Screen

Field	Description
Username	Your operator username (e.g., admin@brasil.treasury.gov). Minimum 3 characters.
Password	Your operator password. Minimum 6 characters.

Click **Sign in**. On success, you are redirected to the Dashboard. If the account does not carry the `ROLE_TREASURY` claim, sign-in will succeed at the authentication layer but the portal will deny access with an authorisation error.

- Session note:** Session credentials are held in memory only and are not persisted to local storage or across browser tabs. Opening the portal in a new browser tab requires a fresh sign-in.
- PKI accounts:** Accounts that require PKI-certificate-based authentication cannot log in through this portal. Use a standard username / password treasury account.

3. Navigation

The sidebar provides access to all currently active screens.

Sidebar label	Route	Notes
Dashboard	/	Landing page after login
Issuance Approvals	/deposits-approval	Approve or reject bank deposit / issuance requests
Tokenisation Approvals	/escrows-approval	Approve or reject burn-to-mint (pledge) requests
Redeem Approvals	/redeems-approval	Approve or reject tCeBM redemption requests
PvP Settlement	/htlc-monitor	Read-only monitor of cross-border HTLC contracts
Transfer Limits	/transfer-limits	Create and remove per-participant daily transfer limits

Any URL that does not match a known route redirects automatically to the Dashboard.

4. Screens

4.1 Dashboard (/)

The Dashboard is the landing page after login. It provides a live summary of supply metrics, pending work, recent mint/burn operations, and a simulated event stream.

Note: The event stream on this Dashboard is a browser-simulated demo ticker. It is **not** backed by the API Gateway or any live backend connection. The supply metrics, pending-request counts, and mint/burn operations cards are sourced from the real backend; the event stream is not.

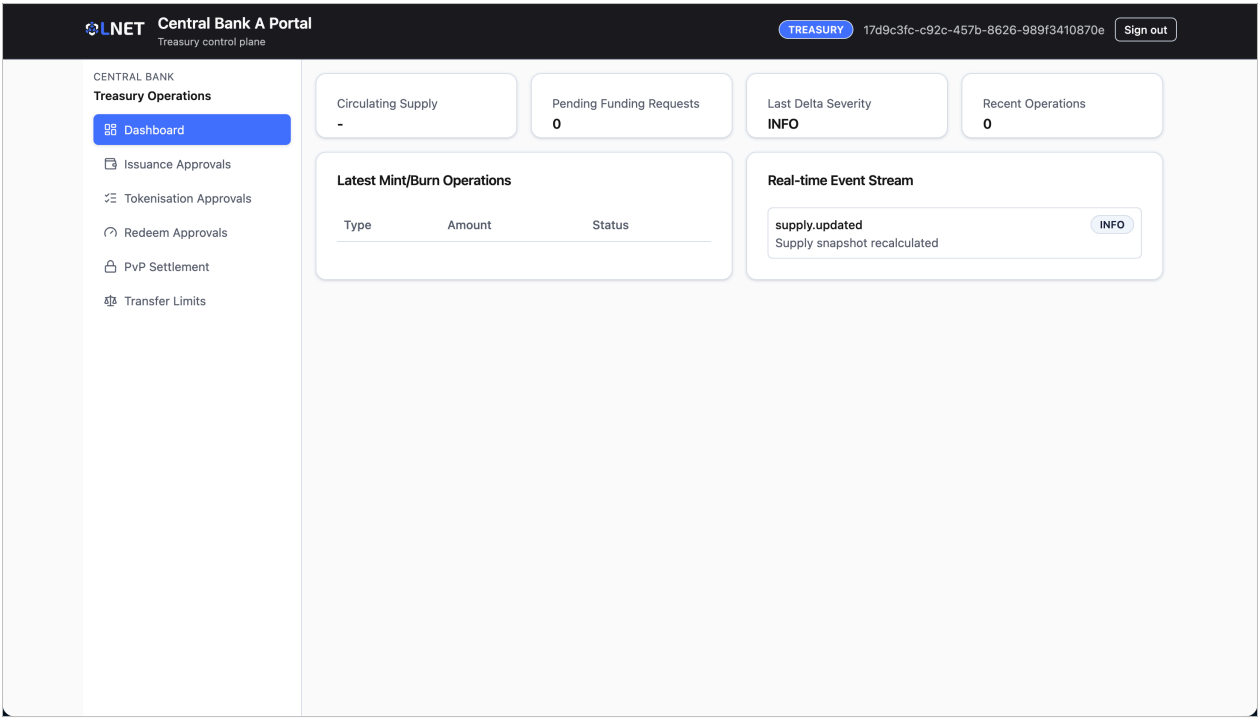


Figure 21: Treasury Dashboard

Summary Cards

Card	Description
Circulating Supply	Current tCeBM total circulating supply across this spoke.
Pending Funding Requests	Number of funding requests with status PENDING awaiting action.
Last Delta Severity	Severity level of the most recent reconciliation delta computation (OK , WARNING , or CRITICAL).
Recent Operations	Count of recent mint and burn operations recorded by the supply audit service.

Latest Mint/Burn Operations Table

Displays the six most recent token operations (mints and burns), including operation type, amount, and confirmation status.

Column	Description
Type	MINT or BURN.
Amount	Token amount involved in the operation.
Status	CONFIRMED (highlighted) or pending/other state.

Real-time Event Stream

This panel is labelled **Real-time Event Stream** on screen, but despite its name it is a browser-simulated demo ticker. It displays the eight most recent events from that ticker. The ticker is generated locally in the portal (a fixed set of sample messages, one emitted approximately every 12 seconds) and is **not** connected to the API Gateway or any backend event feed. Events are classified

by type and severity (INFO, WARNING, CRITICAL). The stream is read-only; no actions are available here.

4.2 Issuance Approvals (/deposits-approval)

When a commercial bank submits a request for the Central Bank to mint tokenised fiat — the first step in the currency lifecycle — that request appears on this page. Approving a request triggers the on-chain mint of tokenised fiat; it does not yet create tCeBM.

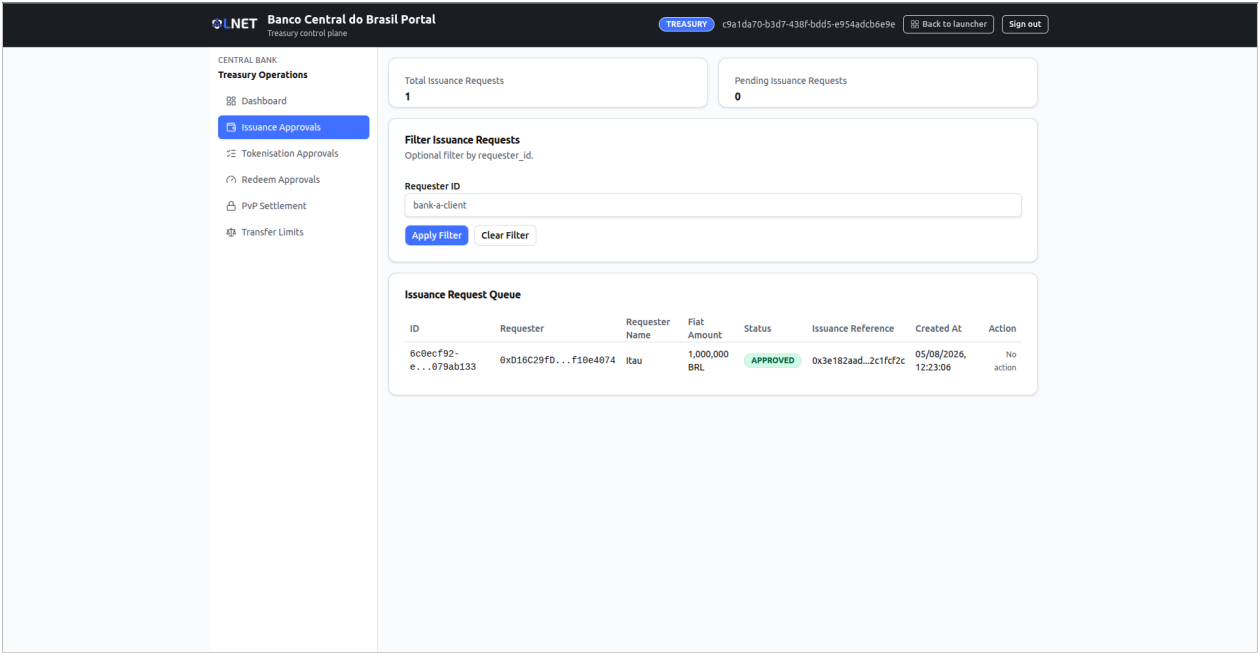


Figure 22: Deposits Approval Page

Summary Cards

Card	Description
Total Issuance Requests	Total count of issuance requests across all statuses.
Pending Issuance Requests	Count of requests currently awaiting your action.

Filter

Enter a **Requester ID** (e.g., bank-a-client) and click **Apply Filter** to narrow the table to a specific institution’s requests. Click **Clear Filter** to reset.

Issuance Request Queue

Column	Description
ID	Unique request identifier (truncated; hover for full value).
Requester	The commercial bank operator’s identity (truncated; hover for full value).

Column	Description
Requester Name	The display name of the requesting institution (falls back to – when unavailable).
Fiat Amount	The fiat collateral amount backing this issuance request.
Status	Current request status — see Status Reference .
Issuance Reference	On-chain transaction hash of the tokenised fiat mint (populated after approval).
Created At	Timestamp when the bank submitted the request.
Action	Available actions based on current status.

Available Actions

Approve Issuance — available for **PENDING** requests only.

1. Click **Approve Issuance** in the row's Action column. A confirmation panel appears at the bottom of the page showing the Request ID.
2. Review the Request ID carefully.
3. Click **Confirm Action** to proceed. The system approves the deposit record and immediately executes a fiat exchange (minting tokenised fiat). On success, a toast notification confirms approval and displays the first and last characters of the mint transaction hash.

Approval is an on-chain action. Confirm the requester and amount before confirming — the action cannot be undone.

Reject — available for **PENDING** requests only.

1. Click **Reject** in the row's Action column. A rejection panel appears at the bottom of the page.
2. Type a rejection reason in the text area (minimum 3 characters).
3. Click **Confirm Reject**. The status changes to **REJECTED** and no tokens are minted.

Retry Mint — available for **MINT_FAILED** requests only.

The approval step succeeded but the on-chain fiat exchange (mint) failed. Click **Retry Mint** to re-attempt the mint operation without requiring a new approval decision.

4.3 Tokenisation Approvals (/escrows-approval)

After the Central Bank approves an issuance request (tokenised fiat minted), the commercial bank submits a **tokenisation request** (also referred to as a pledge). This requests the burn-to-mint operation: tokenised fiat is burned and an equivalent amount of tCeBM (CBDC) is minted to the bank's wallet. This page manages those requests.

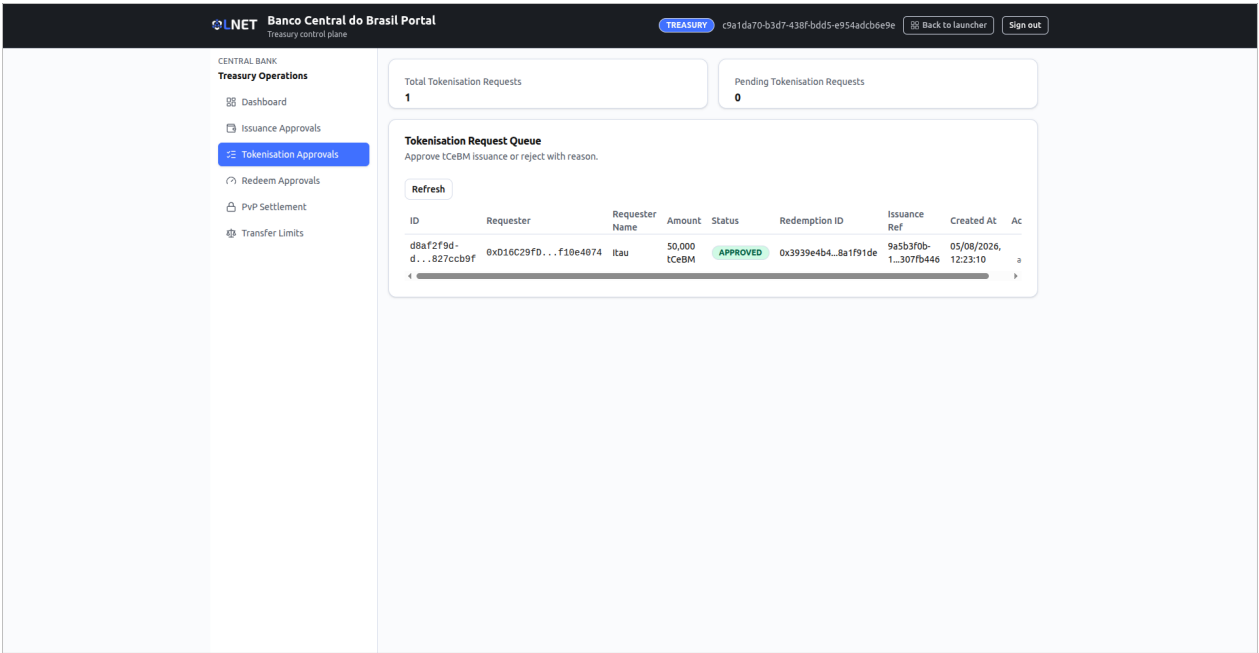


Figure 23: Escrows Approval Page

Summary Cards

Card	Description
Total Tokenisation Requests	Total count of tokenisation requests across all statuses.
Pending Tokenisation Requests	Count of requests currently awaiting your action.

Tokenisation Request Queue

Column	Description
ID	Unique request identifier (truncated; hover for full value).
Requester	The commercial bank operator’s identity.
Requester Name	The display name of the requesting institution (falls back to – when unavailable).
Amount	The amount of tCeBM to be minted (and the equivalent tokenised fiat to be burned).
Status	Current request status — see Status Reference .
Redemption ID	Transaction hash of the burn operation (populated after approval).
Issuance Ref	Transaction hash of the tCeBM mint operation (populated after approval).
Created At	Timestamp of submission.
Action	Available actions.

Available Actions

Approve — available for PENDING requests only.

1. Click **Approve** in the row’s Action column. A confirmation panel appears.
2. Click **Confirm Approve**. The burn-to-mint operation executes atomically: tokenised fiat is burned and tCeBM is minted. On success, a toast notification displays the first and last characters of both the burn transaction hash (Redemption) and the mint transaction hash (Issuance).

Reject — available for PENDING requests only.

1. Click **Reject** in the row’s Action column. A rejection panel appears.
2. Type a rejection reason (minimum 3 characters).
3. Click **Confirm Reject**. The status changes to REJECTED.

Click **Refresh** at any time to manually reload the tokenisation queue from the backend.

4.4 Redeems Approval (/redeems-approval)

When a commercial bank wishes to convert tCeBM back to fiat reserves, it submits a redemption request. This page is where those requests are reviewed and the fiat release is authorised.

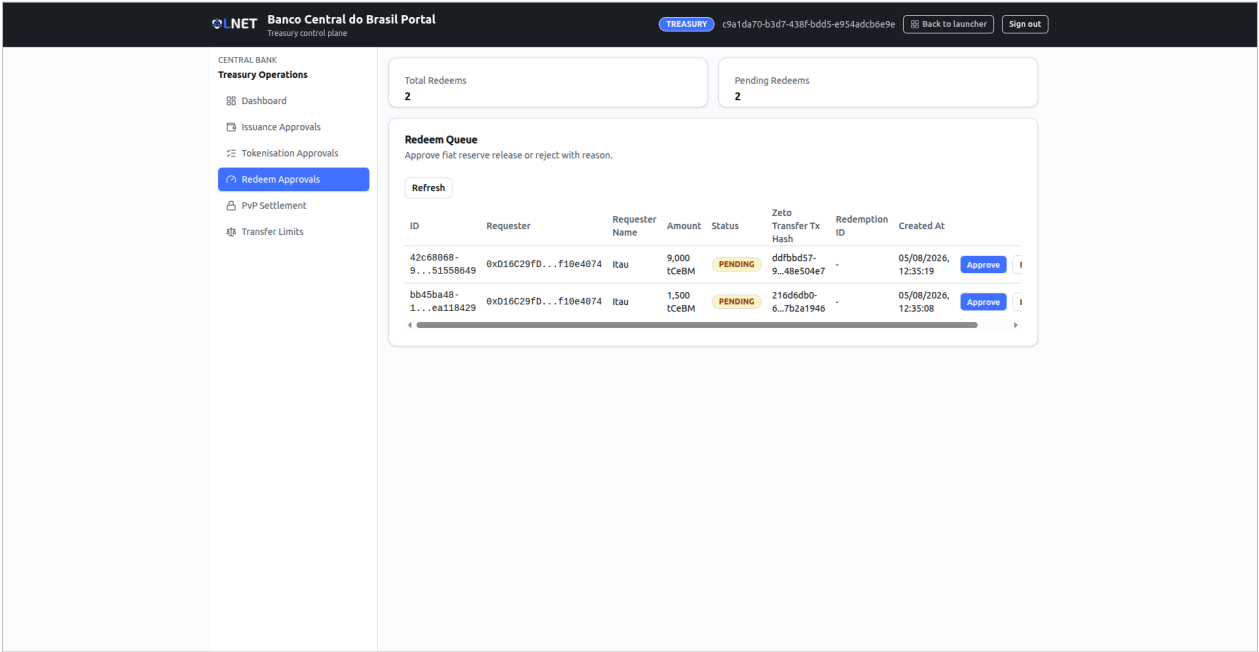


Figure 24: Redeems Approval Page

Summary Cards

Card	Description
Total Redeems	Total count of redemption requests across all statuses.
Pending Redeems	Count of requests currently awaiting your action.

Redeem Queue

Column	Description
ID	Unique request identifier (truncated; hover for full value).
Requester	The commercial bank operator's identity.
Requester Name	The display name of the requesting institution (falls back to - when unavailable).
Amount	The amount of tCeBM to be redeemed.
Status	Current request status — see Status Reference .
Zeto Transfer Tx Hash	Transaction hash of the privacy-preserving Zeto token transfer previously executed by the privacy proxy (system-populated).
Redemption ID	Transaction hash of the fiat mint/release (populated after approval).
Created At	Timestamp of submission.
Action	Available actions.

Available Actions

Approve — available for **PENDING** requests only.

1. Click **Approve** in the row's Action column. A confirmation panel appears showing the Redeem ID.
2. Click **Confirm Approve** to authorise the fiat release. On success, a toast notification displays the first and last characters of the fiat mint transaction hash (Redemption ID).

Reject — available for **PENDING** requests only.

1. Click **Reject** in the row's Action column. A rejection panel appears.
2. Provide a mandatory rejection reason.
3. Click **Confirm Reject**.

Click **Refresh** at any time to manually reload the redeem queue.

4.5 HTLC Monitor (/htlc-monitor)

The in-page title of this screen is **PvP Settlement Monitor**. It provides a **read-only** view of all cross-border atomic settlement contracts (Hash Time Lock Contracts — HTLCs) tracked by the system. No approval actions are available here; it is used for oversight and auditing of the PvP settlement lifecycle.

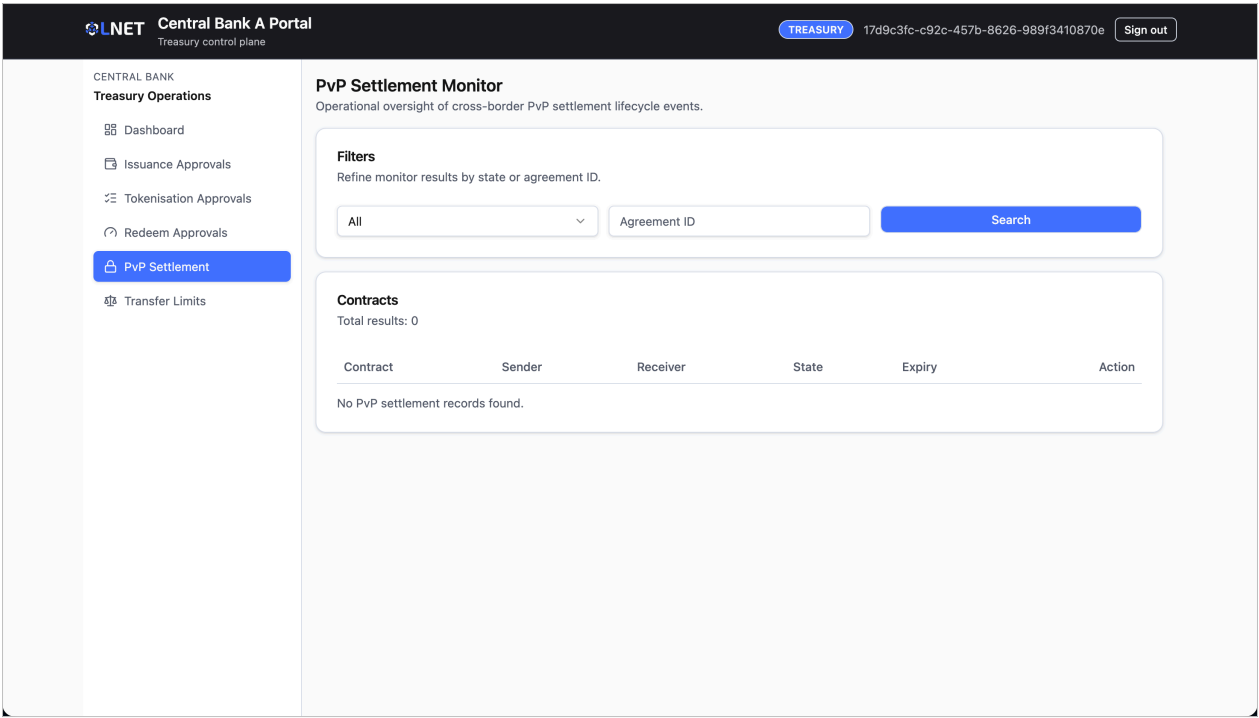


Figure 25: HTLC Monitor

Filters

Filter	Description
State	Dropdown to filter by contract state: All, Pending Settlement, Processing, Settled, Revoking, Revoked.
Agreement ID	Text input to search for contracts linked to a specific FX agreement identifier.

Click **Search** to apply the selected filters.

Contracts Table

Column	Description
Contract	Unique identifier for the HTLC contract.
Sender	Paladin identity of the initiating commercial bank.
Receiver	Paladin identity of the receiving commercial bank.
State	Current contract state — see Status Reference .
Expiry	Timestamp after which the contract can be revoked if not yet settled.
Action	View details button.

The total number of results matching the current filters is displayed above the table.

Contract Detail Panel

Click **View details** on any row to expand a detail panel below the contracts table.

Field	Description
Sender	Initiating bank’s Paladin identity.
Receiver	Receiving bank’s Paladin identity.
Settlement Code	The hash lock used to secure the atomic swap.
Settlement Expiry	The settlement window expiry timestamp.
State	Current contract state.
Settlement Reference	On-chain reference of the Zeto lock (populated when the contract progresses).
Completion Code	The secret pre-image — shown as [restricted] when present.

Click **Close** to dismiss the detail panel.

This screen is for oversight only. Completing or revoking a settlement is performed by the participating banks in the Bank Portal, not here. If a contract appears stuck, refer to [Troubleshooting](#).

4.6 Transfer Limits (/transfer-limits)

This screen allows treasury operators to create and remove daily transfer limits for participants. Limits can be scoped to a specific participant and/or currency, or applied globally.

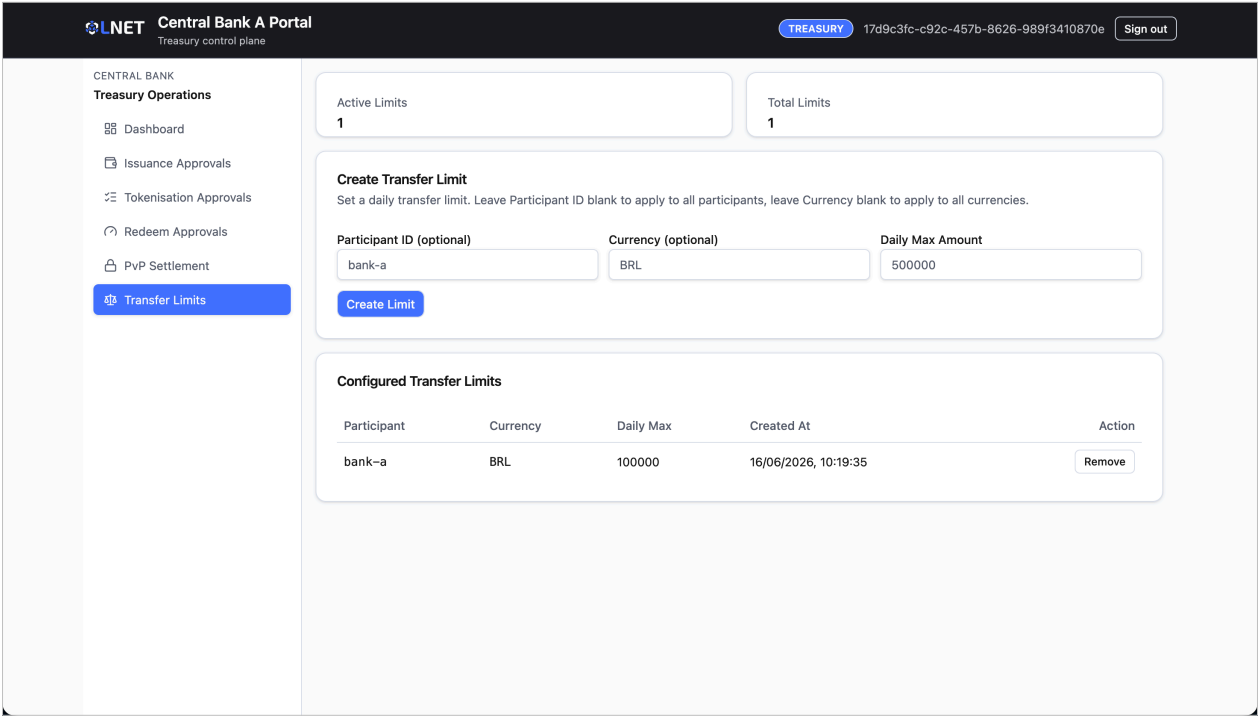


Figure 26: Transfer Limits

Summary Cards

Card	Description
Active Limits	Number of currently active transfer limits.
Total Limits	Total number of configured limits (including inactive).

Create Transfer Limit

Field	Required	Description
Participant ID	Optional	The participant identifier to apply the limit to (e.g., bank-a). If left blank, the limit applies to all participants.
Currency	Optional	The currency to restrict (e.g., BRL). If left blank, the limit applies to all currencies.
Daily Max Amount	Required	The maximum transfer amount per day for the scope defined by the above fields.

Click **Create Limit** to save. On success, a toast confirms creation and the new limit appears in the table below.

A limit with both Participant ID and Currency blank acts as a global cap affecting all participants and currencies. Use this configuration with care.

Configured Transfer Limits Table

Column	Description
Participant	Participant ID the limit applies to, or “all participants” if global.
Currency	Currency the limit applies to, or “all currencies” if unconstrained.
Daily Max	The configured daily maximum transfer amount.
Created At	When the limit was created.
Action	Remove button.

Removing a limit: Click **Remove** in the row’s Action column. A confirmation panel appears. Click **Confirm Remove** to permanently delete the limit. This action is irreversible.

5. Screens Not Currently Enabled

The following pages exist in the codebase but are **not wired into the current application routes** and are therefore inaccessible from the sidebar or by URL. Do not present them as working.

Screen	Route	Status
Issuance (direct mint form)	<code>/issuance</code>	Disabled — direct issuance is handled via Issuance Approvals
Redemption (direct burn form)	<code>/redemption</code>	Disabled — redemptions are handled via Redeems Approval
Funding Requests	<code>/funding-requests</code>	Disabled
Reconciliation	<code>/reconciliation</code>	Disabled
Audit	<code>/audit</code>	Disabled
Settings	<code>/settings</code>	Disabled

These screens may be enabled in future releases.

6. Typical Workflows

6.1 Approving a Bank's Issuance (Deposit) Request

This is the most frequent treasury action. A commercial bank has submitted a deposit request and is waiting for tokenised fiat to be minted.

1. Navigate to **Issuance Approvals** (`/deposits-approval`).
2. Confirm the **Pending Issuance Requests** card shows a non-zero count.
3. Review the request table. Identify rows with status **PENDING**.
4. Optionally filter by **Requester ID** to isolate a specific bank's requests.
5. In the target row, click **Approve Issuance**.
6. In the confirmation panel, verify the Request ID.
7. Click **Confirm Action**.
8. Wait for the success toast confirming approval and the fiat exchange transaction hash.
9. Verify that the row's status has changed from **PENDING** to **APPROVED** and that the **Issuance Reference** column now contains a transaction hash.

If the row transitions to **MINT_FAILED** instead, use the **Retry Mint** action — see [Troubleshooting](#).

6.2 Approving a Tokenisation (Pledge) Request

After an issuance is approved, the bank submits a pledge to convert tokenised fiat into tCeBM.

1. Navigate to **Tokenisation Approvals** (`/escrows-approval`).
2. Identify rows with status **PENDING**.
3. Click **Approve** in the target row.
4. In the confirmation panel, click **Confirm Approve**.
5. Wait for the success toast displaying both the burn transaction hash (Redemption) and the mint transaction hash (Issuance Ref).
6. Verify the row status changes to **APPROVED** and both hash columns are populated.

6.3 Approving a Redemption Request

A commercial bank has submitted a request to redeem tCeBM back to fiat.

1. Navigate to **Redeems Approval** (`/redeems-approval`).

2. Identify rows with status **PENDING**.
 3. Review the **Requester**, **Amount**, and **Zeto Transfer Tx Hash** (confirming that the privacy-preserving transfer has already been executed by the system).
 4. Click **Approve** in the target row.
 5. In the confirmation panel, click **Confirm Approve**.
 6. Wait for the success toast displaying the fiat release transaction hash.
 7. Verify the row status changes to **APPROVED** and the **Redemption ID** column is populated.
-

6.4 Rejecting Any Request

The reject flow is identical across all three approval queues.

1. Navigate to the relevant approvals page.
2. Click **Reject** in the target row's Action column.
3. In the rejection panel, type a clear, descriptive reason (minimum 3 characters).
4. Click **Confirm Reject**.
5. Verify the row status changes to **REJECTED**.

The rejection reason is recorded in the backend and is visible to the requesting institution.

6.5 Monitoring the HTLC Settlement Lifecycle

1. Navigate to **PvP Settlement** (/htlc-monitor).
2. Use the **State** dropdown to filter by contract state (e.g., **Pending Settlement** to see contracts awaiting receiver action).
3. Use the **Agreement ID** input to locate contracts tied to a known FX agreement.
4. Click **Search** to apply filters.
5. Click **View details** on any contract to see the hash lock, expiry, Zeto lock reference, and current state.
6. Click **Close** to return to the contracts table.

If a contract is stuck in **PENDING_SETTLEMENT** past its expiry, the counterparty bank must initiate revocation from the Bank Portal. Refer to [Troubleshooting](#).

6.6 Setting a Daily Transfer Limit for a Participant

1. Navigate to **Transfer Limits** (/transfer-limits).
 2. In the **Create Transfer Limit** form, enter the **Participant ID** (e.g., bank-a) and/or **Currency** (e.g., BRL). Leave either blank for a global scope.
 3. Enter the **Daily Max Amount**.
 4. Click **Create Limit**.
 5. Verify the new limit appears in the **Configured Transfer Limits** table with the correct participant, currency, and amount.
-

7. Status Reference

Issuance / Tokenisation / Redeem Request Statuses

Status	Meaning	Next Step
PENDING	Request submitted by the bank; awaiting your action.	Approve or Reject.
APPROVED	Request approved; on-chain operation completed successfully.	No action required.
REJECTED	Request rejected by a treasury operator; a reason is recorded.	No further action. The bank must submit a new request if needed.
MINT_FAILED	Approval was granted but the on-chain mint transaction failed. Displayed in the status badge as MINT FAILED (no underscore).	Use Retry Mint (Issuance Approvals only). Investigate backend and node connectivity.

HTLC Contract States

State (displayed)	Internal state	Meaning
Pending Settlement	HTLC_STATE_LOCKED	Contract locked; waiting for the receiver to claim by revealing the pre-image.
Processing	HTLC_STATE_SETTLING	Settlement or revocation is being processed on-chain.
Settled	HTLC_STATE_SETTLED	Atomic swap completed successfully; funds delivered to the receiver.
Revoking	HTLC_STATE_REFUNDING	Timelock expired; revocation in progress.
Revoked	HTLC_STATE_REFUNDED	Funds returned to the sender after settlement window expiry.

8. Troubleshooting

Login fails with “unauthorized” or access is denied

- Verify that the Username and Password are correct.
- Confirm the treasury operator account exists and carries the `ROLE_TREASURY` claim. Accounts without this claim will be rejected after authentication.
- Accounts requiring PKI-certificate authentication cannot log in here. Use a standard username/password treasury account.
- Confirm the spoke backend services are running (`docker compose ps` from the relevant `deploy/local/` directory).

An approval request does not appear in the queue

- Click the **Refresh** button (available on the Tokenisation and Redeems pages) or reload the page. The page does not auto-refresh on all queues.
- Confirm the commercial bank has completed their request submission from the Bank Portal and that the request shows a valid Request ID on their side.

- Check the **Requester ID** filter is not limiting the displayed results.

Approval action fails with an error toast

- Note the error message shown in the toast notification — it typically contains the root cause.
- Confirm the API Gateway is reachable.
- If the on-chain mint fails and the request transitions to **MINT_FAILED**, use **Retry Mint** rather than creating a new approval.

A request shows **MINT_FAILED**

1. Note the Request ID.
2. Check the backend API Gateway and Besu node logs for the failed transaction.
3. Confirm the Central Bank signer account has sufficient gas.
4. Once the issue is resolved, click **Retry Mint** in the row's Action column on the Issuance Approvals page.

Status does not update after approval

- Wait several seconds for the on-chain transaction to confirm on the Besu network.
- Click **Refresh** or reload the page. The list will reflect the updated status on the next fetch.

A PvP settlement is stuck in Pending Settlement

- This screen is read-only; treasury operators cannot complete or revoke settlements.
- Check whether the settlement window has expired (the **Expiry** column). If it has, the sending bank must initiate revocation from the Bank Portal.
- If the expiry has not passed, the receiving bank must complete the settlement from their Bank Portal.
- If a cross-spoke settlement appears frozen before expiry, check the relay (Cacti hub-and-spoke) logs for event propagation issues.

The Event Stream on the Dashboard shows no events

- The Dashboard event stream is a browser-simulated demo ticker, not a backend feed. It begins emitting sample events shortly after the Dashboard loads (approximately one every 12 seconds).
- If it appears empty, reload the page to restart the ticker.
- Absence of events here does not indicate a backend outage; the event stream is not connected to the backend.

Supervisor Portal — User Manual

(Scenario A)

Audience: Supervisors and regulators.

Preview status

The Supervisor Portal is currently a **UI preview**. Most data-fetching screens call real backend endpoints, but **the data those endpoints return in the current development environment is demonstration data, not a live production data set**. The authentication flow uses institutional SSO/OIDC credentials, but **no production identity-provider is wired up yet** — the login accepts any credentials that satisfy the minimum-length validation while the backend is running locally.

- **Do not** use any figure, balance, alert, HTLC record, participant entry, or audit log shown in this portal for a real supervisory or regulatory decision.
- **Do not** treat the login as a production security boundary.
- Live data wiring and production Keycloak authentication are **not yet available**.

Every screen that displays data currently sourced from the development/demo environment is labelled **Mock data** in a callout below its description.

1. Overview

The Supervisor Portal is the **read-and-intervene** oversight surface for regulatory authorities and internal auditors in Scenario A (Enhanced Correspondent Banking). It provides visibility into HTLC settlement flows, participant compliance status, and audit records without granting any transactional powers (no minting, burning, or token transfers).

The portal exposes six screens:

Screen	Route	Purpose
Dashboard	/	Network KPIs and live HTLC settlement table
Audit Vault	/audit	Credential verification, transaction decryption, audit log review

Screen	Route	Purpose
Investigation	<code>/investigation</code>	Open, co-sign, and look up disclosure requests
Compliance Registry	<code>/participants</code>	Compliance registry — read-only participant list
Settlement Health	<code>/stability</code>	HTLC expiry risk and active alert counts
Settings	<code>/settings</code>	Session-scoped operational preferences

2. Who uses it

Role	Keycloak claim	Scope
Regional Supervisor	<code>ROLE_SUPERVISOR</code>	Cross-spoke visibility; all modules
Compliance Officer / NOC	<code>ROLE_NOC</code>	All modules; operational compliance
Central Bank Admin	(other)	All modules; administrative oversight

Supervisors cannot mint tokens, burn tokens, or initiate settlements. The only write actions available are:

- Opening or co-signing a disclosure request (Investigation screen).
 - Submitting a transaction decryption request (Audit Vault screen).
 - Saving session-level preferences (Settings screen).
-

3. Access and login

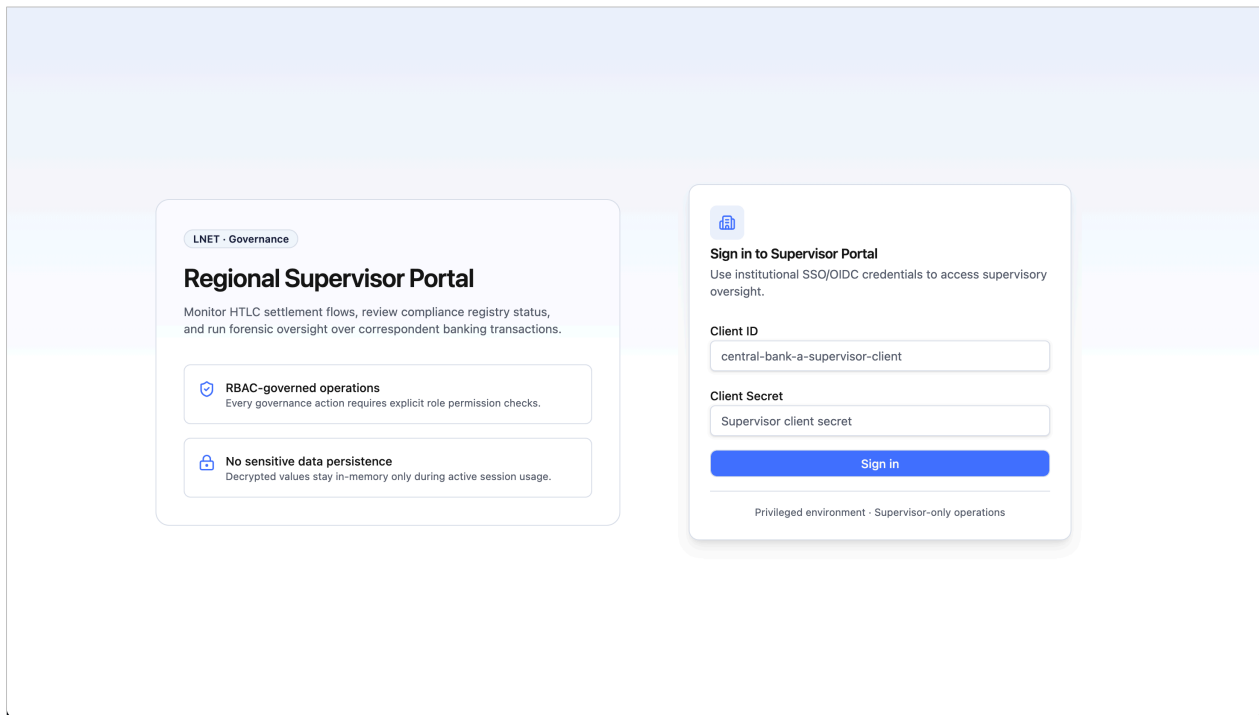


Figure 27: Supervisor Login

Open the Supervisor Portal URL in your browser. Two instances run by default:

Entity	Default URL
Spoke A supervisor	http://localhost:5182
Spoke B supervisor	http://localhost:5183

You will land on the sign-in screen.

Enter your **Client ID** (minimum 3 characters) and **Client Secret** (minimum 6 characters), then click **Sign in**. On success you are redirected to the Dashboard.

Mock data — demo login: The login form calls `/api/v1/auth/login` with the supplied credentials. In the current development environment the backend accepts any locally-configured credentials. Production Keycloak OIDC enforcement is not yet active. Do not treat this login as a security boundary.

Session credentials are never written to `localStorage` or `sessionStorage`; the session is held in an `HttpOnly` cookie set by the backend. Refreshing the browser tab does **not** end the session — on reload the portal automatically restores it from the `HttpOnly` session cookie (via `/api/v1/auth/me`). You are returned to the login screen only when the session/refresh token expires or you explicitly log out.

4. Navigation

After login, a persistent sidebar is shown on the left. It contains links to all six screens. The current screen is highlighted. All routes are protected — unauthenticated users are redirected to `/login`. Unknown routes redirect to the Dashboard (`/`).

5. Screens

5.1 Dashboard (/)

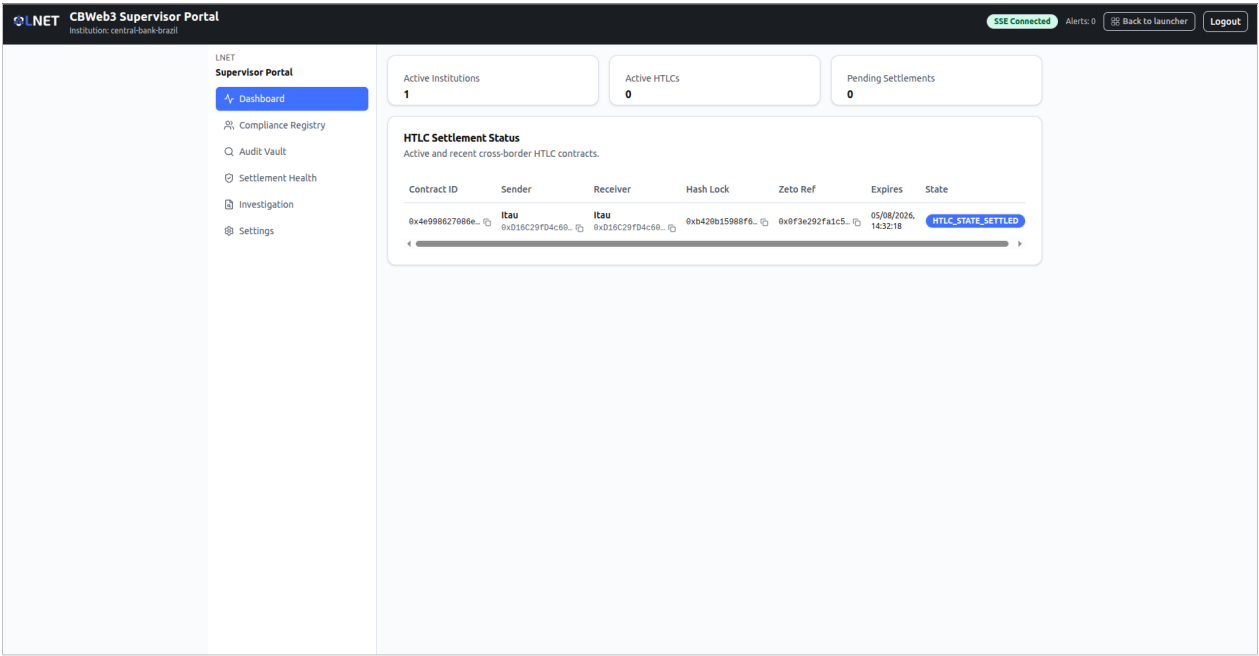


Figure 28: Supervisor Dashboard

The landing page after login. It provides a network-level summary and a full HTLC settlement table.

Summary cards

Three metric cards are shown at the top of the page.

Card	Description
Active Institutions	Count of participants whose status is ACTIVE in the compliance registry
Active HTLCs	Count of HTLC contracts currently in the HTLC_STATE_LOCKED state
Pending Settlements	Count of contracts in HTLC_STATE_LOCKED or HTLC_STATE_SETTLING

HTLC Settlement Status table

Displays all active and recent HTLC contracts retrieved from the backend.

Column	Description
Contract ID	Unique identifier for the HTLC contract (truncated; click to copy the full value)
Sender	The originating party (institution name and/or on-chain address)

Column	Description
Receiver	The receiving party (institution name and/or on-chain address)
Hash Lock	The cryptographic hash commitment securing the HTLC (truncated; click to copy)
Zeto Ref	The Paladin/Zeto privacy-token lock reference (truncated; click to copy)
Expires	Timestamp at which the time-lock expires and a refund becomes claimable
State	Current lifecycle state of the contract (see the status reference table below)

The table is refreshed automatically on page load. If no HTLCs exist, the table shows “No active HTLCs”.

Mock data: In the current development environment, data is sourced from the local backend running with seeded demonstration records. The figures shown are not from a production network.

5.2 Audit Vault (/audit)

Compliance & Audit Vault
Use regulatory view keys to decrypt shielded transactions for forensic review.

Transaction hash
0x8f41b290d90e10a89c90f6d20d9eae1288a8ff1f9d

Regulatory view key (in-memory only)
.....

Reason
AML investigation review

Decrypt Transaction

Sensitive decrypted values are not persisted to browser localStorage/sessionStorage.

Decryption Result
No decrypted payload in session yet.

Immutable Audit Logs
Read-only compliance audit logs from the backend. Page 1 · 20 per page.

Severity (e.g. CRIT) Category Filter Clear

Timestamp	Actor	Action	Category	Severity	Outcome
16/06/2026, 10:19:35	17d9c3fc-c92c-457b...	CREATE_TRANSFER_LIMIT	TRANSFER_LIMIT	INFO	SUCCESS
16/06/2026, 10:11:48	...	SIGN_CSR	CREDENTIAL	INFO	SUCCESS
16/06/2026, 10:11:44	c1bc969c-3c2e-4361...	APPROVE_KYC	CREDENTIAL	INFO	SUCCESS
16/06/2026, 09:25:40	service-account-ce...	GOVERNANCE_BOOTSTRAP	CREDENTIAL	INFO	SUCCESS

Figure 29: Supervisor Audit

The Audit Vault is a three-panel forensic screen:

1. **Credential Verification** — look up a participant’s KYC status by subject identifier.
2. **Compliance & Audit Vault** — decrypt a shielded transaction using a regulatory view key.
3. **Immutable Audit Logs** — paginated, filterable log of all governance actions recorded by the backend.

Mock data: In the development environment, audit log entries and decryption results come from the local backend and are demonstration records. They must not be used as evidence or cited in regulatory proceedings.

Credential Verification panel

Use this panel to look up the KYC/credential lifecycle status of any participant by their subject identifier (a UUID or DID).

1. Enter the subject's ID in the **Subject ID** field.
2. Click **Verify** (or press Enter).
3. The result panel shows the subject identifier, status badge, last-updated timestamp, and — if the credential is frozen — the freeze reason.

Status	Meaning
ACTIVE / APPROVED	Credential is current and the participant is cleared to transact
PENDING	Credential is under review
FROZEN	Credential has been temporarily suspended; freeze reason is shown
REVOKED / REJECTED	Credential has been permanently removed or denied

Transaction Decryption panel

Cross-border value transfers in Scenario A use Paladin/Zeto ZKP-shielded tokens. When legally authorised, a supervisor can request decryption of a specific transaction.

Field	Requirement	Description
Transaction hash	Minimum 8 characters	The 0x... hash of the shielded transaction to investigate
Regulatory view key	Minimum 8 characters	The Paladin/Zeto view key issued to the supervisor; held in memory only, never persisted
Reason	Minimum 5 characters	Mandatory justification for the audit trail

The **Decrypt Transaction** button is disabled until all three fields meet their minimum length requirements. Click it to submit. On success, the decryption result panel (to the right) shows:

- Transaction hash
- Amount and currency
- Sender address
- Receiver address

Decrypted values exist only in the active browser session and are never written to `localStorage` or `sessionStorage`.

Submitting a decryption request automatically refreshes the audit log, adding a new entry that records the action.

Immutable Audit Logs table

A paginated, read-only log (20 entries per page) of all governance actions recorded by the compliance service.

Column	Description
Timestamp	Local date and time of the event
Actor	Subject ID or system identifier that performed the action (truncated; click to copy)
Action	Action type (e.g., DECRYPT_TX, VIEW_REGISTRY, KYC_LOOKUP)
Category	Functional category of the action
Severity	INFO, LOW, MEDIUM, HIGH, or CRITICAL
Outcome	SUCCESS (green) or non-success (red)

Filtering: Enter a severity value (e.g., CRITICAL) or a category string in the filter fields above the table, then click **Filter**. Click **Clear** to remove all filters and reload the unfiltered log.

Every supervisor action — including viewing records and submitting decryption requests — is recorded here and is non-repudiable.

5.3 Investigation (/investigation)

Figure 30: Supervisor Investigation

The Investigation screen manages the **disclosure request workflow** — the multi-party sign-off process required to legally compel disclosure of shielded transaction details under AML/CFT or court-order authority.

The screen contains three panels.

Mock data: In the development environment, disclosure requests are persisted to the local backend database and are demonstration records only.

Open Disclosure Request panel

Initiates a 2-of-N disclosure workflow. The request expires automatically after 72 hours if quorum is not reached.

Field	Description
Transaction Reference	The 0x... reference of the transaction under investigation
Requestor Bank ID	The identifier of the bank initiating the request (e.g., <code>cb_lnet</code>)
Reason Code	Select one of the four statutory reason codes (see table below)

Reason codes:

Code	Use case
AML_ALERT	Anti-money laundering alert
CFT_INVESTIGATION	Counter-financing of terrorism investigation
COURT_ORDER	Legally binding court order
REGULATORY_EXAM	Scheduled regulatory examination

Click **Open Request**. On success, a summary card appears showing the new request ID, status, quorum progress, and expiry time. Record the request ID — it is required for subsequent co-signing and status lookups.

Co-sign Disclosure Request panel

Add your institution's co-signature to an existing **PENDING** disclosure request.

Field	Description
Request ID	The UUID of the disclosure request to co-sign
Signer Bank ID	Your institution's bank identifier (e.g., <code>cb_spoke_a</code>)

Click **Sign Request**. A success toast confirms the signature was recorded. Once quorum is reached the request transitions to **QUORUM_REACHED**.

Disclosure Status Lookup panel

Look up the current state and quorum progress of any disclosure request.

1. Enter the **Request ID** (UUID).
2. Click **Look up**.
3. A summary card is displayed showing: request ID, state badge, target transaction reference, reason code, quorum progress (`quorumReached` / `quorumRequired`), and expiry datetime.

State	Meaning
PENDING	Open; awaiting additional signatures
QUORUM_REACHED	Sufficient co-signatures received; disclosure may proceed
EXPIRED	The 72-hour window closed before quorum was reached

5.4 Compliance Registry (/participants)

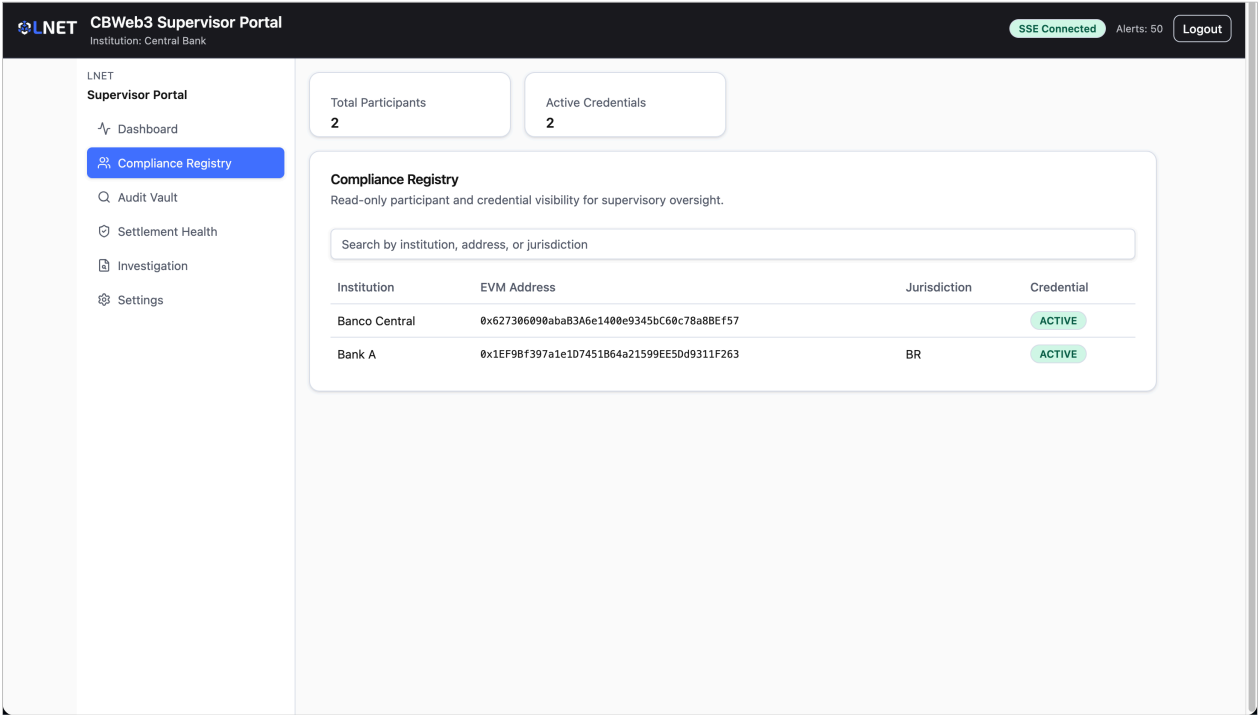


Figure 31: Compliance Participants

A read-only view of the compliance registry for all onboarded participants.

Mock data: In the development environment, the participant list is sourced from the local compliance service and contains demonstration records, not the live registry.

Summary cards

Card	Description
Total Participants	All institutions registered in the compliance registry
Active Credentials	Institutions whose credential status is ACTIVE

Compliance Registry table

Column	Description
Institution	Full legal name of the registered institution

Column	Description
EVM Address	On-chain wallet address (0x...)
Jurisdiction	ISO 3166-1 alpha-2 country code
Credential	ACTIVE (green) or other status (red)

Search: Type in the search box above the table to filter by institution name, EVM address, or jurisdiction code. Filtering is performed client-side in real time.

No write actions are available in this screen. Participant suspension or reinstatement is a governance function handled outside this portal.

5.5 Settlement Health (/stability)

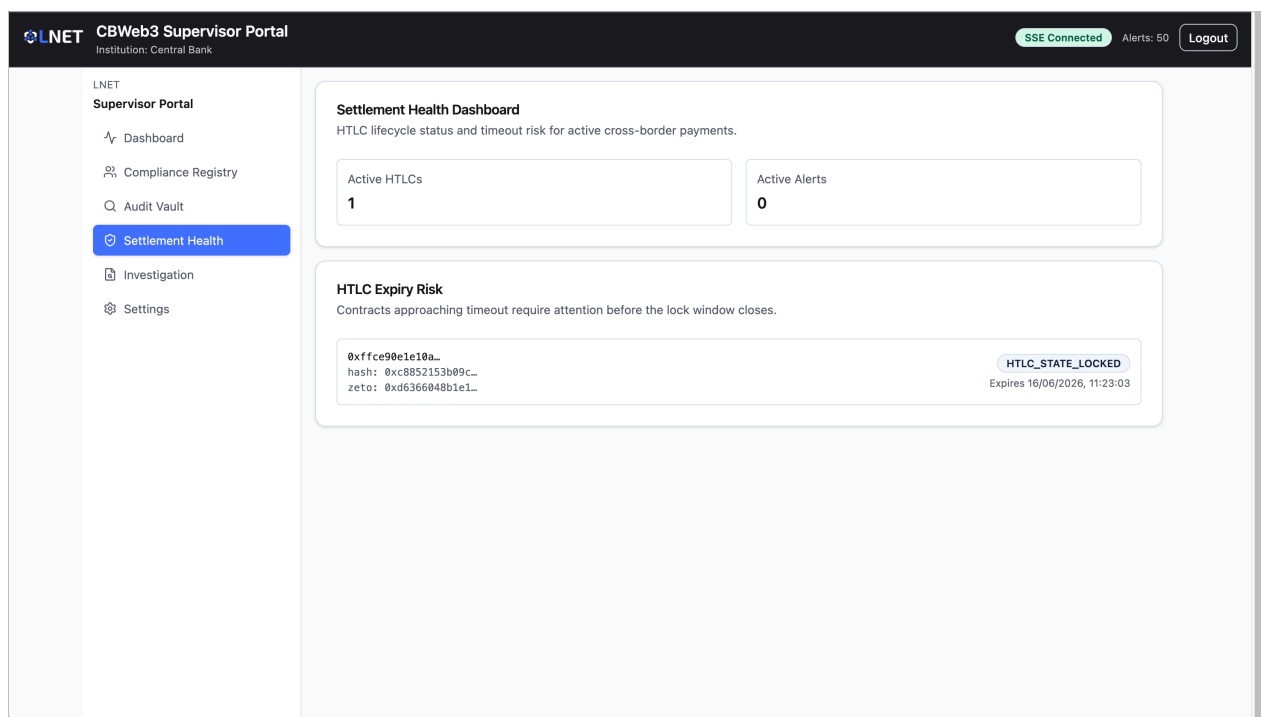


Figure 32: HTLC Health

The Settlement Health screen provides HTLC expiry risk monitoring and a count of active stability alerts. It is intended for senior supervisors watching for systemic settlement risk.

Mock data: In the development environment, HTLC and alert data is sourced from the local backend with seeded demonstration records.

Summary cards

Card	Description
Active HTLCs	Count of contracts currently in HTLC_STATE_LOCKED
Active Alerts	Count of unresolved stability alerts

HTLC Expiry Risk list

Each active HTLC is shown as a card with:

- **Contract ID** (truncated to 14 characters)
- **Hash Lock** (truncated)
- **Zeto Ref** (truncated)
- **State badge** — colour-coded by lifecycle state (see the status reference table)
- **Expires** — local datetime at which the time-lock closes

Contracts approaching their expiry time require attention before the lock window closes. Once a time-lock expires, the HTLC transitions to a refundable state and the corresponding cross-border payment must be re-initiated.

This screen is read-only. Circuit-breaker pause and resume actions (if applicable) are owned by the Governance Portal and are not accessible here.

5.6 Settings (/settings)

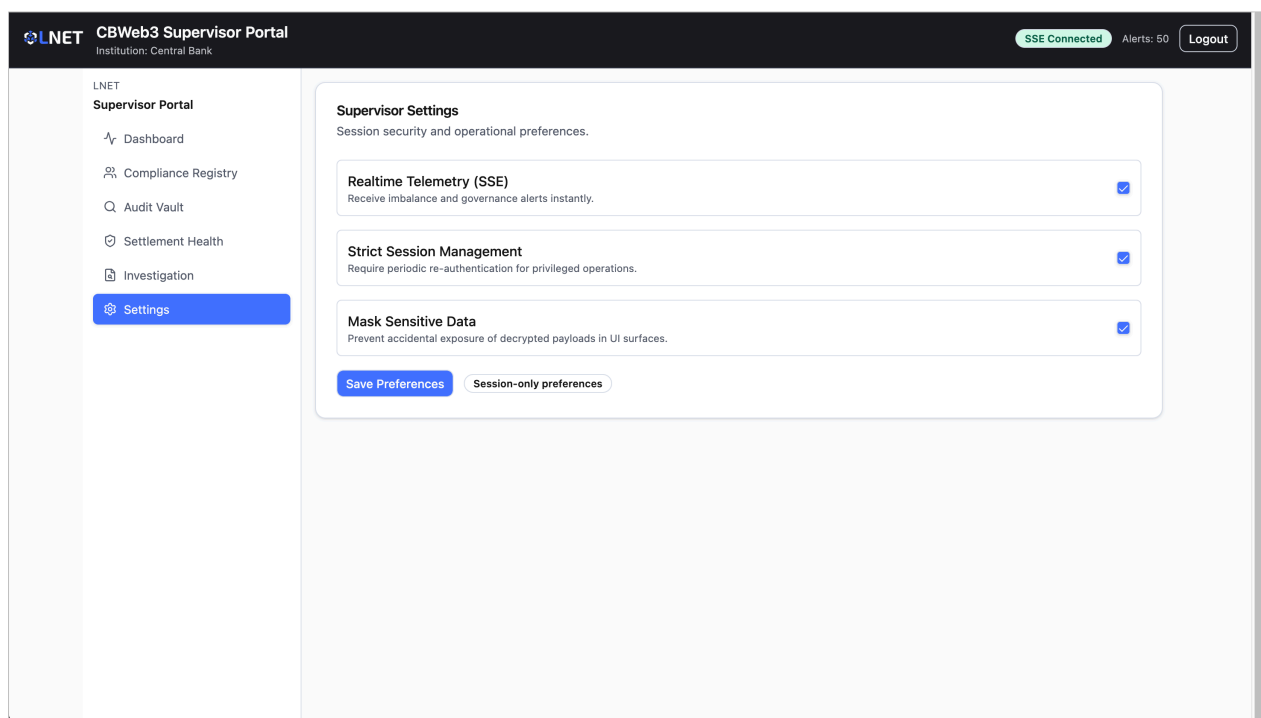


Figure 33: Settings

Session-scoped operational preferences. Changes are saved for the current browser session only and are lost on logout or tab close.

Setting	Default	Description
Realtime Telemetry (SSE)	On	Receive imbalance and governance alerts instantly via Server-Sent Events
Strict Session Management	On	Require periodic re-authentication for privileged operations

Setting	Default	Description
Mask Sensitive Data	On	Prevent accidental exposure of decrypted payloads in UI surfaces

Click **Save Preferences** to apply. A confirmation toast is shown. The **Session-only preferences** label is a reminder that these settings are not persisted to the backend.

6. Typical workflows

Investigate a flagged transaction

1. Open the **Audit Vault** (/audit).
2. In the **Credential Verification** panel, verify the credential status of the parties involved using their subject IDs.
3. In the **Compliance & Audit Vault** panel, enter the transaction hash, your regulatory view key, and a documented reason. Click **Decrypt Transaction**.
4. Review the decrypted amount, currency, sender, and receiver in the result panel.
5. The decryption action is automatically recorded in the **Immutable Audit Logs** table.

Open a multi-party AML disclosure request

1. Navigate to **Investigation** (/investigation).
2. In the **Open Disclosure Request** panel, enter the transaction reference, your requestor bank ID, and select the appropriate reason code.
3. Click **Open Request** and note the returned **Request ID**.
4. Share the Request ID with the other required co-signers.
5. Each co-signer goes to the **Co-sign Disclosure Request** panel, enters the Request ID and their signer bank ID, and clicks **Sign Request**.
6. Once all required signatures are collected, use the **Disclosure Status Lookup** panel to confirm the state has changed to QUORUM_REACHED.

Monitor settlement health

1. Open the **Dashboard** (/) to see a summary of active HTLCs and pending settlements.
2. Open **Settlement Health** (/stability) for a detailed list of HTLC contracts with their expiry times.
3. Focus attention on contracts whose expiry timestamp is approaching. Coordinate with the relevant bank operators to ensure secrets are revealed and settlements are completed before the time-lock closes.

Review the compliance registry

1. Navigate to **Compliance Registry** (/participants).
2. Note the **Total Participants** and **Active Credentials** counts.
3. Use the search field to locate a specific institution by name, wallet address, or jurisdiction code.
4. Review the credential status badge. A red badge indicates a non-active credential; escalate through your governance channel to investigate.

7. Status reference

HTLC lifecycle states

State	Badge colour	Meaning
HTLC_STATE_PENDING	Grey	Contract created; not yet locked
HTLC_STATE_LOCKED	Grey	Funds locked; awaiting secret reveal
HTLC_STATE_SETTLING	Yellow	Secret revealed; settlement in progress
HTLC_STATE_SETTLED	Green	Settlement complete; funds released
HTLC_STATE_REFUNDING	Yellow	Timeout or cancellation; refund in progress
HTLC_STATE_REFUNDED	Red	Funds returned to sender
HTLC_STATE_INVALID	Red	Contract is in an unrecoverable error state

Disclosure request states

State	Meaning
PENDING	Open; waiting for co-signatures to reach quorum
QUORUM_REACHED	Sufficient co-signatures received; disclosure authorised
EXPIRED	72-hour window elapsed before quorum; request is closed

Credential / KYC states

Status	Meaning
ACTIVE	Credential is current; participant is cleared to transact
APPROVED	Credential has been approved (equivalent to ACTIVE)
PENDING	Credential is under review
FROZEN	Credential temporarily suspended; freeze reason provided
REVOKED	Credential permanently removed
REJECTED	Credential application was denied

Audit log severity levels

Level	Meaning
INFO	Routine informational event
LOW	Minor event; no immediate action required
MEDIUM	Event warranting attention
HIGH	Significant event; prompt review recommended
CRITICAL	Urgent event; immediate attention required

8. Security notes

Concern	Current behaviour
Authentication	Institutional SSO/OIDC credentials submitted to <code>/api/v1/auth/login</code> . Production Keycloak enforcement is not yet active.
No token issuance	The Supervisor Portal has no mint, burn, or transfer capabilities by design.
View key handling	Regulatory view keys are never written to <code>localStorage</code> or <code>sessionStorage</code> ; they exist in memory only for the active browser session.
Audit trail	Every action — including decryption requests and registry views — is logged in the immutable audit log.
Session persistence	The session lives in an <code>HttpOnly</code> cookie, not <code>localStorage/sessionStorage</code> . Refreshing the tab restores the session automatically via <code>/api/v1/auth/me</code> ; the user is returned to login only when the session/refresh token expires or logout is invoked.
Role-based access	Three roles are supported: <code>SUPERVISOR_ROLE</code> , <code>COMPLIANCE_OFFICER</code> , and <code>CENTRAL_BANK_ADMIN</code> . All currently share the same set of portal permissions.

9. Troubleshooting

Symptom	Likely cause	Resolution
Login fails immediately	Client ID or Client Secret too short (minimum 3 / 6 characters)	Check the minimum-length error message shown below the field
Login fails with an authentication error from the server	Backend auth service is not running, or credentials are not recognised in the local configuration	Confirm the backend stack is up (<code>make scenario-a.up</code> or equivalent) and verify credentials
Dashboard cards show -	The compliance or HTLC backend service is unreachable	Check that all backend containers are running and the API base URL is correctly configured
HTLC table is empty	No HTLC records seeded or created yet	Run the scenario-a demo scripts to create sample HTLC locks
Decrypt button stays disabled	One or more fields are below the minimum length	Ensure tx hash ≥ 8 characters, view key ≥ 8 characters, reason ≥ 5 characters

Symptom	Likely cause	Resolution
Audit log table shows “No audit logs found”	No governance actions have been performed yet, or the backend audit service is unreachable	Perform any action (e.g., a KYC lookup) to generate the first log entry; verify backend connectivity
Disclosure request fails to open	Missing transaction reference or requestor ID	Ensure both required fields are filled before clicking Open Request
Status lookup returns “not found”	The request ID is incorrect, or the request has expired	Verify the request ID and check whether the 72-hour expiry has elapsed
Participant list is empty	The compliance participants endpoint is unreachable or returned no data	Verify backend connectivity; check if participants have been registered
Data looks static or unrealistic	Expected — the portal runs on demonstration data in the current development environment	No action required; see preview status notice at the top of this manual

When the portal is wired to live production data and production Keycloak authentication, this manual will be updated to remove the mock-data warnings and document the verified behaviour of each screen.

NOC Portal — User Manual

(Scenario A)

Audience: Network operations / SRE engineers. **Scenario:** A — Enhanced Correspondent Banking (dual-layer HTLC). **Data source:** The portal connects to a live NOC backend service. The backend exposes real component health, alert, log, and topology data. The WebSocket telemetry stream is not yet implemented; the portal falls back to HTTP polling automatically.

Read first — current status

The NOC Portal is wired to a real backend API (Keycloak auth + NOC backend service). Component health, alerts, and logs reflect actual spoke state when the stack is running. The **WebSocket push stream** is a stub — the portal always runs in HTTP polling mode until the stream is implemented. Authenticated sessions are stored as JWT tokens in `localStorage`.

1. Overview

The NOC (Network Operations Center) Portal is the monitoring surface for engineers responsible for the health of Scenario A's spoke network. It provides:

- Real-time health status for blockchain nodes (Hyperledger Besu), Paladin privacy nodes, payment orchestrators, and CACTI relay containers.
- An active alert feed with acknowledge/dismiss workflow.
- Container log inspection for any registered component.
- An interactive topology diagram showing spoke layout and link health.
- An immutable audit trail of operator actions.

Actor	Portal	Role
Network Operations Engineer / SRE	NOC Portal	Infrastructure health, alert triage, relay monitoring, log review, topology

The portal is **read-only with two exceptions**: operators can acknowledge and dismiss alerts. No destructive infrastructure controls (token issuance, configuration changes) are exposed here.

2. Access and Login

URL: The NOC Portal URL provided by your system administrator.

Authentication: Keycloak OIDC. The portal authenticates against the `cbweb3` realm with client ID `noc-portal`. The portal itself does not enforce the `SYS_ADMIN` role at login — after a successful Keycloak authentication the frontend assigns the `SYS_ADMIN` role to the session. Role-based access control is enforced server-side by the NOC backend, which rejects API calls from accounts that do not carry the required role.

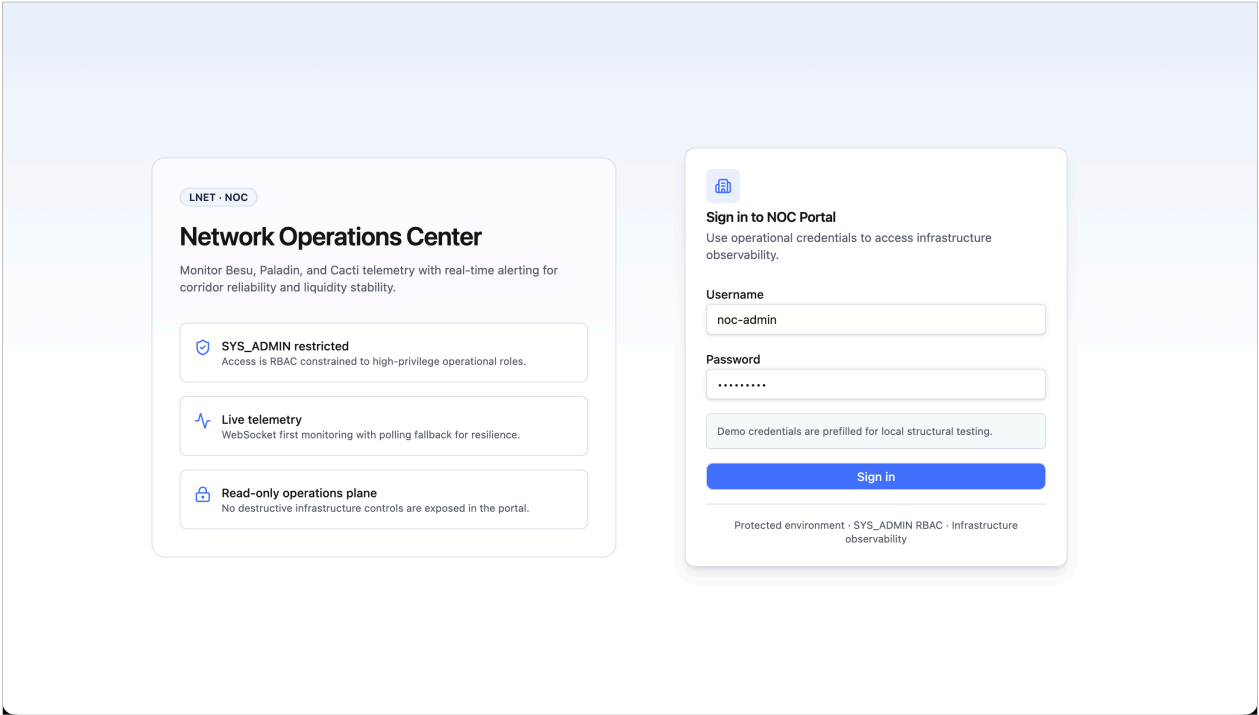


Figure 34: Login Page

Login steps

1. Open the NOC Portal URL in your browser.
2. Enter your `SYS_ADMIN` username and password.
3. Click **Sign in**.
4. On success, you are redirected to the Dashboard.

Login page layout

The login page has two panels on desktop:

- **Left panel** — feature summary (`SYS_ADMIN` restriction, live telemetry, read-only operations plane).
- **Right panel** — sign-in form (username, password, submit button).

Field	Notes
Username	Your <code>SYS_ADMIN</code> Keycloak username.
Password	Your <code>SYS_ADMIN</code> Keycloak password.

Standard bank-portal or governance-portal credentials cannot be used here. If login fails immediately, verify the account has the `SYS_ADMIN` role in Keycloak.

Session persistence: On successful login the portal stores `noc_access_token` and `noc_refresh_token` in `localStorage`. The access token is short-lived (5 minutes by realm default), so the portal renews it automatically — shortly before expiry and again if a request is still rejected — using the refresh token. A portal left open therefore keeps working without re-authentication up to the realm’s SSO session limit, and a page reload within that window restores the session instead of returning to the login page.

When renewal is no longer possible (refresh token expired, session revoked), the portal drops the session and returns to the login page with “*Session expired. Sign in again.*”. **Sign out** clears both tokens and returns to the login page; use **Back to launcher** to leave the portal for the entity launcher.

3. Navigation

The portal uses a persistent left sidebar. All routes are protected — the login page is the only publicly accessible screen.

Sidebar label	Route	Description
Dashboard	/	Platform-wide health snapshot and alert feed.
Infrastructure	/infrastructure	Per-component health table (nodes, orchestrators).
Relays	/relays	CACTI relay container health per spoke.
Topology	/topology	Interactive network diagram.
Audit	/audit	Operator action history and resolved alerts.
Settings	/settings	Polling interval and notification preferences.

The **Log Viewer** (/logs/:componentId) has no sidebar entry. It is reached by clicking the **View Logs** button on the Infrastructure or Relays pages.

Unknown routes redirect to / (Dashboard).

4. Screens

4.1 Dashboard (/)

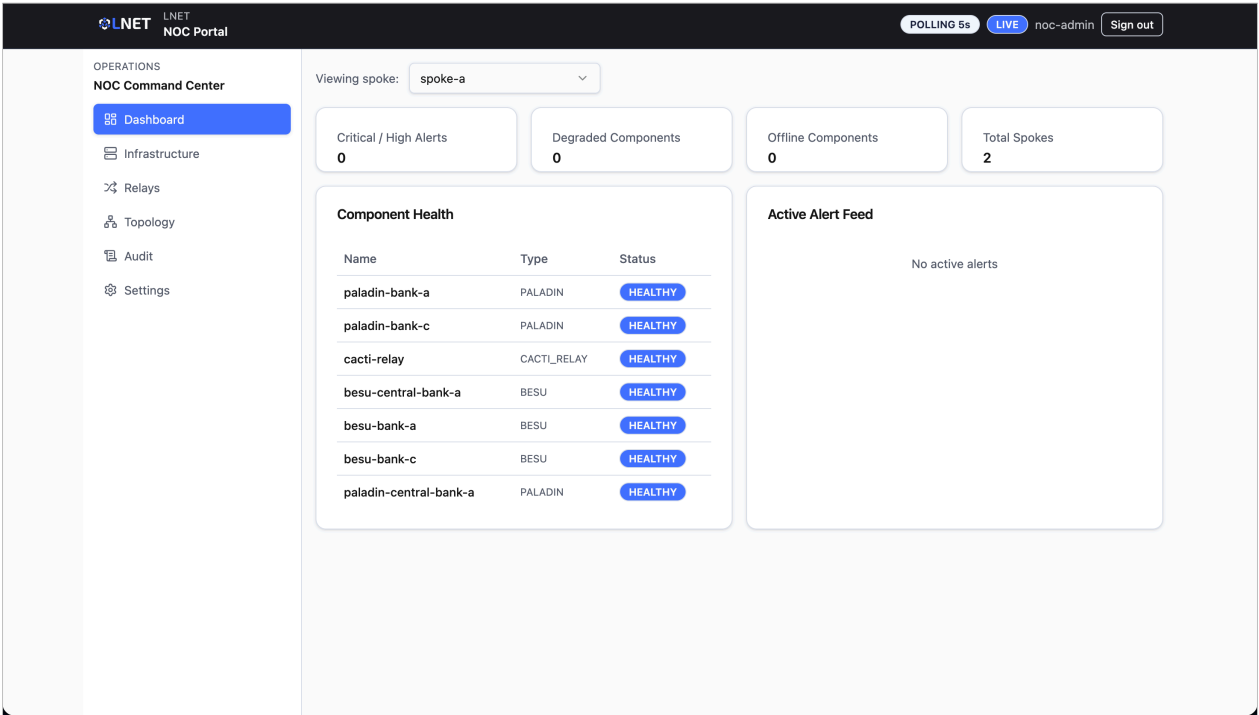


Figure 35: Dashboard

The Dashboard is the primary monitoring view. It aggregates platform-wide health data across all spokes (or a single selected spoke) and displays the active alert feed.

Spoke Selector

A dropdown at the top of the page. On load, the first available spoke is selected automatically. Select **a specific spoke** to filter both the Component Health table and the Alert Feed to that spoke only.

The dropdown lists only registered spokes: the platform-wide “all spokes” mode exists in the code but has no option to select it, so it cannot currently be reached from the UI.

When **Critical Alerts Only** is enabled in [Settings](#), the feed lists only HIGH and CRITICAL alerts and a CRITICAL ONLY badge appears next to the feed title. The Summary Cards are not affected by that setting.

Summary Cards

Four cards provide an instant platform status snapshot:

Card	Colour when non-zero	Description
Critical / High Alerts	Red	Active alerts with severity HIGH or CRITICAL. Requires immediate attention.
Degraded Components	Orange	Components not in HEALTHY state (includes DEGRADED, OFFLINE, UNKNOWN).

Card	Colour when non-zero	Description
Offline Components	Red	Components in OFFLINE state. Requires immediate attention.
Total Spokes	—	Count of active spokes registered in the system.

Component Health Table

Shows the first 8 components for the selected scope. Columns: **Name**, **Type**, **Status** badge. For the full component list, navigate to [Infrastructure](#).

Active Alert Feed

Lists up to 8 active alerts. Each alert row shows:

Element	Description
Title	Short description of the alert condition.
Severity badge	INFO, WARNING, HIGH, or CRITICAL.
ACK badge	Shown when the alert has been acknowledged by any operator.
Dismiss button	X icon on the right — click to dismiss directly from the feed.

Click any alert row (not the dismiss button) to open the **Alert Detail Modal**.

Alert Detail Modal fields:

Field	Description
Title	Alert title.
Severity	Severity badge.
Affected component	The component that raised the alert (name, type, endpoint, health status).
Created	Timestamp of alert creation, shown in browser local time.
Root cause sig	System-generated root-cause signature.
Acknowledge button	Marks the alert as seen; keeps it in the feed.
Dismiss button	Removes the alert from the active feed.

Polling

The Dashboard refreshes automatically at the interval configured in [Settings](#) (default: 15 seconds, minimum: 5 seconds). The WebSocket stream is not yet active; the portal always uses HTTP polling.

4.2 Infrastructure (/infrastructure)

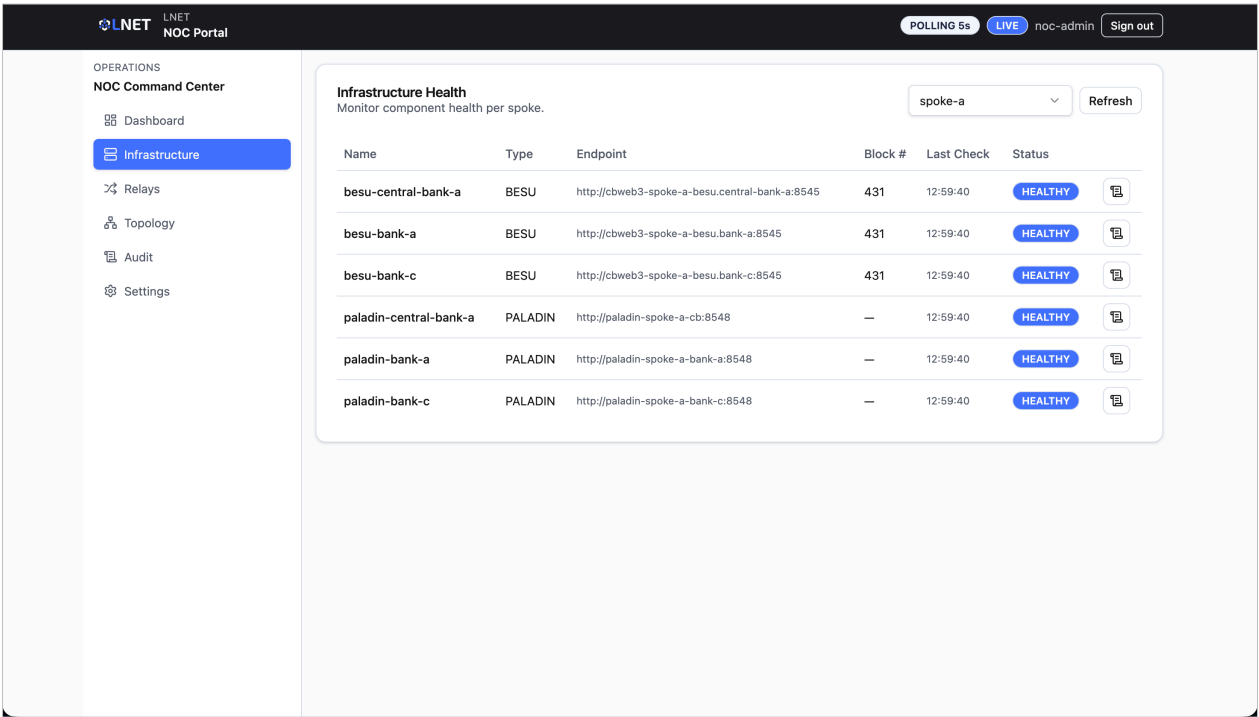


Figure 36: Infrastructure

Lists the health status of all non-relay infrastructure components for the selected spoke: Hyperledger Besu blockchain nodes, Paladin privacy nodes, and payment orchestrators. CACTI relay containers are excluded here — use [Relays](#) for those.

Controls

Control	Description
Spoke Selector	The first available spoke is auto-selected on load, so the table is generally populated on arrival. Selecting a different spoke reloads the table for that spoke.
Refresh button	Force an immediate health poll; disabled while loading.

Component Health Table

Column	Description
Name	The component’s display name.
Type	Component category: BESU, PALADIN, or PAYMENT_ORCHESTRATOR.
Endpoint	The component’s API or RPC endpoint (truncated).
Block #	Latest block number — populated for BESU nodes only. - for other types.
Last Check	Timestamp of the most recent health poll (shown as local time).

Column	Description
Status	Health status badge. See Status Reference .
View Logs	Icon button — navigates to the Log Viewer for this component.

The table auto-refreshes at the configured polling interval whenever a spoke is selected.

4.3 Relay Status (/relays)

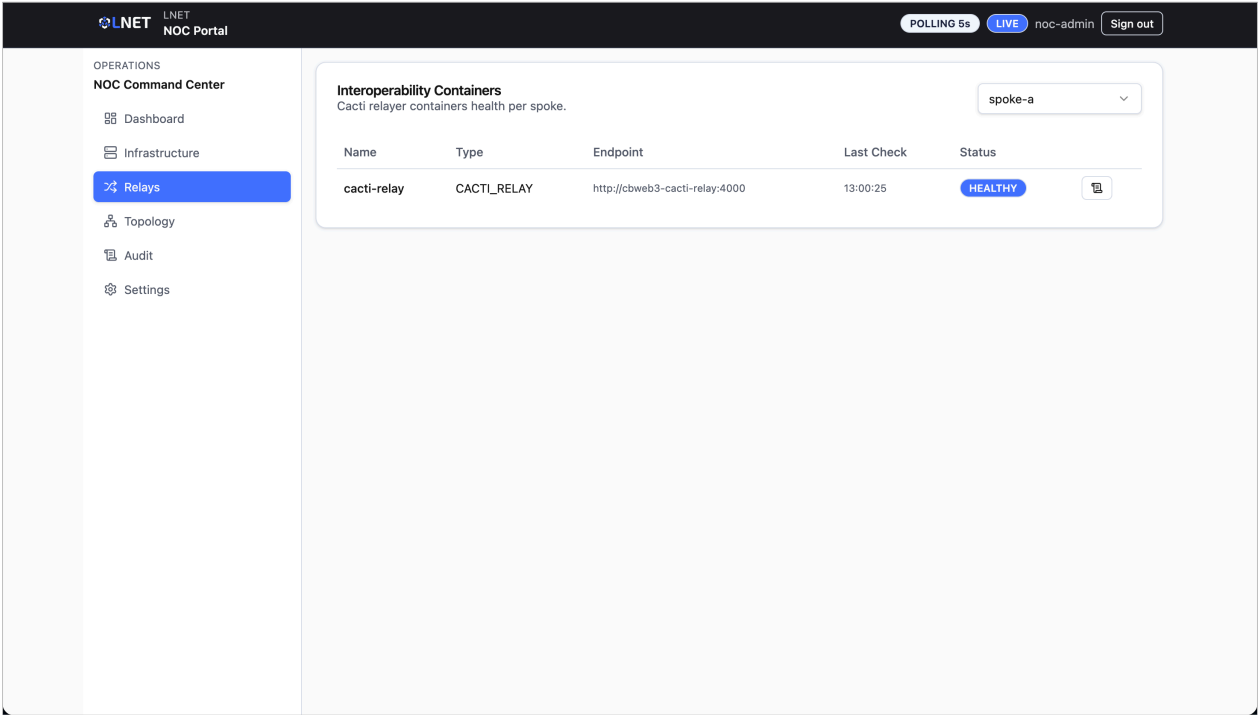


Figure 37: Relay

Shows the health of **CACTI interoperability relay containers** — the components responsible for cross-spoke event propagation. When a cross-border settlement appears stuck, this page is the first place to check.

The sidebar label for this screen is **Relays**; the in-page title is **Interoperability Containers**.

Controls

Control	Description
Spoke Selector	The first available spoke is auto-selected on load; its relay containers appear on arrival. Selecting a different spoke reloads the table for that spoke.

Relay Components Table

Displays only CACTI_RELAY type components for the selected spoke.

Element	Colour	Description
Green edge	Green line	A healthy link between two nodes.
Red edge	Red line	An unhealthy link between two nodes.

Each node card shows: a colour-coded dot, the node name, its kind label, a health status badge, and the name of the spoke it belongs to.

Nodes are grouped by spoke in columns. The layout is generated automatically based on spoke membership.

Diagram Controls

Control	Action
Scroll wheel	Zoom in / out.
Click and drag (on the canvas)	Pan the diagram.
Mini-map (bottom-right corner)	Navigate quickly across a large topology.
Controls panel (bottom-left)	Zoom in, zoom out, fit view, lock layout buttons provided by ReactFlow.

Topology Health Summary Card

Located to the right of the diagram:

Metric	Description
Total Nodes	Number of nodes registered in the topology.
Total Links	Number of directed edges between nodes.
Healthy Links	Count of edges marked as healthy (green).
Unhealthy Links	Count of edges marked as unhealthy (red).

A colour legend below the metrics identifies each node type.



4.5 Log Viewer (/logs/:componentId)

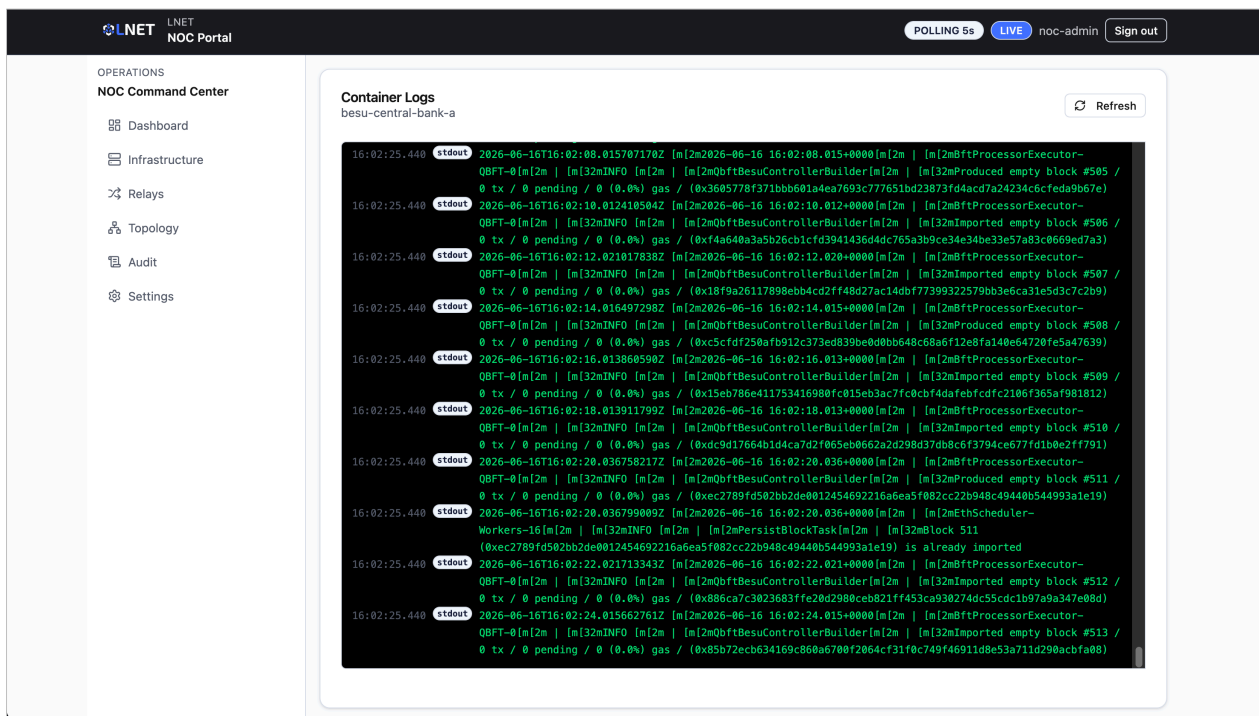


Figure 39: Logs

Displays container logs for a specific infrastructure or relay component. There is no sidebar entry for this screen — navigate here by clicking the **View Logs** button from the [Infrastructure](#) or [Relays](#) pages.

The component name is shown in the page subtitle. The `componentId` in the URL is the internal component ID; the human-readable name is passed as a query parameter (`?name=...`).

Controls

Control	Description
Refresh button	Force an immediate log fetch. Spinner is shown while loading.
SNAPSHOT <n>s OLD badge	Appears when the newest line collected is more than 60 seconds old — see <i>Snapshots, not a live tail</i> below.

Log Panel

Terminal-style viewer (black background, green text).

Column	Description
Timestamp	Time portion of the log line's <code>occurred_at</code> timestamp (HH:MM:SS.mmm). This is when the agent collected the batch, so a whole batch shares one timestamp; each line's own time appears inside the log text.

Column	Description
Stream badge	<code>stdout</code> (normal output) or <code>stderr</code> (error output). <code>stderr</code> lines appear in red.
Log line	Raw log text from the container.

- Displays the **last 500 lines**.
- Auto-scrolls to the bottom on initial load and on each refresh.
- Refreshes automatically every **15 seconds**. This cycle is fixed and is not affected by the Settings polling interval.

Snapshots, not a live tail

Logs are collected by the spoke's NOC agent and pushed to the backend; the viewer reads what the backend stored. If the container stops, or its agent stops collecting, the screen keeps showing the last snapshot with no other visible change — the **SNAPSHOT <n>s OLD** badge is what tells you the lines are not current.

4.6 Audit (/audit)

The screenshot displays the NOC Portal interface. The top navigation bar includes the LNET NOC Portal logo, a polling interval of 5s, a LIVE status indicator, the user name 'noc-admin', and a Sign out button. The left sidebar lists navigation options: Dashboard, Infrastructure, Relays, Topology, Audit (highlighted), and Settings. The main content area features a search bar for actors, actions, targets, and details. Below the search bar, there are two sections: 'Audit Trail' and 'Resolved Alerts'. The 'Audit Trail' section shows an empty table with columns for Timestamp, Actor, Action, Target, and Detail, indicating 'No audit records'. The 'Resolved Alerts' section shows a table with columns for Resolved At, Title, Severity, and Root Cause Sig, listing two resolved alerts for 'besu-bank-a OFFLINE' and 'besu-bank-c OFFLINE' with a severity of 'CRITICAL'.

Figure 40: Audit

Provides an immutable record of all operator actions taken within the NOC Portal and a history of resolved alerts.

A **search box** at the top filters both tables simultaneously. The filter matches against actor, action, target ID, and detail text in the Audit Trail table; and against title, severity, and root cause signature in the Resolved Alerts table.

Audit Trail Table

Every alert acknowledgement and dismissal performed by any NOC operator is recorded here.

Column	Description
Timestamp	When the action was performed (local time).
Actor	Username of the NOC operator who performed the action, taken from the preferred_username claim of the token used. Deployments whose token carries no username fall back to the account's internal ID, and a request with no readable identity is recorded as dev-user .
Action	The action type: Acknowledge (from ACKNOWLEDGE_ALERT) or Dismiss (from DISMISS_ALERT).
Target	The internal ID of the alert that was acted upon.
Detail	Any additional context recorded at the time of the action.

Resolved Alerts Table

A searchable history of alerts that have been resolved (automatically or via dismissal).

Column	Description
Resolved At	Timestamp when the alert transitioned to resolved (local time). – if not yet recorded.
Title	The alert title.
Severity	Severity badge at the time of resolution.
Root Cause Sig	System-generated identifier for the root cause pattern (monospaced, truncated).

4.7 Settings (/settings)

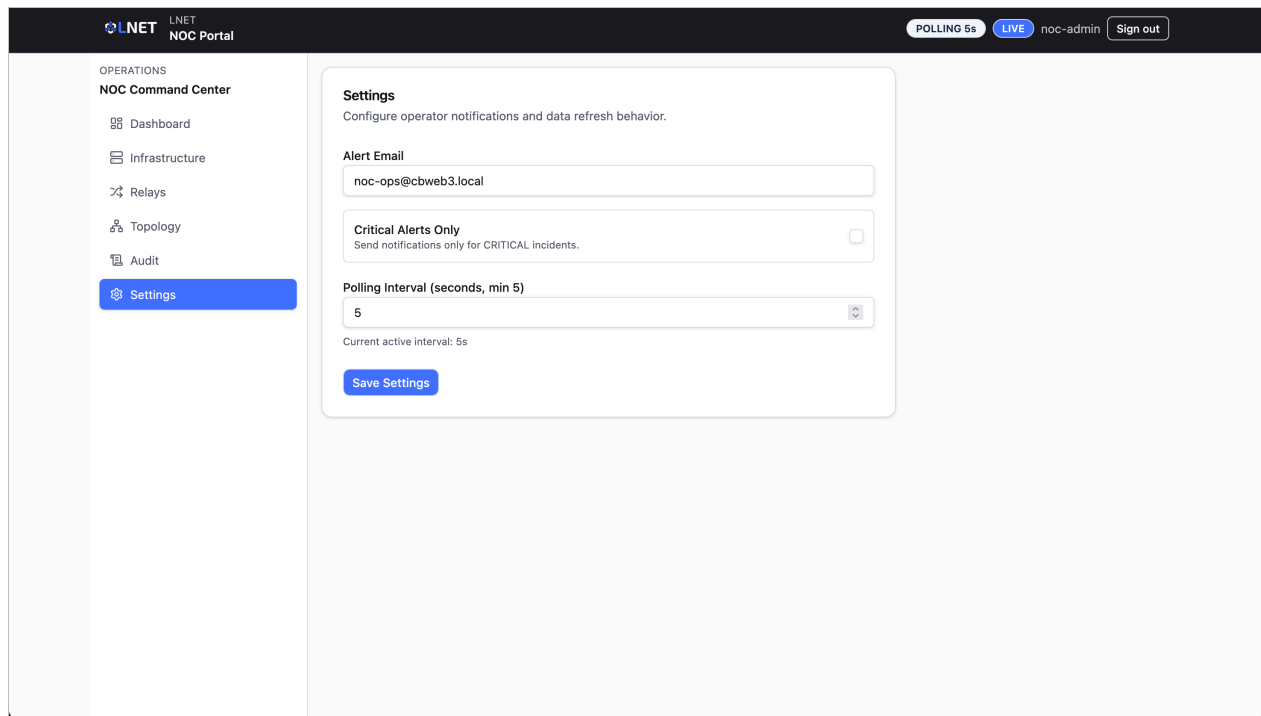


Figure 41: Settings

Configures operator-level monitoring preferences. These preferences are persisted to `localStorage` (key: `noc-ui-settings`): the **Alert Email** (`alertEmail`), the “Critical Alerts Only” toggle (`muteAlerts`), the display timezone (`timezone`), and the polling interval (`fallbackPollingSeconds`). They survive page reloads within the same browser and reset when the browser storage is cleared. The Alert Email and Polling Interval are committed when you click **Save Settings**; the Critical Alerts Only toggle takes effect immediately. No preference is sent to the backend.

Setting	Default	Description
Alert Email	<code>noc-ops@cbweb3.local</code>	Email address for alert notifications. Stored locally — not sent to the backend.
Critical Alerts Only	Off	Applied on Save Settings . When checked, hides <code>INFO</code> and <code>WARNING</code> severity alerts from the Dashboard’s Active Alert Feed. Only <code>HIGH</code> and <code>CRITICAL</code> alerts appear, and a <code>CRITICAL ONLY</code> badge is shown on the feed header.
Polling Interval	15 seconds	Applied immediately on toggle. How frequently the auto-refreshing pages reload data. Minimum: 5 seconds. Values below 5 are rejected. Applied on Save Settings .

Click **Save Settings** to apply the Alert Email and Polling Interval. A toast confirms success, and

the active polling interval shown below the input updates immediately. If the interval is below 5 seconds or not a number, an error toast appears and **nothing is saved** — including the Alert Email. The **Critical Alerts Only** checkbox does not require Save; it takes effect as soon as it is toggled.

The Polling Interval governs every auto-refreshing page in the portal — [Dashboard](#), [Infrastructure](#) and [Relay Status](#) — not the Dashboard alone.

Note that the **Critical / High Alerts** counter in the Dashboard KPI row always counts **HIGH** and **CRITICAL** active alerts regardless of this setting.

5. Alert Management

NOC operators can take two actions on active alerts: **Acknowledge** and **Dismiss**. Both actions are logged in the [Audit](#) page.

Acknowledge vs. Dismiss

Action	Meaning	Effect
Acknowledge	An operator has seen the alert and is actively investigating.	Alert remains in the active feed. An ACK badge appears on the row. Action logged in Audit.
Dismiss	The alert is not actionable, is a false positive, or has been resolved externally.	Alert is removed from the active feed and moved to Resolved Alerts on the Audit page. Action logged in Audit.

How to Acknowledge

1. On the Dashboard, locate the alert in the **Active Alert Feed**.
2. Click the alert row to open the **Alert Detail Modal**.
3. Review the alert title, severity, affected component, and description.
4. Click **Acknowledge**.

The ACK badge appears on the alert row. The action appears in the Audit Trail.

How to Dismiss

Option A — from the alert feed directly:

1. Click the **X (dismiss) button** on the right side of the alert row.
2. The alert is removed immediately.

Option B — from the Alert Detail Modal:

1. Click the alert row to open the modal.
2. Click **Dismiss** in the modal.

In both cases the alert disappears from the feed and is logged in the Audit Trail. Dismissed alerts cannot be re-activated from the portal.

Dismissing does not silence an ongoing fault. If the component is still unhealthy, the next agent report raises a new alert for it — dismissal clears the row you acted on, not the condition

behind it. The alert stops coming back once the component reports **HEALTHY**, which also resolves it automatically. Use **Acknowledge** for a fault you are actively working on: it keeps the alert in the feed and marks it as taken.

An agent that stops reporting is itself alerted on: its components move to **UNKNOWN** after the grace window and a **HIGH** alert is raised for each, because a NOC that cannot see a component must say so rather than show a quiet screen.

6. Typical Workflows

6.1 Morning health check

1. Log in to the NOC Portal.
2. On the **Dashboard**, check the four summary cards. Any non-zero values in **Critical / High Alerts** or **Offline Components** require immediate action.
3. If critical/high alerts are present, click each alert row to open the Alert Detail Modal and review the affected component and its health status.
4. Navigate to **Infrastructure** and select each spoke in turn. Confirm no unexpected **DEGRADED** or **OFFLINE** components.
5. Navigate to **Relays** and select each spoke. Confirm all CACTI relay containers show **HEALTHY**.
6. Navigate to **Topology** and verify that all edges are green (healthy links).

6.2 Investigating a stuck cross-border settlement

1. Navigate to **Relays** and select the spoke involved in the settlement.
2. Identify any relay in **DEGRADED** or **OFFLINE** state.
3. Click **View Logs** for the suspect relay to open the Log Viewer.
4. Review **stderr** lines (highlighted in red) for error messages. Look for connection failures, timeout errors, or panics.
5. If the relay log shows no recent activity (no entries within the expected settlement window), the relay may be stalled — not the contract.
6. Escalate to the infrastructure team with the component ID, the log excerpt, and the last-checked timestamp from the Relays table.

6.3 Acknowledging and tracking an active alert

1. On the **Dashboard**, click the alert row in the Active Alert Feed.
2. Read the Alert Detail Modal fully: title, severity, affected component, created time (local time), and root-cause signature.
3. If you are taking ownership, click **Acknowledge**. The ACK badge appears.
4. Investigate using the **Log Viewer** for the affected component.
5. Once resolved externally, return to the Dashboard and **Dismiss** the alert.
6. Verify the action appears in **Audit** > Audit Trail.

6.4 Checking container logs for a specific component

1. Navigate to **Infrastructure** (for nodes/orchestrators) or **Relays** (for CACTI containers).
2. Select the appropriate spoke from the Spoke Selector.
3. Locate the component in the table and click the **View Logs** icon button.
4. The Log Viewer opens showing the last 500 lines.
5. Look for **stderr** lines (red) — these typically contain error or warning messages.
6. Click **Refresh** to force an immediate fetch if the 15-second auto-refresh has not triggered yet.

6.5 Adjusting the polling interval during an incident

1. Navigate to **Settings**.
2. Lower the **Polling Interval** to 5 seconds (the minimum) for faster alert and health updates during active incident investigation.
3. Click **Save Settings**.
4. Return the interval to the normal value (15–30 seconds) after the incident is resolved to reduce backend load.

7. Alert Reference

Severity Levels

Severity	Badge colour	When to act
INFO	Grey	Informational — no action required. Review periodically.
WARNING	Yellow/orange	Potential issue — monitor. Investigate if condition persists or worsens.
HIGH	Red	Significant problem — investigate as soon as possible.
CRITICAL	Red	Severe failure — requires immediate action.

Recommended Response by Severity

Severity	Recommended response
INFO	Acknowledge to mark as seen; dismiss if it is a known benign event.
WARNING	Acknowledge and investigate root cause. Check component logs. Dismiss once confirmed non-critical.
HIGH	Acknowledge immediately. Investigate component logs and relay status. Escalate if not resolved within SLA.
CRITICAL	Page on-call engineer. Acknowledge immediately. Isolate affected spoke if necessary. Escalate to lead.

8. Status Reference

Component Health Statuses

Status	Badge colour	Meaning
HEALTHY	Green (default)	Component is responding correctly to health checks.

Status	Badge colour	Meaning
DEGRADED	Yellow/orange (warning)	Component is responding but with errors or elevated latency. Investigate promptly.
OFFLINE	Red (destructive)	Component is not responding to health checks. Requires immediate attention.
UNKNOWN	Grey (secondary)	Health check has not been performed yet, or the last result could not be determined.

Relay-Specific Interpretation

Status	Relay meaning
HEALTHY	CACTI relay container is up and forwarding events between spokes.
DEGRADED	Relay is responding but may be experiencing partial failures, high latency, or reconnection loops. Cross-spoke settlements may be slow.
OFFLINE	Relay container is not responding. Cross-spoke settlements will stall. Investigate relay logs and container state immediately.
UNKNOWN	Relay has not been polled yet, or the NOC backend cannot reach the container endpoint.

Topology Link Colours

Colour	Meaning
Green edge	Link between nodes is healthy.
Red edge	Link between nodes is unhealthy.

9. Troubleshooting

Login fails

- Verify the username and password are correct for a **SYS_ADMIN** Keycloak account. Standard bank-portal credentials will not work.
- Confirm the Keycloak service is running (`docker compose ps` in the deploy directory — look for the `keycloak` container).
- Confirm the NOC backend service is running and reachable.
- Check the browser console for the Keycloak error description; the portal surfaces the `error_description` field from the Keycloak token endpoint.

Dashboard shows no data or all components show UNKNOWN

- Verify the NOC backend service is running and that the spoke services are up.
- Navigate to **Infrastructure**, select a spoke, and click **Refresh** to trigger a manual health poll.

- Check the NOC backend service logs for database connectivity issues or spoke-endpoint unreachability.

Relay page shows an empty table

- A spoke must be selected in the **Spoke Selector** before relay data is shown. Select a specific spoke — the table will populate with `CACTI_RELAY` components for that spoke.
- If the table remains empty after selecting a spoke, no CACTI relay components are registered for that spoke in the NOC backend. Verify the spoke was onboarded with relay container registration.

Log Viewer shows no logs or an error

- Verify the component is running and that its endpoint is reachable from the NOC backend host.
- Click **Refresh** to force a new fetch.
- Some components produce minimal stdout output; check the `stderr` stream for error messages.
- If the backend returns an error, check the NOC backend logs for the `/components/:id/logs` request.

Alerts are not updating

- Check the **Polling Interval** in [Settings](#). If set to a high value, new alerts will appear infrequently.
- The WebSocket stream is not yet active — the portal always polls. This is expected behavior and not an error.
- Click **Refresh** on the Infrastructure page or navigate away and back to the Dashboard to trigger an immediate data pull.

Alert dismissed by mistake

- Dismissed alerts are moved to the **Resolved Alerts** table on the Audit page and cannot be re-activated from the portal.
- Contact the system administrator to re-raise the alert at the backend level if re-activation is required.

Topology diagram shows no nodes or all links red

- Navigate to **Infrastructure** and verify that component data is loading (indicates the backend is reachable and spokes are registered).
- If components load but the topology is empty, the topology endpoint (`/topology`) may have no data registered. Check the NOC backend for topology seeding.
- All-red links indicate that the NOC backend's link health checks are failing for every pair. Check backend logs for the health check job errors.

Returned to the login page while working

- The portal renews its token automatically, so this normally only happens when the realm's SSO session limit is reached or the session was revoked in Keycloak. The message shown is *"Session expired. Sign in again."*
- If it happens after only a few minutes, the renewal call is failing: check the browser network tab for a rejected request to the Keycloak token endpoint, and confirm the portal's `VITE_KEYCLOAK_URL` points at a realm reachable from the browser.

Relay logs are empty or a log screen looks frozen

- Container logs are agent-pushed snapshots, not a live tail. A **SNAPSHOT <n>s OLD** badge in the viewer header means collection stopped — check the container and its NOC agent, not the viewer.
- Log collection requires the component to be registered with its container name; a component registered with an endpoint only reports “*No logs available.*” even while healthy.

Settings do not persist after page reload

- Settings are stored in `localStorage` under the key `noc-ui-settings`. Verify the browser is not running in a private/incognito mode (which clears storage on close) or that storage is not being cleared by a browser extension.
- Nothing is written until a setting actually changes, so an untouched Settings screen leaves no entry under that key — that is expected, not a failure.
- If **Save Settings** appears to do nothing, check for the error toast: an interval below 5 seconds (or a non-numeric one) aborts the whole save, including the Alert Email.

End-to-End Walkthrough (Scenario A)

Audience: Anyone running or demonstrating a complete Scenario A cross-border PvP settlement — operators, integrators, reviewers, and demo drivers. **Scenario:** Scenario A — Enhanced Correspondent Banking (dual-layer HTLC Payment-vs-Payment settlement). **Scope:** This document ties the five per-portal manuals together into one “golden path”: the ordered sequence of actions, across every portal, that atomically settles a cross-border currency exchange between Bank A (BRL leg) and Bank B (ARS leg) under an agreed FX rate — either both legs settle or neither does.

Each per-portal manual documents its own screens in isolation (see the “Typical Workflows” section in each). This walkthrough is the cross-portal view: who does what, in which portal, in what order, and what a successful result looks like at each hand-off.

This is the **happy path** — the successful case with no exceptions. Timeout, revocation (refund), rejection, and account-freeze paths are covered per-screen in the individual manuals and in the Troubleshooting section of each.

1. The cast

Actor	Portal	Manual
Central Bank A / B treasury (issuance, tokenisation, redeem approvals)	Treasury	treasury.md
Central Bank A / B governance (registry KYC, account freeze, audit)	Governance	governance.md
Commercial Bank A operator (originator)	Bank	bank.md
Commercial Bank B operator (counterparty)	Bank	bank.md
Supervisor / regulator (read-only monitoring, disclosure)	Supervisor	supervisor.md
Network operations engineer (infrastructure and relay health)	NOC	noc.md

Running example used throughout: **Spoke A settles in BRL, Spoke B settles in ARS**. Bank A originates the settlement and locks its BRL leg; Bank B is the counterparty and locks its ARS

leg. At settlement each side receives the counterpart currency per the agreed FX rate. Substitute your own corridor's currencies as needed.

Unlike Scenario B (one-directional payment through a Hub AMM), Scenario A is a **bilateral, symmetric** exchange: there is no shared liquidity pool, so **both** banks must hold their own tCeBM and **both** lock a leg before settlement can complete.

2. Prerequisites

Before you start, confirm:

- **Access.** You have the portal URL and sign-in credentials for each role you will act as, provided by your administrator or Central Bank. Each manual's "Access and Login" section describes the sign-in screen for its portal.
- **A healthy network.** Both spoke networks and the relay must be operational so the two HTLC legs can be observed and settled atomically. A network operations engineer can confirm this in the NOC portal: all spokes report healthy on the Dashboard, and the relay is running. See [noc.md](#) → [Relay Status](#).

If a gateway or the relay is unavailable, the lock and settlement steps stall in a pending state rather than fail loudly — check relay health first.

3. The golden path

The path has seven stages. Stages A–B are one-time set-up per bank; Stage C is per-corridor negotiation; Stages D–F are the settlement itself; Stage G is routine liquidity management.

3.1 Stage A — Onboard the commercial banks

Each commercial bank registers with its own central bank and is approved through KYC. Both banks must be onboarded, because the settlement resolves each leg's receiver by bank identity.

#	Actor	Portal → screen	Action	Expected result
A1	Bank A operator	Bank → Onboarding	Complete the registration wizard (Steps 1–4)	Onboarding submitted to CB-A
A2	CB-A governance	Governance → Registry	Approve Bank A's pending KYC request	Bank A status becomes ACTIVE
A3	Bank B operator	Bank → Onboarding	Complete the registration wizard	Onboarding submitted to CB-B
A4	CB-B governance	Governance → Registry	Approve Bank B's pending KYC request	Bank B status becomes ACTIVE

Both banks can now sign in and reach the operational screens (see [bank.md](#) → [Access & Login](#)).

3.2 Stage B — Fund both banks with tCeBM

Both banks need transferable CBDC (tCeBM) to lock, because each locks its own currency leg. Funding is a two-step, central-bank-approved flow per bank: issue the fiat reserve, then tokenise it. Repeat the whole stage for Bank A (BRL) and Bank B (ARS).

#	Actor	Portal → screen	Action	Expected result
B1	Bank operator	Bank → Issuance Requests	Submit an issuance (deposit) request for the fiat amount	Request created, status PENDING
B2	CB treasury	Treasury → Issuance Approvals	Approve the request	Reserve minted to the bank; request APPROVED
B3	Bank operator	Bank → Reserve Tokenisation	Request tokenisation of the approved reserve	Escrow created, status PENDING
B4	CB treasury	Treasury → Tokenisation Approvals	Approve the escrow	tCeBM minted to the bank
B5	Bank operator	Bank → Dashboard	Refresh	tCeBM balance is non-zero — the bank is ready to transact

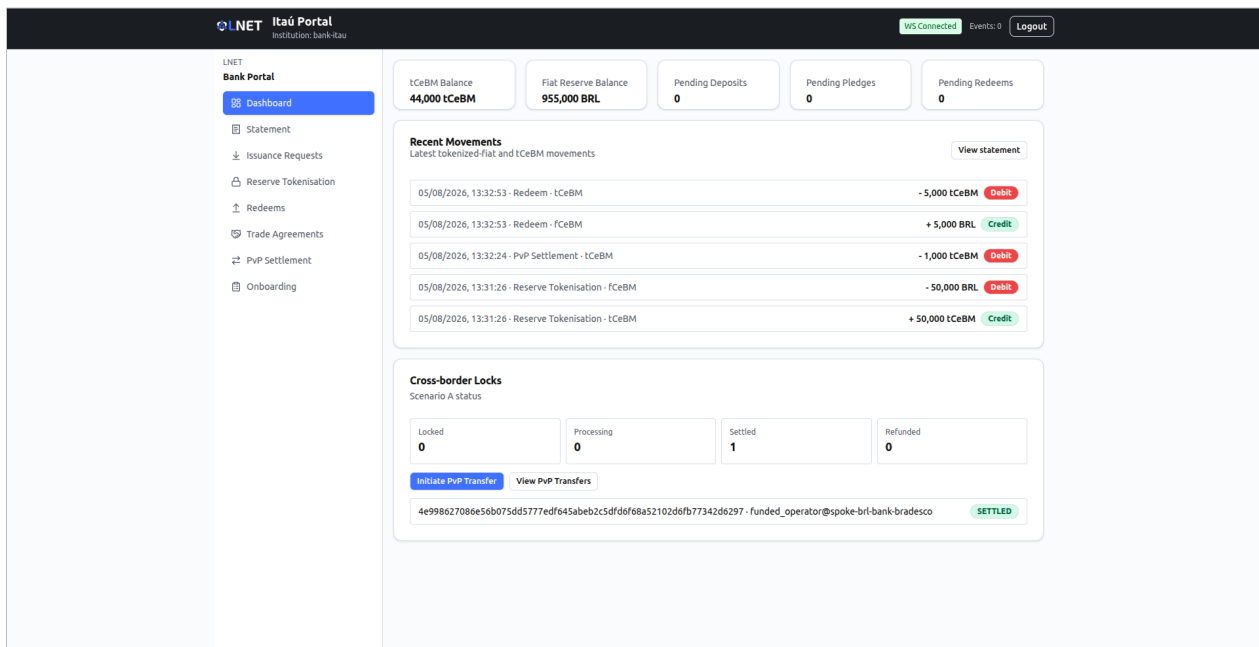


Figure 42: Bank Dashboard with a non-zero tCeBM balance after issuance and tokenisation

3.3 Stage C — Negotiate the FX trade agreement

The two banks agree bilaterally on the exchange: who sends what, in which currency, at which rate, and by when. Bank A proposes; Bank B accepts.

#	Actor	Portal → screen	Action	Expected result
C1	Bank A operator	Bank → Propose Agreement	Fill parties (counterparty, settlement agent, custodian, beneficiary) and trade terms (send BRL, receive ARS, rate, expiry ≥ 5 min ahead); Review then Confirm Proposal	Agreement created in PROPOSED state
C2	Bank B operator	Bank → Agreement Inbox → Agreement Detail	Open the PROPOSED agreement, review parties and terms, click Accept	Agreement moves to ACCEPTED — ready for PvP Settlement

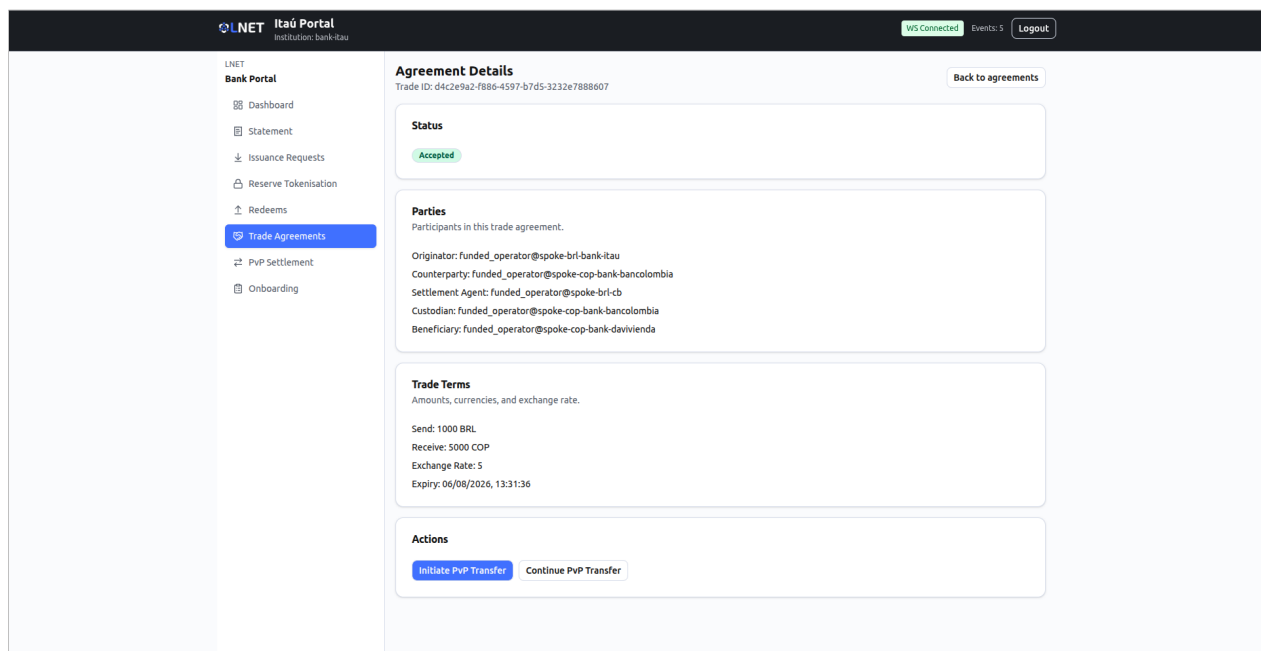


Figure 43: Bank Agreement Detail with the trade agreement in ACCEPTED state, showing parties and trade terms

3.4 Stage D — Originator locks the first leg

Bank A locks its BRL-side tCeBM in an HTLC. A Settlement Code (the hash lock) is generated and must be shared with Bank B out-of-band so it can lock the matching leg.

#	Actor	Portal → screen	Action	Expected result
D1	Bank A operator	Bank → Agreement Detail	On the ACCEPTED agreement, click Initiate PvP Transfer	New Transfer form opens pre-filled as Originator
D2	Bank A operator	Bank → New Transfer	Verify receiver and amount, set a timelock if required, Review then Confirm PvP Transfer	Contract created in LOCKED state; a 64-char Settlement Code is generated
D3	Bank A operator	Bank → Settlement Details	Copy the Settlement Code and share it with Bank B out-of-band (secure channel)	Settlement Code delivered to the counterparty

3.5 Stage E — Counterparty locks the second leg

Bank B uses the Settlement Code to lock its ARS-side tCeBM on Spoke B. Both legs are now locked against the same hash.

#	Actor	Portal → screen	Action	Expected result
E1	Bank B operator	Bank → Agreement Detail	On the same ACCEPTED agreement, click Continue PvP Transfer	New Transfer form opens pre-filled as Counterparty
E2	Bank B operator	Bank → New Transfer	Paste the Settlement Code, verify receiver and amount, Review then Confirm PvP Continuation	Counterparty contract created in LOCKED state on Spoke B

Once both legs are locked, the relay reflects `counterparty_locked = true` on the originator's contract, and Bank A's **Complete Settlement** action becomes available.

3.6 Stage F — Atomic settlement

With both legs locked, the originator reveals the secret to finalise the swap. Completion is atomic — it delivers tCeBM to both receivers at once.

#	Actor	Portal → screen	Action	Expected result
F1	Bank A operator	Bank → Settlement Details	Once the Completion Code is available and the counterparty leg is locked, click Complete Settlement and confirm	Both legs move to SETTLED; Settlement Reference and Completion Code populate
F2	Bank B operator	Bank → Settlement Details	Refresh (the page auto-polls every 5 s)	Counterparty contract reaches SETTLED; the BRL counterpart is delivered to Bank B's receiver
F3	CB treasury	Treasury → HTLC Monitor	Review	Both contracts show as SETTLED end-to-end

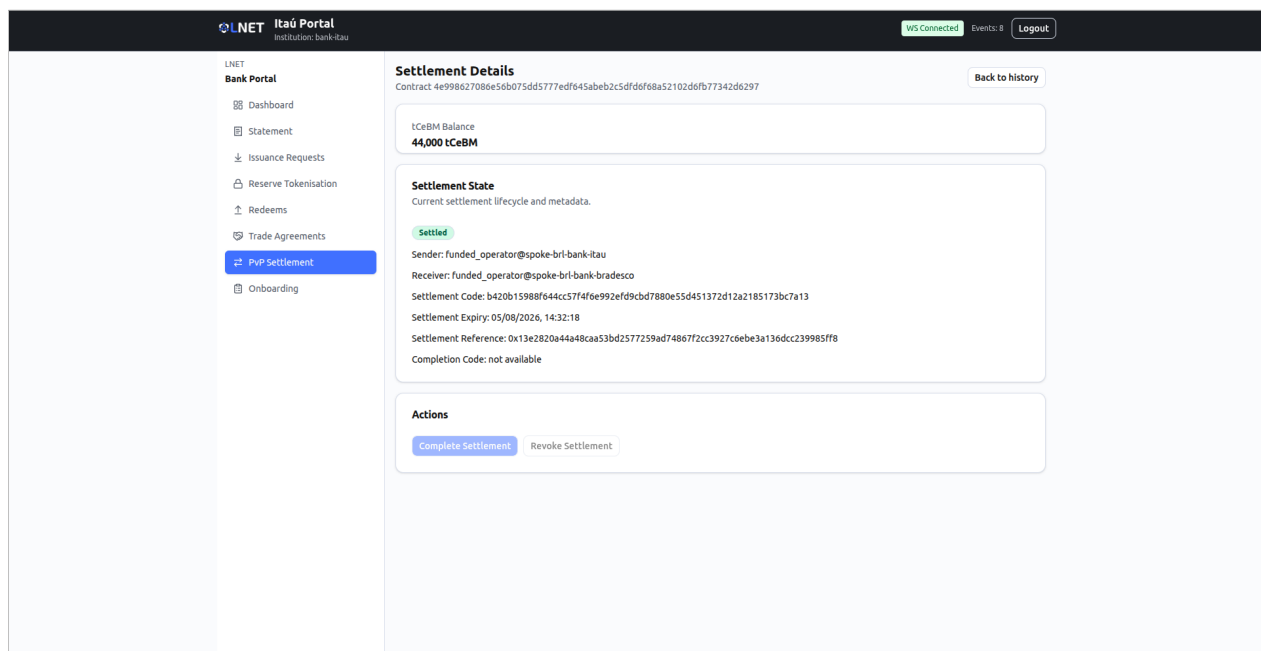


Figure 44: Bank Settlement Details with status SETTLED and the Settlement Reference populated

3.7 Stage G — Redeem back to fiat

Routine liquidity management: a bank that received tCeBM can redeem it back to fiat with its central bank's approval.

#	Actor	Portal → screen	Action	Expected result
G1	Bank operator	Bank → Redeems	Submit a redemption request for the tCeBM amount; Review then Confirm	Redeem request created, status PENDING (Zeto transfer executes)
G2	CB treasury	Treasury → Redeems Approval	Approve the fiat release	Redemption approved; redemption ID populated

4. Oversight overlay (optional)

Central-bank and supervisory oversight runs alongside the settlement flow. These actions do not block the settlement but exercise the control surface:

- **Account freeze / unfreeze.** A governance operator can freeze a participant account (with a reason of at least 10 characters) and later unfreeze it. On the successful path all participants stay ACTIVE throughout. See [governance.md → Accounts](#).
- **Transfer limits.** A treasury operator can set a daily transfer limit per participant, which the settlement paths respect. See [treasury.md → Transfer Limits](#).
- **Multi-party AML disclosure.** A supervisor opens a 2-of-N disclosure request for a specific transaction and a second party co-signs it to reach quorum, authorising decryption of shielded details for investigation. See [supervisor.md → Investigation](#) and [supervisor.md → Audit Vault](#).
- **Continuous monitoring.** Throughout the flow, the Supervisor portal's [Settlement Health](#) view tracks HTLC expiry risk, and the NOC portal's [Relay Status](#) confirms the relay is observing both spokes.

5. Success checklist

The walkthrough is complete when all of the following hold:

- ☐ Bank A and Bank B are both **ACTIVE** in the Registry (Governance → Registry).
- ☐ Both banks hold a non-zero **tCeBM** balance after tokenisation (Bank → Dashboard).
- ☐ The FX trade agreement is **ACCEPTED** (Bank → Trade Agreements).
- ☐ The originator leg is **LOCKED** and the Settlement Code has been shared (Bank → Settlement Details).
- ☐ The counterparty leg is **LOCKED** (`counterparty_locked = true` on the originator contract).
- ☐ Both legs reach **SETTLED** with a Settlement Reference and Completion Code (Bank → Settlement Details).
- ☐ No expired settlement window and no stuck relay events (Treasury → HTLC Monitor; NOC → Relay Status).